

FLAVOR FOR ALL

Everyday Recipes & Creative Pairings Inspired by THE FLAVOR MATRIX

By James Briscione and Brooke Parkhurst

Introduction

For both of us, sharing THE FLAVOR MATRIX has been an incredible journey. James created an exceptional reference book that is recognized as much for its innovation as for its thought-provoking graphics. He distills the complex chemistry of flavor and gives everyone a lesson in food pairing.

The diversity of our readers surprises and excites us: food geeks, culinary students, dialed-in soccer moms, mixologists, and “weekend warriors.” They might study a matrix-a-week or tackle twenty pages of chemistry and cooking in a forty-eight-hour period. No matter their background, they all ask us the same question: “How do you cook at home?”

That oft-repeated question spurred us on to write a companion book, an index of no-fail, kick-ass, chemistry-enabled recipes for everyday cooking.

While we are always thinking about how flavors and pairings can make every meal as delicious as possible, the reality is, we’re busy. We have a new restaurant. We have two young children at home. We have around one hour at the end of the day to make something tasty. That means that every dish on our table cannot be a deeply researched masterpiece. The matrices are always in our heads, but we also have a “cheat sheet” of greatest hits.

THE FLAVOR MATRIX taught us a new way to think about cooking, but not every recipe can be Roast Turkey with Apricot, Sage, and Pistachio Chutney. Sometimes a subtle twist on a classic or a gentle pop of unexpected flavor is enough. The recipes in FLAVOR FOR ALL reflect our simple, thoughtful, flavor-focused approach to home cooking. They’re deeply delicious, yet easy; unexpected, but still familiar. Dishes like:

- Asparagus and Potato Chip Gratin
- Roast Fish with Creamy Tomato and Bacon Faro

- Red Wine and Chocolate Bread Pudding

And we never ignore the classics. THE FLAVOR MATRIX revealed why some of our favorite flavor combinations—think cheese pizza or pork chops and applesauce—work so well together.

Margherita pizza contains the compound 4-methylpentanoic acid, a chemical commonality found in cheese, tomato sauce, and baked wheat, which serves to enhance the flavor of all three when combined. In the pairing of pork and apples, 2-acetylfuran is one of the compounds that links this classic combination together. You always knew that these dishes tasted great, and THE FLAVOR MATRIX helps you understand why. FLAVOR FOR ALL, in turn, builds on the science behind some of these classic combinations and fine tunes the technique and ingredients to create 100+ epic recipes.

For example, the aforementioned pork and apple pairing leads us to create Apple-Brined Pork Chops with Caramelized Onion and Bourbon Gravy. The compound 2-acetylfuran found in pork and apples is also prevalent in bourbon and alliums. This recipe kicks off with a simple brine made from items you already have in your fridge or pantry—apple juice, soy sauce, and mustard—so you can mix it up before work or school and be ready to cook the juiciest, most flavorful pork you’ve ever tasted as soon as you get home.

The strong pairing between green beans, mushrooms, and toasted nuts shows that the classic Thanksgiving green bean casserole is well loved for a reason. In FLAVOR FOR ALL, however, we pinpoint what’s been missing all along: bacon, mustard, and dill.

FLAVOR FOR ALL is a book about how we live—to eat! We are working parents first, food and wine-obsessed innovators second. We want this book to be functional, fun, and indispensable. The breakdown of the book is just like us—simple and focused on the food. We begin with a brief introduction to review the concept of flavor pairing and to set the scene for the recipes to come.

Next, we wanted to create section of the book that would act as a culinary cheat sheet. Something that could help new and experienced cooks alike understand how to build maximum flavor into every single dish they prepare. This next level approach is something James has taught to cooks around the world through his role as the Director of Culinary Research at the Institute of Culinary Education and now as a Food Network personality. Whether it's about how proteins react to salt, the right way to boil an egg, or the proper technique behind a perfect bowl of spaghetti, this insider knowledge will make everyone a more confident and capable cook. Then we decided not to do that.

Instead, we took those nuggets of culinary wisdom gained over a lifetime in the kitchen and put them where they would be most helpful- in the recipes themselves. All of *our* best-loved cookbooks are full hand written notes in the margins. Helpful reminders, or brief explanations of why something is done a specific way. We love those little margin notes, so we decide to go ahead and create them for you- right there in the margins as if you had written them yourself. It's our way of saying, "Hey, here's something you should know about this recipe" without interrupting the cooking process. These notes range from the science behind what is happening in a particular step of the recipe to gentle reminders about basic technique.

Beyond the science of cooking, we also wanted to pull the curtain back on the theory of taste development. Creating a balance of acidity, heat, umami, and the other elements of taste make for the most delicious dishes. Within each savory recipe (everything up to Drinks), the ingredients that directly contribute each of these tastes are annotated. We believe this will help readers understand what each ingredient brings to a dish and therefore encourage experimentation. For example, if a recipe calls for soy sauce (salt, umami) but last night's take-out sushi bender left your bottle empty, you know that you can deliver those same tastes to the recipe with any of the following: fish sauce, Worcestershire sauce, parmesan cheese, tomato paste, mushrooms, seaweed, or prosciutto. Acidity in a dish can come in many forms, as well. If an ingredient list will show that acid is important in the recipe. However, where that acid comes from is less so; lemon juice, lime juice, vinegar, wine all contribute the taste of acidity,

each with its own flavor. When it comes to the construction of recipe, acidity is essential for the finished dish but the flavor that accompanies that acidity can be up to you. Basically, if you see a taste in parentheses in the ingredient, you know that you are free to swap that ingredient for another of your choice that provides the same taste. (Just don't blame us if it doesn't taste perfect!) In the following chapter you will find a brief discussion of each of the tastes, what they typically contribute to a recipe and the things you should keep in mind when substituting ingredients.

The thrust of this book, however, is its recipes. The elevated yet everyday dishes engineered for real life.

In FLAVOR FOR ALL we organized recipes the same way we eat—by dish, not by meal. In our house, the only difference between lunch and dinner is the time of day it is consumed. And yes, cheese is its own food group in our house. We are equal opportunity eaters—eggs for dinner and steak for breakfast? If the mood strikes us, why not? We simply desire food that is interesting and delicious. Readers can mix and match dishes to suit their own tastes.

Not only can readers mix and match different recipes from within the book, we empower readers to customize each recipe to perfectly suit their own tastes. Every recipe chapter begins with an introduction to the foods contained in that section. These introductions are filled with information that will give cooks of any skill level more confidence and competence in the kitchen. In the introduction to the chicken chapter you will learn the truth about cooking meat on the bone, why 165° F is an important temperature for chicken, and the best way to approach cooking different cuts of poultry.

After the introduction, readers will find protein-based (chicken, fish, beef, etc.) recipes designated by cooking method. We chose this grouping so readers can make their own choices—some love grilled chicken breasts, others prefer chicken thighs when they fire up the

coals. With the advice from the introduction, readers can confidently approach any recipe with their favorite cut for a given cooking method.

FLAVOR FOR ALL is thoughtful food for everyone. Our book is for those who want to continue THE FLAVOR MATRIX journey. It's also for those who never quite joined the journey, the cooks who might have been intimidated by illustrations that looked better suited for a chemistry book than a cookbook. FLAVOR FOR ALL combines the brains of THE FLAVOR MATRIX with the feel of your favorite sauce-stained, "staple" cookbook. Fans of THE FLAVOR MATRIX have asked us for just one thing—more recipes. Here they are.

As you skim the recipes in the following pages, you will immediately see how it is that we devised each dish. It's the same way that James approaches the menus in the kitchen of our restaurant Angelena's. It begins with 'ingredient inspiration' each recipe starts with an interesting combination of ingredients which THE FLAVOR MATRIX suggested will taste great in a dish. In case you need a little more insight, we also show the specific compound that contributes to that pairing and list the aroma that compound exhibits in the given ingredients. We give that information for two reasons- first, to show how our previous research lead to the creation of this specific recipe. Second, and more importantly, we think this makes the creative part of cooking more transparent and will hopefully inspire further creations and variations on what we have begun here.

Once our 'ingredient inspiration' was determined, we would write down those 3-5 ingredients on a note pad and treat them like a low-pressure mystery basket on an episode of Chopped. Sometimes the idea for the dish would come to us instantly, other times that combo rolled around the back of our heads for days or weeks. We would study the ingredients, think about them- what form could this ingredient take? Say for example we found eggs were a good pairing. Should we use boiled eggs? Maybe a hollandaise would be better? A super-egg aioli might be easier to make. Cured egg yolks, could be grated over the dish at the last minute. The possibilities for each ingredient were many and varied. We chose one solution to the puzzle of

those ingredients. It doesn't mean it was only way to put the pieces together, just the one that we liked best. Each set of ingredients might inspire you in a slightly different way. That's ok. Our goal is to make this book a conversation, not a monologue.

With all of that said, if you're like us many days of the week and you need to just turn your brain off and cook because it's time for dinner, you can do that too. We tried to create these recipes so that they were accessible and approachable. Well suited for Wednesday dinner, with flavors that fit at a holiday table.

Ingredients

Flavor Pairing, in 500 words or less

Our previous book, THE FLAVOR MATRIX, was developed around and dedicated to the art of flavor pairing. A relatively new concept in the culinary world, flavor pairing begins with understanding where the flavors in food come from. It's a process that requires examining food on a molecular level. But don't worry, there's no need to dust off a microscope before you get into the kitchen. You simply need to know that there are thousands of different compounds naturally present in all foods and each of those compounds has its own unique aroma. The combination of all of these aromatic compounds in a single bite is what we call flavor. What you taste when you bite into an apple is the result of over 400 different compounds all working together to create the unique flavor of that specific apple. Not mention the sugars, acids, and textures that all contribute the perception of flavor—but more on that in a bit.

Understand that these compounds create the flavor of each ingredient is step one in flavor pairing. In the THE FLAVOR MATRIX we did a deep dive breakdown of 58 different ingredients groupings to illustrate how flavor was constructed in food. With that under our belts, we turned to the flavor pairing theory, a concept set forth by Heston Blumenthal and his R&D team at *The Fat Duck*. The flavor pairing theory states that if two ingredients share a significant number of flavor compounds or if they have a single similar compound in high concentration, then we can predict that those two ingredient will taste good when combined in a dish. Every recipe we create in this book is based on this idea that the ingredients are chemically matched for maximum deliciousness.

Well matched flavors are essential in any great dish, but taste cannot be left as an after thought. The most perfectly matched flavors will fall flat if not accompanied by dynamic and balanced tastes. Flavors come from aromatic compounds that are detected by the nose. Taste comes from elements of food that are detected by the tongue. There are currently 6 tastes that we are

aware of: fat, sour, salt, umami, bitter, sweet. Each of these is recognized by dedicated receptors on the tongue. To be a great cook, it is important to understand what each of these elements does in food both in the structure of a recipe and its sensual qualities in a finished dish.

Fat

Fat, an essential nutrient, has only recently been recognized a taste. It was always known to contribute to mouth-feel, the physical sensation created in the mouth by a bite of food, but in the past few years scientists have discovered that there are cells on the tongue that respond to the presence of free fatty acids in food. Fat enhances the tastes of Umami, Sweet and Salty and balances the taste Spiciness in food.

In most recipes, fat is relied on to coat foods, prevent sticking, promote browning and/or provide moisture. Any fat can provide these abilities, however the type of fat you choose will have effect on the final flavor of the dish. The best demonstration of this is scrambled eggs. Imagine you make three batches of scrambled eggs, one with butter, one with olive oil and one with sesame oil. They will all cook the same and will the finished product of all three will look identical. However, the flavor of each egg will be dramatically different. Eggs have very little flavor on their own, so the flavors of the 3 different fats will really come through. No one fat is better than the other, it is simply a matter of preference.

In something simple like a scrambled egg, the type of fat used has little impact on the finished dish. Eggs are very quick cooking and never reach high temperatures. If instead of eggs, you were sautéing a chicken breast, you need to consider not only the flavor of the fat you wanted to use, but also how it holds up to temperature. This quality of fats, known as the 'smoke point,' is the temperature at which the fat begins to break down. When a fat reaches it's smoke point, it will begin to impart off flavors to the food and cause the food to brown or burn more quickly. Foods will not begin to brown in a saute pan or on a grill until their surface temperature surpasses 300°F, meaning the external temperature should be at least 350°F to create browning without completely drying out the item. As you will see from the chart below, this means that

butter, extra virgin olive oil, coconut oil and sesame oil should not be the primary fat used for browning meats because those fats will break down at those temperatures. You will also note that this chart dispels the common cooking myth that you should never fry in olive oil. Frying in olive oil is fine and it is done commonly throughout Italy. What you should consider however, is that the olive oil will impart a specific flavor into the food while frying. Also, olive oil is not nearly as durable as other oils so it will need to be replaced more often if frying large quantities, which gets expensive. A final word about olive oil: the compounds that give extra virgin olive oil its unique flavor will break down even before the oil reaches its smoke point. This means that expensive oils with deep complex flavors should never be heated. They should be reserved for finishing and dressing items just before serving.

Whole butter is more than just fat. Most butters are around 80% fat, the other 20% is a combination of water and milk solids. Milk solids contribute a significant amount of flavor, but they are delicate and will burn well before the fat in the butter reaches the smoke point. This is why melted butter and clarified/ghee (where the milk solids have been removed) have different smoke points. Butter is best to promote browning of other foods, but it can burn at high temperature so it is not ideal for searing. In baking, fats are not as easily swapped for one another. Because of its unique composition butter can provide structure in baked goods that simple oil does not. In regular recipes you can use any type of oil you'd like, as long as the temperature capacity is appropriate. Just remember that there is a flavor effect to the oil you chose.

Type of Fat	Smoke Point
Safflower Oil	510°F/265°C
Light/Refined Olive Oil	465°F/240°C
Soybean Oil	450°F/230°C
Peanut Oil	450°F/230°C
Clarified Butter/Ghee	450°F/230°C
Corn Oil	450°F/230°C
Sunflower Oil	440°F/225°C
Vegetable Oil	400-450°F/205-230°C
Canola Oil	400°F/205°C
Grapeseed Oil	390°F/195°C
Lard	370°F/185°C
Vegetable Shortening	360°F/180°C
Sesame Oil	350-410°F/175-210°C
Butter	350°F/175°C
Coconut Oil	350°F/175°C
Extra-Virgin Olive Oil	325-375°F/165-190°C

Acid/Sour

Acid's main job in a recipe is to punctuate (provide a little punch) and help balance other tastes. Specifically, acid is the counter to Sweet and Fat. Though it works with these tastes in different ways. Sour is the taste opposite of sweet, so adding acidity to a dish will effectively reduce the amount of sweetness perceived in a dish. Fat can overwhelm the palate, coating the tongue and making the all other tastes more difficult to perceive. Acidity in a dish will help cut through that fat, in a sense cleaning the palate and preparing it receive new tastes.

Acid can also work to *denature* proteins, as in ceviche (denaturation is the more accurate name for what happens to a protein in the both the cooking and curing processes). Acid essentially 'cooks' the protein. This is why marinades that include vinegar or citrus juice (specifically lemon or lime) should only be used for short periods, otherwise the acid in the marinade will begin to 'cook' the protein before your ever get it on the grill which can leave the meat dry or tough.

Acid can have a similar effect on plant cells. Acid in one ingredient can break down the cell walls in plants, collapsing the structure of that item. This is wilted salad effect. When combining an acid with crisp raw fruits or vegetables, it should be done at the last minute to preserve the texture. Salads should tossed with dressing just moments before serving, while coleslaw is best dressed well in advance so that the crunchy cabbage can soften. Acid will also break down chlorophyll in green vegetables, causing them to turn from vibrant green to a dull olive green color. This effect is amplified under heat, so if you want your sautéed green beans to taste bright and lemony, you should only add lemon zest while they cook, and give them light squeeze of fresh lemon after they are on the plate.

Salt

In food, salt works enhance the perception of all other tastes. Salt is like an alarm clock for tastes buds, it makes sure they are on their toes and ready to grab whatever tastes are coming.

In low concentrations salt will suppress the perception of bitter tastes, effectively making foods tastes sweeter. This is why a small amount of salt is essential in any sweet dish. In higher concentrations, typical of savory dishes, salt enhances the perception of umami.

Salt also helps to draw moisture out of food, as it does in the curing process. However, when foods are exposed to lower concentrations of salt (than those used in curing) the results can greatly enhance a dish. Lightly salting a protein before cooking can help remove moisture from the surface of meat, allowing it to develop a better sear when cooking. It can also draw excess moisture from fruits or vegetables to help concentrate their flavor.

While salt helps to draw moisture out of food, it can also help proteins retain moisture when cooked (read: be juicier). When salt is dissolved into water, as in a brine, muscle fibers in proteins will absorb the excess moisture through the process of osmosis in an effort to reach equilibrium. When the proteins in the muscle denature in cooking, that moisture is retained in a matrix created by the unraveling proteins and your meat is juicier.

When you salt your food matters. Salt should be added gradually throughout the cooking process. This will help all components of the recipe to be properly seasoned. The best illustration of this is cooking rice. Boil one cup of rice in salted water so that the rice absorbs salt throughout the cooking process. Boil a second cup of rice with no salt and stir the salt into the rice once it is finished cooking. The difference in that taste of the two will be dramatic. The rice cooked in salted water will have a more rounded, 'full' flavor, because the salt has been completely absorbed during the cooking process and slowly releases to the tongue. The rice salted after cooking will taste salty because the salt remains on the surface of the grains hitting the tongue immediately and overwhelming salt sensors. Yet, as you chew the grains of rice they will taste flat and bland.

Umami

Umami is that what-makes-this-taste-so-good element in food that can sometimes be hard to identify. Umami was recognized as a taste in 1908 by a researcher at Tokyo University, but it did

not gain wide recognition or acceptance until nearly 100 years later. Umami is sometimes described as savoriness, and is identified on the tongue by detection of glutamate, an amino acid. Umami cannot be created in the cooking process in the same way that sweetness can be developed by slowly caramelizing onions, rather it is naturally present in a ingredients. However, umami can develop over time and is found in all fermented and aged ingredients. Think of umami as a flavor secret weapon. Anytime you can add umami-containing ingredients to a recipe—savory or sweet—it will help to enhance all other tastes.

Umami Rich Foods

Seaweed

Tomatoes, Garlic, Asparagus, Mushrooms, Truffle

Scallops, Shrimp, Uni, Clams Mussels

Anchovy

Dried Hams

Parmesan cheese

Miso

Soy Sauce, Fish sauce, Oyster Sauce

Kimchi

Eggs

Bitter

Foods that are overly bitter are generally not palatable. However, bitterness in food can be desirable when properly crafted. Bitterness in food generally comes from tannins, naturally occurring phenolic compounds found in a variety of ingredients like coffee, tea, plants in the Brassicaceae family, olives, some nuts, and cocoa. Bitterness is a smaller, but important part of the taste puzzle and should not be ignored in recipes. Without the pleasant bitterness of cocoa a pan of brownies would be more like a big block of sugar cookie.

Bitter is also believed to have some psychological effects on appetite. In many cultures a sip of something bitter- Campari, pernod or ouzo is consumed before a meal to help stimulate that

appetite. Research has shown that bitter compounds do, in fact, stimulate the release of hunger hormones. Bitter foods can also stimulate the production of saliva and stomach acid, aiding the digestion process.

Sweet

The human body is programmed to crave sweet foods. They are instantly recognized as a good source of quick energy. In food sweet plays well with all other tastes: enhancing salty, umami, and fat and balancing sour and bitter. While sweet tends to be the dominate taste in desserts, it remains an important component of savory recipes as well. As with all other tastes, the goal with sweet should be to create balance. Rarely when constructing a recipe do you want to allow one taste to dominate, in the same way that a symphony dominated by brass instruments won't sound as pleasing.

When looking to build flavor in a dish, we prefer to find ways of developing sweetness that go beyond simply adding sugar. Sweeteners come in many forms and sugars are naturally in most fruits and vegetables. Careful cooking can enhance the natural sugars in most foods. There are also ingredients that enhance the perception of sweetness. Citrus zest, vanilla and warm spices with cinnamon, nutmeg and allspice can all make foods taste sweeter without adding sugar in any form.

Vegetables

Salad-y things

Umami in a Bottle *Dijon, soy sauce, fennel seed, red wine vinegar*
Grilled Leeks Vinaigrette *mushroom, almonds, feta, tarragon*
Avocado Toast *fennel, apple, cocoa, ancho*
Avocado 'Succotash' *corn, basil, tomato, lime*
Roasted Beet Salad *orange, fennel, hazelnuts, yogurt*
Coco-Slaw *coconut milk, cabbage, cilantro, carrot, lime, chile*
A Different Kind of Green Salad *cucumbers, grapes, honeydew, jalapeno, mint*
Watermelon Bruschetta *tomato, lemon zest, garlic, basil, pecorino*
Sicilian-Style Peach Salad *peach, fennel, olive, mint, pepper*

Sides

Brussels Sprouts Kabobs *chipotle, yogurt, dill, garlic*
Asparagus Gratin *parmesan, mushrooms, potato chips*
'Caesar' Artichoke Dip *anchovy, egg yolk, lemon, Worcestershire*
Pumpkin Pilaf *jalapeno, cinnamon, fennel seed, pumpkin, Israeli couscous*
Whole Roasted Cauliflower *honey mustard, cilantro, India pale ale*
Grilled Broccoli *pistachio, lemon, parmesan*
Eggplant Dip *yogurt, dill, sesame, pomegranate*
Squash Skewers *bacon, ginger, carrot, vinegar*
'Scalloped' Parsnips *lemon, cream, cheddar*
Summer Squash Gratin *tomato, mint, ricotta*

Grains & Pasta

Spaghetti *asparagus, peas, madeira, brown butter, lemon*
Mac and Cheese *creamed kale, caramelized onion*
Rigatoni alla Gricia with Mushroom 'Pancetta' *maple syrup, soy sauce, vinegar*
Quinoa Salad *hazelnuts, vinegar, sweet potatoes, honey*
Kale Salad *buckwheat, bell pepper, tomato, feta*
Summer Spaghetti Vongole *clams, corn, tomato, dill*
Sweet Pea and Mushroom Rigatoni Carbonara *egg, sweet pea, mushroom, pasta*
Vegetable Bolognese *tomato, fennel, olive, sage, orange, almond*

Say Cheese

Buffalo Bake *tomato, cauliflower, blue cheese*
Baked Brie *cranberry, Dijon, orange*
Cheese Plate *cherry, olive, sage*
Grilled Cheese Sandwich *kiwi, jalapeno, bacon*
Grilled Cheese Sandwich *kimchi, broccoli, cheddar*
Grilled Cheese Sandwich *rye bread, gouda, cumin, cream cheese, hot sauce*
'To-Mayo' Sandwich *bacon, lettuce, tomato*
Frico aka Crispy Cheese Cake *broccoli rabe, onion, potato*

Land

Chicken

Pan Roast Chicken with Creamed Green and Potato *mustard greens, cream, lemon, potato*
Pan Roast Chicken with Herbed Carrots and Mustard *dill, butter, lemon, carrot*
Ultimate Honey Mustard Chicken *yogurt, mustard, basil, honey*
Southwest Grilled Chicken Salad *grapes, grilled corn, tomato, cilantro*
Creamy Tomato Chicken Ragu *tomato, cream, onion, vodka*
Ginger Beer Battered Chicken Fingers *ginger beer, lime, chili powder*
Indian Chicken and green Vegetable Stir Fry *bell pepper, zucchini, asparagus, cilantro*
Spicy Chicken and Pomegranate Lettuce Wraps *pomegranate, chili, lemongrass*

Beef

Slow Cooker Sweet Potato and Lager Beef Stew *sage, sweet potato, beer*
Pan Seared Steak with Bourbon Raisin Sauce *raisins, bourbon, mustard, Worcestershire*
Seared Steak with Almond-Cherry Pesto *almond, garlic, oregano, cherry*
Grilled Beef and Eggplant with Espresso Butter *eggplant, espresso*
Chianti Braised Beef with Grits *red wine, corn, butter*

Pork

Pan Seared Pork Medallions with Apples and Balsamic *coffee, balsamic, apple, bourbon*
Slow Cooker Pork with Mustard, Orange and Sage *mustard, orange, sage*
Apple-Brined Pork with Caramelized Onion and Bourbon Gravy *apple, onion, whiskey, cream*
Grilled Pork with Spicy Pineapple and Basil Relish *pineapple, corn, jalapeno*
Brown Butter, Maple and Pear Pork Roast *brown butter, maple, pear*

Lamb

Braised Lamb with Blue Cheese and Mushroom Potato Gratin *blue cheese, mushroom, potato, brandy*
Braised Lamb with Spicy Peach Chutney *polenta, corn, peaches, Grenache*
Roast Lamb and Carrots with Mint and Lemon *carrot, mint, lemon, garlic*
Horseradish and Citrus Crusted Lamb Roast *orange, basil, horseradish*
Grilled Lamb with Anchovy Vinaigrette *radish, anchovy, lettuce*

Sea

Fish

Mild (Bass, Halibut, Snapper, Grouper, Flounder)

Pan Seared Fish with Broccoli, Balsamic and Garlic Croutons *breadcrumbs, broccoli, balsamic*
Pan Seared Fish with Sweet Pea and Ham Quinoa Pilaf *peas, mushroom, ham*
Pan seared Fish with Bacon and Creamy Tomato Farro *tomato, faro, bacon cream*
Pan Seared Fish with Lemon and Dill Lettuce Saute *lettuce, lemon, dill*

Better-than-it-sounds Milk Poached Fish with Basil *milk, onion, basil*

Full Flavored (Tuna, Salmon, Cod)

Miso and Maple Glazed Fish *maple syrup, miso, vinegar*

Pan Seared Fish with Brown Butter and Grape Vinaigrette *brown butter, sesame, grapes*

Fennel Roasted Fish with Chilled Green Beans, Sesame, Dill *fennel, green beans, sesame, dill*

Pan Seared Fish with Sweet Potato, Mushroom and Poblano Hash *sweet potato, mushroom, Pan*

Roast Fish with Spiced Potatoes and Avocado Cream *potato, tomato, avocado*

Shrimp & Shellfish

Poached Shrimp Salad with Chilled Carrot Soup *carrot, ginger, red wine*

Creamy Shrimp Pasta with Charred Tomato and Pumpkin p *pumpkin, cream, bread, pasta*

Brazilian Shrimp and Coconut 'Stew' *tomato, coconut, pepper*

Lump Crab Bruschetta *tomato, sesame, toast*

Crab and Crisp Apple Salad *apple, mayonnaise, chive, lime*

Grilled Lobster and Butternut Squash with Tomato, Olive and Preserved Lemon Relish *butternut squash, olive, tomato*

Drinks

Cocktails

Grapefruit Cucumber Spritz *grapefruit, gin, Aperol*

Cucumber, Apple, Sauvignon Blanc Sangria *cucumber, sauvignon blanc, grape, apple*

Peach, Basil, Rosé Sangria *peach, basil, lime rosé*

Cocoa and Lemon Old Fashioned *cocoa, whiskey, lemon soda*

Spiced Butternut Squash Old Fashioned *bourbon, butternut squash*

Mezcal Michelada *mezcal, beer, coriander*

Non-alcoholic

Raspberry Vanilla Soda *vanilla, raspberry, thyme*

Grapefruit Shrub *grapefruit, honey, vinegar*

Spicy Peach Soda *peach, pepper, ginger*

Homemade Cola *citrus, star anise, cinnamon, mint*

Baked

Savory

Walnut and Olive Focaccia *olive, walnut, rosemary*

Cherry Tomato and Nectarine Crostata *cherry tomato, nectarine, balsamic, mint*

Sweet

Nutty Better Cookies *peanut, fish sauce, cayenne*

Tropical Caramel Upside Down Cake *citrus, banana, pineapple, caramel*

Sweet and Sour Plum Upside Down Cake *orange, honey, vinegar, pound cake, cornmeal*

Bourbon-Pecan and Coffee Sweet Rolls *bourbon, pecan, coffee*

Spicy Chocolate Crinkle Cookie *chile, coffee, cocoa*

Lime & The Coconut Popcorn Mousse *popcorn, coconut, lime*
Carrot Cake Creme Brulee *carrots, walnut*
Rhubarb and Bourbon Zabaglione *rhubarb, bourbon*
The “King” of Brownies *peanut butter, banana, chocolate*
Not-That-Kind-of-Mushroom Brownies *mushroom, cocoa*
Brown Butter Espresso Carrot Cake *walnut, espresso*
Brown Butter Ice Cream Sundae *brown butter, vanilla*
Nutella and Brown Butter Squash Panini *vanilla, squash, Nutella*
Chocolate and Red Wine Bread Pudding *merlot, chocolate, whiskey*
Vernaccia-Poached Pears with Pecorino *Vernaccia, pecorino*

Vegetables

Preparing vegetables and fruits can be a tricky undertaking. Taste and texture are extremely important for fruits and vegetables. Think of that perfectly ripe summer strawberry, still warm from the sun. You bite into it, it's juicy and sweet with just a touch of acidity. The creamy flesh yields to tiny, crunchy seeds as its perfect perfume fills your nose. For those few, fleeting weeks when strawberries are at their peak, nothing could be better. But for the rest of the year strawberries just aren't the same- too firm, too much acidity, too soft, no aroma- any number of short comings leave you underwhelmed because there is no way to replicate what nature does so perfectly. However, we can get close- adding a few drops of vinegar to overly sweet strawberries, a pinch of sugar for one that are too tart. With a little time, that sugar will also help soften their texture, though left to marinate too long and they turn to mush. If you learn to think about each ingredient in it's individual components of taste, you will understand how to best prepare it. In this chapter, we try to find ways to get the absolute best out of each ingredient. Whether it's a salad dressing carefully assembled to full complement the green aromas lettuces or carefully coaxing a sweet, smokiness from the simple leek.

One of the most common questions we get from home cooks is: *"Why do vegetables at restaurants taste so much better than the vegetables I cook at home?"*

The answer is simple, the majority of people over cook and under season vegetables. Couple that with the fact that restaurants are buying their vegetables directly from the same people that sell to your local grocery store. Restaurants are typically serving you vegetables that are days fresher than they are when purchase them from a grocery store and prepare them yourself. So how do you get better vegetables at home?

- 1.. cook your vegetables less
2. season them more- that means salt, spices and other aromatic ingredients

3.shop at a farmer's market or learn how to pick out the freshest vegetables at your local grocery store.

Cook Vegetables Less

Texture is one of the most overlooked factors in taste. Research has shown that crispy textures greatly enhance both the perception of freshness in food as well as enjoyment. An Oxford study which had participants wear headphones while eating from the same bowl of potato chips found that those with loud crunching noises playing in their headphones perceived the potato chips as fresher and more flavorful than those with muffled sounds playing in their headphones.

When vegetables are fresh and properly prepared they should retain some of their natural crunch, making them taste better. This chapter will explore a number of different methods for preparing vegetables, many of which involve cooking, some will not. Heat, salt and acid are the main methods of softening a vegetable's texture. This is extremely important to keep in mind in the kitchen. If you want crisp greens and a lively crunch in your salad you should not season or dress it until just moments before serving. On the other hand if you find yourself with tomatoes that are a bit bland and mealy, dress them with vinegar and salt early on to help soften their texture and enhance their flavor.

When it comes to cooking, the best methods varies depending on the item, but high heat and short cooking times are generally best for most vegetables. Cooking vegetables quickly with high heat allows you to soften and flavor the exterior while maintaining interior texture.

Season More

Salt matters. Salt's role in food is to enhance flavor. Though we refer to it as seasoning, salt is a flavor enhancer. It is not adding any flavor to your food, just giving more oomph to the flavors that are already there. It does this in two ways, by drawing water out of foods therefore concentrating that flavor naturally present. Secondly salt acts directly on your tongue, waking

up your taste buds. Our bodies are also programmed to naturally crave salt, because it is essential for normal function in our cells. Crunch and salt are the reason we love snack foods so much, it's programmed into our brains.

However, salt is only one part of the seasoning equation. The dishes we all love to eat have their salty elements but also provide complex flavors through aromas found in herbs and spices, match sweetness with bracing acidity and punctuate with heat or spiciness. When creating a new recipe for our restaurant or this book we try to look at how many different elements of taste—Salt, Sweet, Sour, Spicy, Bitter, Umami and Fat—we can incorporate through creative (not complex) seasoning.

Seek Freshness

It's quite simple, fresh food tastes better. The moment a fruit or vegetable is picked from its plant, it begins to change. In the case green peas, they immediately begin to convert their sugar into starch to preserve the energy source. Meaning that peas quickly transform from tender and sweet to firm and starchy. This is why frozen green peas are often the best option, they are frozen immediately after harvest, at their peak of freshness. Other vegetables are harvested anticipating the changes that will under go. Tomatoes are often picked in their under ripe, green state so that then can survive the days or sometimes weeks long journey to your grocery store shelves and then ultimately your kitchen.

Fruits and vegetables that allowed to fully ripen on the plant are always are allowed more time to develop taste and flavor from their environment. The items must then be transported to market and consumed more quickly. However, there are some fruits that will continue ripen and improve in taste after they are harvested. These include apricots, avocados, bananas, cantaloupe, kiwi ,mangoes, nectarines, peaches, pears, plantains, plums and tomatoes. Fruits that should picked only when ripe, because they do not improve are apples, cherries, corn, cucumbers, eggplant, grapefruit, grapes, oranges, pineapple, strawberries, tangerines,

watermelon, zucchini and squash. When selecting vegetables, look for items that are firm to the touch, have smooth, tight skin and show little signs of scars or bruising.

When storing fruits and vegetables, the general rule is that the refrigerator should be used for preservation—to either slow the ripening process of items that will be consumed in the future, or allow the ripe state to be extended. Tomatoes are a great example, they should be stored at room temperature until they are fully ripe. Once ripened, if they are not ready to be consumed, they may be moved to the refrigerator to extend their shelf life, safely holding there for 3 or more days. Though you should be very careful with refrigerating tomatoes. Some of the compounds responsible for aroma in tomato will be destroyed when they reach temperatures below 55°F. If you've ever felt like refrigerated tomatoes don't quite taste the same- you're right! Other produce that can be sensitive to refrigeration include stone fruits (peaches, plums, apricots), melons, avocados and citrus.

Asparagus is also best kept at room temperature, when you get asparagus home, trim a thin slice from the base of the stalks and stand them up on water as you would for fresh flowers. This will prevent the delicate tips of the asparagus from collecting water and going bad. Most other green vegetables are sensitive to excess moisture as well. They should never be stored in closed plastic bags or sealed containers, which trap moisture that plants release and cause the plants to break down (go 'bad') more quickly. A bowl with dampened paper towels (to preserve moisture) over the top or another breathable container are the ideal storage containers for delicate greens.

Salad-y Things

Umami in a Bottle

Inspiration ingredients- Lettuce, Dijon, fennel, vinegar

Compound: tridecane

Aroma: alkane

This is the ultimate salad dressing. We selected each of the ingredients in this dressing for its chemical affinity for one another and because each of them pair extraordinarily well with the green aromas found in leafy greens.

Makes 1 cup dressing

3 garlic cloves, minced or finely grated

1 tablespoon Dijon mustard (spicy)

1 tablespoon soy sauce or fish sauce (salt, umami)

½ teaspoon ground fennel seed (aroma)

¼ cup red wine vinegar (sour)

½ cup olive oil (fat)

□ *The standard ratio for salad dressings is 3 parts oil to 1 part vinegar. Here, we reduce the amount of oil for a little more ‘punch.’ But if you find the dressing too tart, add more oil to taste.*

Combine the garlic cloves, mustard, soy sauce, fennel seeds, and red wine vinegar in a small jar or bottle. Place the lid on and mix vigorously. Add the olive oil and mix again. Adjust seasoning and balance to taste. Keeps refrigerated up to 4 weeks. Shake vigorously before using.

□ *When tasting your dressing, dip a leaf of lettuce, carrot stick, or other vegetable into the mixed dressing. This will give you more accurate representation of what your salad will taste like. Dressings will always seem too strong if you taste them directly from a spoon.*

Grilled Leeks Vinaigrette

Inspiration ingredients- allium, mushroom, almonds, feta, tarragon

compound: 1-octen-3-one

Aroma: earth, metal, mushroom

This is our fresh take on the classic French dish Leeks Vinaigrette. When alliums of any variety are cooked, pungent sulphur compounds diminish and they become sweet. At the same time, charring or cooking over open flame adds smoky depth. Sour flavors are a natural pairing to the earthiness of leeks and mushrooms. Yet the strongest pairing in the recipe is the combination of mushrooms, tarragon and allium.

¼ cup White wine vinegar (sour)

2 cloves Garlic, finely grated

1 tablespoon minced tarragon, plus more for garnish

3-4 medium sized leeks

3 portabella mushrooms, wiped clean and gills removed (umami)

1 teaspoon kosher salt (Salt)

¼ cup Extra virgin olive oil (fat)

2 tablespoons good quality balsamic vinegar (acid)

¼ cup sliced almonds, toasted (fat)

½ cup feta cheese crumbles (salt, sour, umami)

In a small bowl, combine the vinegar, garlic and tarragon. Mix well and set aside for the flavors to develop.

To prepare the leeks, trim away the dark green leafy portion (light green is ok). There is a natural separation where leek turns from solid and onion-like to dark green and looser; this is where you want to cut. Discard the dark green portion or reserve for making stocks/soups. *the reason for the difference in color is chlorophyll. The parts of the plant that grow above ground*

are exposed to the sun and therefore able to develop chlorophyll. When looking at a leek, the dark green top has spent a lot of time in the sun, above ground, the pale green middle recently reached above the soil and has developed little chlorophyll, and the white stalk remained underground through it's life span.

Do not trim the root end of the leek- this will help it hold together during cooking. Split the leeks in half lengthwise, cutting through the root. Rinse the leeks thoroughly, pulling back the layers under running water to wash away any trapped dirt or sand. Pat the leeks dry and place them in a 13x9 baking dish along with the cleaned mushrooms. Pour the vinegar mixture over the vegetables and turn them to coat well. Sprinkle with salt and drizzle over the olive oil. Leave at room temperature to marinate or refrigerate up to 36 hours before cooking.

□illustrate the cut on the leek

□the marinade will not only flavor the leek before it cooks, it will also help to soften both the texture of the leek and its strong allium aroma.

Heat a grill to medium-hot. Place the leeks and mushrooms, cut side down onto the hot grill, making sure the leeks are at a forty five degree angle to the grates. Reserve the pan and the marinade.

Leave the vegetables to cook undisturbed 3-4 minutes, then rotate each item 90°. Cook 3-4 minutes, then flip and repeat the process on the second side. At this point the mushrooms should be tender all the way through and give easily to the tip of knife. Return the mushrooms to the pan with the marinade.

Depending on the size of the leeks, they may need to cook 4-8 minutes more. The leeks should be tender all the way through and well charred on the cut side. When the leeks are done, return them to the pan with the mushrooms and turn well to coat with the remaining marinade.

To serve, slice the leeks cross-wise into 1 inch sections, this time discarding the root. Cut the mushrooms in half vertically, then thinly slice horizontally. Return the cut vegetables to the pan with the marinade one more time and gently toss to coat. Season to taste with salt. Arrange the dressed vegetables on a platter and drizzle with balsamic vinegar. Top with almonds, feta and chopped tarragon.

Avocado Toast

Inspiration ingredients- avocado, fennel, apple, cocoa, ancho

compound: linalool

Aroma: bergamot, citrus, coriander, floral, lavender, lemon, rose

Avocado and fennel are great match, so are fennel and apple. Think we're joking? Try a little minced apple and fennel in guacamole some time. But the best pairing for avocado is deep, earthy flavors like cocoa and chili.

4 slices of rustic sourdough or 7 grain bread (umami)

2 tablespoons extra virgin olive oil (fat)

1 garlic cloves, cut in half

1 teaspoon fennel seeds

2 avocado (fat)

Juice of ½ lime (sour)

¼ cup diced apple* (sweet)

1/8 teaspoon ancho chile powder (or dark chili powder) (spicy)

1/8 teaspoon cocoa powder (bitter)

1 teaspoon kosher salt (salt)

*For the apple, you want mildly sweet and crunchy like a Honeycrisp or Braeburn.

Lightly coat the slices of bread with the olive oil and toast in a large skillet until golden brown and slightly dark at the edges. Work in batches if necessary. When all the bread is toasted rub one side with the cut garlic. *□this is the ideal method for bread that is toasted and crisp on the exterior with a soft, chewy interior- aka perfect toast.*

Remove the skillet from the heat and add the fennel seeds. Swirl the pan until the seeds become very fragrant and immediately transfer them to a cutting board so they do not burn.

Press the seeds with the back of a large spoon or the side of knife to crush and break into smaller pieces.

Cut the avocados in half and remove the pit. Scoop the flesh from the the skin and cut into 1 inch chunks. Place the avocado pieces in a bowl, add the crushed fennel seeds, lime juice and apples. Mix well.

Divide the avocado mixture evenly between the pieces of toast gently press to spread evenly over the surface.

In a small dish combine the chile powder, cocoa powder and salt. Sprinkle evenly over the toast and serve immediately.

Avocado ‘Succotash’

Inspiration ingredients- avocado, corn, tomato, basil

Compound: ethyl acetate

Aroma: brandy, contact glue, grape, sweet

Our succotash, really our kids’ favorite version of a Mexican salsa, balances a variety of tastes, textures and different notes within one flavor category. It’s beautiful in its simplicity and as delicious served with a bowl of tortilla chips as it is with grilled white fish or chicken. All of these ingredients have strong vegetal/fruit-like aromas that make them an ideal match.

Serves 4

3 ears fresh yellow corn (sweet)

1 avocado, seed and skin removed and diced (fat)

1 cup tomato, diced (umami)

2 tablespoons basil leaves, chopped

juice and zest of ½ lime (sour)

¼ teaspoon Basque pepper powder or cayenne pepper or red chili flakes (spicy)

3 tablespoons extra virgin olive oil, plus more as needed (fat)

kosher salt, to taste

Lightly drizzle corn with olive oil. Rub the oil to distribute and season well with salt and pepper. Place corn on a hot grill or onto a baking sheet and then under the oven’s broiler. Turn frequently. Remove once kernels appear golden brown and toasted. Once the corn is cool enough to handle, shave the kernels off the cob with a sharp chef’s knife. *☐Take your time to properly char the corn. As the kernels blacken, they will become smoky and sweet. Don’t worry, they will not get tough or bitter at this stage. You WANT to burn them.*

In a large bowl, combine the diced avocado, tomato, basil, lime, chili, and olive oil. Season with salt and pepper to taste.

Serve immediately, either as an appetizer, with tortilla or baked pita chips. Or as a condiment for grilled seafood or meats.

Roasted Beet Salad

Inspiration Ingredients- beet, orange, mint, hazelnuts, yogurt

Compound- 2-isopropyl-3-methoxypyrazine

Aroma- bell pepper, earth, green, hazelnut, pea

Beets and orange are a phenomenal pairing. THE FLAVOR MATRIX scored them as a nearly 60% match, which is exceptional. Beets become very sweet when roasted, so when pairing them with orange, you need plenty of acidity. In this recipe we get that from vinegar, citrus and yogurt. Both parts of this dish- the roasted beets and yogurt dressing can be made in advance and combined just before serving.

2 lbs beets (umami)

1 fl oz olive oil (fat)

1 tablespoon kosher salt (salt)

½ teaspoon ground black pepper (spicy)

4 branches fresh thyme

1 bay leaf, optional

½ cup white wine vinegar (sour)

Zest of 1 orange (sweet)

1 cup Greek yogurt (sour, umami)

Zest and juice of 1 lemon (sour)

2 tablespoons chopped mint, reserve smaller leaves for garnish

½ teaspoon ground coriander

½ cup toasted hazelnuts (fat)

preheat oven to 350°F

Trim and rinse the beets; toss with olive oil and salt and pepper. Place the beets in an oven proof container just large enough to hold them in a single layer. Add the thyme, bay leaf and

vinegar. Cover with aluminum foil and roast in the oven until tender, 25 minutes for baby beets, 40 minutes for medium beets, up to 75 minutes for larger.

When tender- the beets should be easily pierced all the way through with the tip of knife. Use a paper towel or hold kitchen towel to wipe the skins from the beets. Cut each beet into bite-sized pieces, either quarters or large cubes depending on the size of the beets. Place the cut beets in a bowl and pour over some of the juices from the roasting pan. Toss to coat and season to taste with salt and pepper. Grate the zest of 1 orange over the beets and mix again. Store in a covered container until ready to serve.

□ Peel the beets as soon as they are cool enough to handle when holding with a towel. The skins can be easily rubbed away at this point. When the beets cool, the skins will become much more difficult to remove.

□ adding the zest to the beets after roasting means it will keep it's bright, fresh aroma.

In a separate bowl, combine the yogurt, lemon zest, juice, mint, coriander and cayenne. Mix well.

To serve, spread the yogurt over the base of a large platter. Mound the beets over the yogurt. Garnish with toasted hazelnuts and reserved mint leaves.

Coco-Slaw

Inspiration Ingredients: coconut milk, cabbage, cilantro, carrot

Compound 1-tetradecanol

Aroma- coconut

Coleslaw with a tropical twist. Coconut and cabbage are an unexpected and delicious combination. The thing we love about this recipe (besides how good it is) is that it makes a rich, creamy slaw that is cook out approved- not mayo to sit in the sun and get funky. Added bonus, you can omit the cabbage and make just the dressing. It makes a delicious creamy vegan dressing that pairs nicely with all dark leafy greens (like kale).

Serve with: fish tacos, grilled/roasted pork, grilled/roasted pineapple

1 head green cabbage, shredded (about 12 cups)

2 teaspoons kosher salt (salt)

1 teaspoon chili powder (spicy) *□ use chili powder instead of ancho because it adds a touch of cumin, which really compliments the other ingredients*

15 fl oz coconut milk (fat)

2 cups carrots, shredded (umami) *□ It is worth it to freshly grate the carrot so that some of the carrot juice makes it into the dressing.*

1 jalapeno, diced (spicy)

½ cup cilantro

2 limes, juiced (sour)

Place the cabbage in a large bowl. Add the salt and chili powder and mix well. Continue mixing and tightly squeezing handfuls of the cabbage for 1 minute, to help the salt penetrate and soften the cabbage. Transfer the cabbage to colander and wipe the bowl dry.

In the clean bowl, combine the coconut milk, carrot, jalapeno, cilantro, and lime juice. Mix well.

Add the drained cabbage back to the bowl and toss to combine. Refrigerate until ready to serve.

Keeps up to 3 days.

A Different Kind of Green Salad

Inspiration Ingredients- cucumber, melon, jalapeno

Compound- Hexanal

Aroma- fresh, fruit, grass, green, oil

This one might sound a little crazy but Hexanal—a chemical compound that smells like freshly-cut grass—is responsible for this big, beautiful salad and how well these seemingly disparate ingredients come together. Plus, this is textbook taste development—you get sweet, salty, savory, spicy, and fresh all in one amazing package.

Serves 4 as a side

2 cups Kirby cucumbers, thinly sliced (3-4 cucumbers)

1 cup green grapes, cut in half (sweet)

2 cups honeydew melon cut into 1-inch cubes (sweet)

½ green jalapeno, thinly sliced (spicy, strong)

juice and zest of 1 lime (sour)

1 cup baby arugula, coarsely chopped (spicy, mild)

2 teaspoons kosher salt (salt)

½ teaspoon ground black pepper

¼ cup extra virgin olive oil (fat)

1 teaspoon toasted fennel seeds

2 tablespoons shaved parmesan cheese (salt, umami)

In a large bowl, combine the cucumbers, grapes, melon, lime juice, lime zest, and mint leaves. Season with salt and pepper and toss well to combine. Transfer to a large platter or serving bowl. Refrigerate until ready to serve. □ *this can be done up to 4 hours in advance, if let too long, the acid in the lime juice will start to make the fruits mushy. If you need to prepare*

everything in advance, mix the elements for the dressing, but keep them separate from the fruits and mix just before serving.

Just before serving, drizzle the salad with olive oil, sprinkle with the fennel seeds and parmesan shavings.

Watermelon Bruschetta

Inspiration Ingredients- watermelon, tomato, pecorino

Compound- 1-penten-3-one

Aroma- green, herb, mustard, pungent

The beauty of this is the visual trick it plays: when mixed together, the cubes of watermelon and tomato are nearly indiscernible, while they alternate savory and sweet bursts on the palate.

You're probably not thinking garlic, basil and cheese with your watermelon, but trust us on this one.

2 cups finely diced watermelon (sweet)

1 cup diced tomato (umami)

½ teaspoon fine sea salt (salt)

4 slices rustic sourdough (umami)

¼ cup extra virgin olive oil, divided (fat)

2 cloves garlic

1 tablespoon chopped basil

Freshly ground pepper, for garnish

Shaved pecorino cheese, for garnish (umami, salt)

Combine the watermelon and tomato in a bowl and add the salt. Gently mix to combine, then transfer to a colander and leave to drain for 10 to 20 minutes. *the tomatoes and watermelon will expel a good deal of excess water at this point, helping to intensify their flavor*

Brush the slices of sourdough with approximately half of the olive oil. Toast in a skillet or on a grill until lightly charred on the edges. Rub the toasted bread garlic and set aside.

Return the drained watermelon and tomato to a clean bowl. Add the remaining olive oil and basil. Gently toss to combine. Spoon the watermelon mixture on to the toasted bread and

garnish with a little freshly ground pepper and a generous grating of pecorino cheese.

Sicilian-Style Peach Salad

Inspiration Ingredients- peach, fennel, olive, mint, pepper

Compound- α -phellandrene

Aroma- citrus, mint, pepper, turpentine, wood

This salad celebrates the two places we love most- the American South and Coastal Italy. This recipe is based on a classic Sicilian salad made with oranges, but substitutes the quintessential Southern peach. Try to select firm peaches that are slightly under ripe. They add great crunch and a touch of tartness.

3 tablespoons white wine vinegar (sour)

1 teaspoon honey (sweet)

½ teaspoon kosher salt, plus more as needed (salt)

½ jalapeno, sliced (spicy)

¼ cup red onion, thinly sliced

2 firm peaches, pit removed and sliced (sweet)

1 head fennel, thinly sliced □*for the onions, peaches and fennel: slicing on a mandolin will yield the best results*

½ cup castelvetro green olives, pitted and cut in half (salt, umami)

2 tablespoons fresh mint leaves, chopped, plus more reserved for garnish

¼ cup extra virgin olive oil (fat)

(optional) ¼ cup salty cheese (ie feta or pecorino) (salt, umami)

Combine the vinegar, honey and salt in small bowl. Mix well until dissolved. Add the sliced jalapeno and onion and set aside to marinate while you prepare the salad. □*the vinegar will pull some of the heat (capsaicin) from the pepper and distribute more evenly throughout the salad, while also mellowing the pungency of the onions*

In a larger bowl, combine the peaches, fennel, olives and mint. Add the olive oil and season lightly with salt. Gently toss to combine. Add the vinegar mixture toss gently. Transfer to a platter and garnish with extra mint leaves and cheese if desired. *▯ Adding the vinegar after everything is coated with oil, will slow the softening, keeping the peach and fennel from becoming 'mushy' after mixing.*

Sides

Brussels Sprouts Kabobs

Inspiration Ingredients- Brussels sprouts, chili, yogurt, dill

Compound- methyl octanoate

Aroma- fruit, orange, sweet, wax, wine

Brussels sprouts are so good when charred on a grill. The only problem is they're really hard to cook that way. Thus we resolved to line them up on skewers so they could be easily cooked with a pair of grill tongs in one hand and a drink in the other. Brussels sprouts pair really well with green and dairy flavors making the yogurt dill sauce the perfect thing to dunk them in.

For the sauce:

1 clove garlic

1 teaspoon chipotle based hot sauce (spicy)

1 tablespoon lemon juice (sour)

½ cup Greek yogurt (sour, fat)

1 tablespoon fresh dill, chopped □ *instead of fresh dill, you can mix 1 teaspoon dried dill with 1 tablespoon fresh parsley leaves and chop the two together. This will give you the fresh green aromas that dry herbs typically lack and deliver distinctive dill aroma.*

18 brussels sprouts

½ teaspoon ground cumin

1 teaspoon garlic powder

2 teaspoons honey (sweet)

2 tablespoons extra virgin olive oil (fat)

Finely grate the garlic with a rasp style grater/zester, or use a garlic press. Combine the garlic, hot sauce and lemon juice in a small bowl and stir well. Leave 5 minutes for the flavors to

develop. Add the yogurt and dill and mix well. Store refrigerated until ready to serve. Keeps up to 3 days.

Bring a large pot of salted water to a boil.

Trim the ends from the brussels sprouts and cut each one in half. Drop the brussels into the boiling water and cook for 1 minute. Remove with a spider or slotted spoon and spread the brussels into a single layer on a baking sheet to air dry.

Reserve 1 tablespoon of the cooking water and in a large bowl, mix it with the cumin and garlic powder. Once the spices have absorbed the water and form a thick paste, stir in the honey and olive oil. *□mixing the spices with water before using makes them more aromatic and flavorful.*

When the brussels sprouts are cool enough to handle, but still a little warm, add them to the bowl and toss well to coat with the spices.

Arrange the brussels sprout halves onto skewers so that the flat, cut sides line up on each skewer. Season each skewer with kosher salt and cook on a hot, well-oiled grill. Grill 2-3 minutes on the first side so that the vegetables begin to lightly char, then turn and cook on the second side until tender through. Remove the skewers from the grill and arrange on a platter. Serve with yogurt sauce. *□illustration of brussels on skewers*

Asparagus and Potato Chip Gratin

You might be blown away by this. While Brooke thought the potato chip's best friend was Chardonnay, it turns out asparagus is its rockstar pairing. Pyrroles and Pyrazines are responsible for the hunger-inducing roasted and toasted aromas present in cooked—particularly fried—foods like potato chips. While writing THE FLAVOR MATRIX, we discovered that these compounds are naturally present in asparagus and we knew that we had to find a way to incorporate these two ingredients into one dish. Bet you can't bake just one.

Serves 4

2 tablespoons canola or vegetable oil (fat)
1 lb. cremini mushrooms, cut into quarters (umami)
2 garlic cloves, peeled and crushed
1 shallot, chopped
3-4 branches fresh thyme
1 tablespoon soy sauce (umami)
1 teaspoon red wine vinegar (sour)
4 tablespoons unsalted butter (fat)
3 tablespoons all purpose flour (for thickening, use gluten free alternatives)
3 cups chicken stock
½ cup grated parmesan cheese (umami, salt)
2 bunches asparagus (umami)
½ cup sour cream or crème fraîche (fat)
2 tablespoons thinly sliced chives
1 tablespoon chopped tarragon or dill (aroma)
1 cup grated gruyere or swiss cheese (fat)
2 cups crushed potato chips

Set a large pot of salted water on a back burner to come to a boil.

Heat the oil in a large sauté pan over high heat. When the oil begins to shimmer, add the mushrooms and stir briefly. Allow the mushrooms to cook, undisturbed for 1 minute, then add the garlic, shallot, and thyme. Reduce the heat to medium high (about 60%). Stir well and cook another 3-4 minutes, stirring occasionally. Season to taste with salt and pepper. When the mushrooms are tender and browned, and all the liquid has evaporated from the pan, deglaze with the soy sauce and red wine vinegar. Cook until dry.

Stir in the butter. When the butter is melted, scatter the flour over the mushrooms and stir well to create a roux. Cook the roux until thickened, about 2 minutes.

Stir the chicken stock into the pan with the mushrooms. Continue stirring constantly until the mixture comes to a simmer. Cook 2 minutes and adjust the seasoning to taste. Turn the heat off and whisk in the grated parmesan cheese. Set the mushroom velouté aside and cool slightly.

Trim the tough ends from the asparagus and drop them into the boiling water. You may have to do this in batches, depending on the size of the pot. Cook the asparagus until nearly tender: 1½ to 2 minutes for thin asparagus, 2 minutes for standard or 3 minutes for jumbo asparagus *(you will need to peel the stalk end of the jumbo asparagus as well)*.

When the asparagus is finished cooking, carefully remove it from the pot and allow it to drain on a sheet pan lined with paper towel or a clean kitchen towel. Keep the pieces in a single layer, so they can cool properly. Stacking or piling the asparagus will cause them to overcook. *(don't shock asparagus, the leaves and tips tend to trap water and dilute the flavor of the asparagus)*

Assemble the gratin immediately or refrigerate the asparagus and mushroom velouté separately and assemble when ready to serve.

Arrange the asparagus in a large oval or 13x9 baking dish so that they cover the entire dish in even layers. Whisk the sour cream, chives, and tarragon or dill into the velouté. Pour the velouté over the asparagus and gently spread to distribute evenly.

Scatter the gruyere cheese, then the potato chips over the gratin. Bake in a 425°F oven until golden brown on top and bubbling on the sides. Cool 5 minutes before serving.

‘Caesar’ Artichoke Dip *anchovy, egg yolk, lemon, Worcestershire*

Inspiration Ingredients- artichoke, anchovy, lemon

Compounds- heptanal- anchovy, lemon, artichoke, garlic

Aroma- citrus, dry fish, fat, green, nut, rancid

This is party food taken to the next level: Spinach artichoke dip meets Caesar salad. We were inspired by the pairing of artichoke, anchovy and lemon which is basically the beginning of every meal we’ve ever eaten in Rome. But we wanted to bring to a more playful and comfortable place and let’s face it- a creamy, cheesy dip is never a bad idea.

2 egg yolks (fat, emulsify)

3 flat anchovy fillets, or more to taste, rinsed and drained (umami)

¼ teaspoon black pepper (spicy)

¼ teaspoon smoked paprika

Pinch cayenne pepper (spicy)

4 garlic cloves

½ cup grated Parmigiano Reggiano, divided (umami, salt)

1 tablespoon fresh lemon juice (sour)

1 tablespoon red wine vinegar (sour)

2 teaspoons Worcestershire sauce (umami)

3 dashes Tabasco (spicy)

1 tablespoon Dijon mustard (spicy, emulsify)

4 ounces cream cheese, softened (fat)

½ cup sour cream (fat, sour)

15 ounce quartered artichoke heart

1 (8-9 oz) bag spinach, chopped (about 4 cups)

½ baguette, thinly sliced

Combine the egg yolks, anchovy filets, pepper, paprika, cayenne, garlic cloves, ¼ cup parmesan cheese, lemon juice, vinegar, Worcestershire, tobasco, mustard and cream cheese in a food processor and blend until smooth.

Transfer the mixture to a bowl, add the sour cream and whisk to combine.

Drain the artichoke hearts and press lightly to extract any excess moisture. Slice the artichokes crosswise into ¼ to ½ inch slices. Gently fold together the artichokes and spinach into the caesar sauce. Mix well and season to taste. Refrigerate at least 2 hours, up to 36 hours, before serving.

Scatter the remaining cheese over the top of the dip and bake 375°F until bubbling and browned on top.

Brush the sliced baguette with olive oil and toast in the oven. Serve warm with the dip.

Pumpkin Pilaf

Inspiration Ingredients- jalapeno, cinnamon, fennel seed, pumpkin

Compounds- phenethyl alcohol

Aroma- fruit, honey, lilac, rose, wine

Jalapeno and pumpkin are flat out delicious together. It just so happens that they both pair really well with warm spices like cinnamon and fennel as well. Pilaf is style of cooking of grains that is usually used for just rice. Here we chose to swap the rice for something heartier. Fregola is a small pasta from Sardinia that is small balls of semolina dough that roasted as part of the drying process, give them a delicious flavor and distinctive chew. If you can't find fregola, Israeli couscous is fine substitute.

2 tablespoons butter (fat)
2 cloves garlic, smashed
2 cups diced pumpkin, acorn squash or butternut squash
½ jalapeno, seeds removed, minced (spicy)
½ teaspoon fresh thyme leaves or ¼ teaspoon dry thyme
1 teaspoon kosher salt, plus more as needed (salt)
½ teaspoon fennel seeds
1 cinnamon stick
2 cups fregola or Israeli couscous
4 cups chicken/vegetable stock or water (umami)
¼ cup grated pecorino or parmesan cheese (umami, salt)
Toasted pumpkin seeds for garnish (optional)

Melt the butter in a medium sauce pot over medium high heat. As the butter melts, add the garlic and cook until it begins to sizzle and brown lightly around the edges.

Stir in the pumpkin, jalapeno and thyme. Season with the salt and saute 2-3 minutes, or until the pumpkin begins to soften.

Add the cinnamon, fennel seeds, and couscous stir well and continue to cook together, stirring constantly for 1 minute. When the spices become aromatic, pour in the chicken stock and bring to a boil. Taste the liquid for seasoning. *□ it should taste like a fully seasoned soup at this point- as the couscous cooks, it will absorb the majority of the salt from cooking liquid. If the liquid is not properly seasoned at this point the cooked couscous will remain bland, no matter how much season you add after cooking.*

Place a lid on the pot or cover with aluminum foil and reduce the heat to low. Simmer gently on the stove top until all of the liquid is absorbed, about 15 minutes. Do not stir the pot at any point during the simmering time. *□stirring will release starches from the pasta and make your finished dish more like risotto than a fluffy pilaf.*

When the couscous is done cooking, remove the pot from the heat and gently stir from the bottom of the pot folding motion to bring the couscous from the bottom to the top of the pot. Replace the lid and let stand for 5 minutes before serving.

Serve in bowls topped with grated pecorino and toasted pumpkin seeds.

Whole Roasted Cauliflower

Inspiration Ingredients- cauliflower, mustard, citrus

Compound- Indole

Aroma- burnt, fecal, medicine, mothball

Cauliflower and citrus are a really great pairing. So we thought we would take just about every ingredient with citrus aromas that we could think of put them together. The hops in IPA, coriander, honey, lime, cilantro are all different sources of citrus flavor.

- 1 head cauliflower, stem trimmed
- 2 tablespoons extra virgin olive oil (fat)
- 6 fl oz india pale ale (sour, bitter)
- 1 teaspoon salt, plus more as needed (salt)
- ½ teaspoon ground coriander
- 1 tablespoon Dijon mustard (spicy)
- 2 teaspoons Honey (sweet)
- 1 teaspoon Lime juice (sour)
- 1 tablespoon chopped cilantro

Preheat oven to 375°F

Keeping the head of cauliflower in tact, trim the stem from the bottom of the cauliflower so that it sits evenly in a square baking pan (8x8 or 9x9).

Coat the outside of the cauliflower with oil. *□this is best done by getting your hands dirty.*
Drizzle the oil all over the cauliflower, then use your hands to rub it into all of the nooks and crannies.

In a small bowl, mix together the salt and coriander. Sprinkle the seasoning evenly over the surface of the cauliflower.

Place the whole head of cauliflower in the baking dish and pour the beer around the sides to cover the base of the pan.

Tightly cover the pan with aluminum foil and place in the oven to cook for 30 minutes.

After 30 minutes, remove the aluminum foil and cook 20 minutes more.

While the cauliflower is roasting combine the mustard, honey, lime juice and a pinch of salt in a small bowl. Mix until smooth and even.

When the cauliflower has cooked for 20 minutes uncovered, remove it from the oven and brush the surface with the honey mustard mixture, you will not use it all. Return to the oven for 10 minutes more.

After 15 minutes, check the cauliflower for doneness. The tip of a knife should easily pierce through the center of the cauliflower. If it does not, return to the oven in 5 minute increments until the knife easily passes through.

When the cauliflower is done, brush another layer of the mustard mixture over the cauliflower and transfer to a plate to rest for 5 minutes.

Sprinkle the cauliflower with fresh cilantro and serve.

Broccoli and Pistachio Pesto Salad

Inspiration ingredients: broccoli, pistachio, basil

Compound- alpha- pinene

Aroma- cedar, pine, resin, sharp, turpentine

File this under our monochromatic phase—broccoli, basil, pistachio, olive oil, arugula—green on green on green... on green. Alpha-pinene is one of the most prominent aroma compounds in pistachio and a unique aroma that distinguishes pistachios from other nuts, it is also prevalent in dark green vegetables (like broccoli and arugula). IF you're not up for making pistachio pesto, the pine nuts in traditional pesto are also high in alpha-pinene so try a simpler combination of roasted broccoli and store-bought pesto garnished with chopped pistachio.

1 cup shelled pistachio (fat)

1 clove garlic

2 tablespoons grated parmesan cheese (salt, umami)

2 cups picked basil leaves

¾ cup good quality extra virgin olive oil (fat)

2 heads broccoli (umami)

2 tablespoon vegetable oil, plus more as needed (fat)

1 teaspoon kosher salt (salt)

½ teaspoon granulated garlic

½ lemon

Arugula leaves, for garnish (bitter)

Shaved parmesan cheese, for garnish (salt, umami)

To prepare the pesto, combine the pistachio, garlic and parmesan cheese in a food processor. Blend on high until a smooth paste forms. Remove the top and add the basil, process in short bursts until the basil is finely chopped. Transfer to a bowl and stir in the olive oil. Adding more if needed to form a smooth sauce. Store refrigerated up to 1 week in an air tight container.

Preheat the oven to 425°F.

Peel the broccoli stems and slice ½ to 1 inch from the bottom of the stalk to remove any tough, fibrous portion. Cut each head of broccoli in half lengthwise through the stem. Brush the cut side of the broccoli with olive oil and season each piece with salt and garlic. Arrange the pieces on a baking pan cut side down and brush the backside and florets with remaining oil.

Roast in the oven the edges are crisped and brown and the stems tender, about 18 minutes. Cool briefly on the pan before serving.

To serve, place two large spoons of the pesto in the center of each plate and spread around with the back of a spoon. Place one broccoli piece in the center of each plate. Scatter a few leaves of arugula and shavings of parmesan over each broccoli. Give a few drops of lemon over each plate and serve immediately. *□adding lemon at the last moment will balance the rich pesto and preserve the brilliant green color in the dish*

Eggplant Dip

Inspiration Ingredients- yogurt, dill, sesame, pomegranate, eggplant

Compounds- β -bergamotene- eggplant, pomegranate, dill

Aroma- tea

Pomegranate and eggplant are an exceptional pairing. Both have a pleasant bitterness and take well to sour and dairy flavors. Yogurt is an excellent bridge between the two ingredients. In this recipe it helps to make things creamy and gives just the right amount tart. Enjoy as dip with raw vegetables or toasted pita chips, as a spread on pita sandwiches or serve with roasted fish or meats.

2 large eggplant (bitter)
1 head roasted garlic (*see note below) (sweet)
1½ teaspoons smoked paprika
2 teaspoons kosher salt (salt)
¼ cup olive oil, plus more as needed (fat)
¼ cup Greek yogurt (fat, sour)
2 tablespoons chopped dill, plus more as needed
2 teaspoons toasted sesame seeds
2 tablespoons pomegranate arils (bitter, sweet)
Preheat the oven to broil.

Adjust the oven rack to the center position. Place the whole eggplants on a baking pan and roast underneath the broiler. As the skin blackens, turn the eggplant until all sides are well-charred. Remove from the oven and cool. □*You can also do this on the grill. The goal is to completely blacken the skin which will give the flesh a nice smoky flavor, cooking on a grill will further enhance that flavor.*

When the eggplant are cool enough to handle, split each one open and scoop out the flesh with a spoon. Place the scooped eggplant in a colander and press lightly to extract any excess liquid.

Transfer the eggplant flesh to a food processor; add the roasted garlic cloves, smoked paprika, salt and olive oil. Puree until the mixture is smooth. Transfer the puree to a bowl and press plastic wrap onto the surface. Place in the refrigerator until cooled. This step may be done up to 2 days in advance.

When cooled, add the yogurt and dill and whisk until smooth.

To serve, Drizzle the top with olive oil and sprinkle with sesame seeds, pomegranate and dill.

**To roast the garlic: cut the top (about 1/8 inch) off a whole head of garlic, so the cloves are just exposed. Drizzle the exposed cloves with a small amount of olive oil and season with salt and pepper. Wrap the garlic in aluminum foil – making it look like a large, fat Hershey's Kiss, and bake in the oven at 350° F until very aromatic and golden brown inside, about 40 minutes. When the garlic is roasted and cool enough to handle, squeeze the head of garlic from the root end to pop the roasted cloves out of the top. (illustrate cut on garlic)*

Squash Skewers

Inspiration Ingredients- squash, bacon, carrot, vinegar

Compound- 4-vinylguaiacol

Aroma- clove, curry, smoke, spice

One of the fun things about this recipe is that once the skewers are cooked the carrots and squash look almost exactly alike. But each retains it's own unique character—the carrot firm and slightly sweet, the squash, tender and lightly bitter—and they accentuate the supporting ingredients in different. Ok, fine... maybe we define fun in different ways. Grilled vegetables + bacon = yum. Is that better?

2 teaspoons olive oil (fat)

2 cups carrots, peeled and cut into 1.5 inch pieces

¼ lb thick cut bacon cut into 2 inch (fat, umami)

1 inch piece of ginger, sliced into rounds

1 teaspoon kosher salt, plus more as needed (salt)

Pinch red pepper flakes

2 cups butternut squash cut into 1.5 inch cubes, about ½ of a medium-sized squash

½ teaspoon smoked paprika

1 tablespoon red wine vinegar (sour)

1 tablespoon chopped basil leaves

¼ cup crumbled feta cheese (sour, fat, salt)

Preheat oven to 425°F

Place the oil, carrots, chopped bacon, ginger, salt and red pepper flakes in a bowl and toss well to coat. Transfer to an aluminum foil-lined baking sheet and spread into a single, even layer.

Roast in the oven for 6 minutes, then remove, add the squash to the tray and mix well with a spatula to coat. Return to the oven for another 6 minutes. Remove the vegetables from the oven and set aside to cool slightly. *□add a bit more oil along with the squash if needed to coat evenly*

When cool enough to handle, arrange the vegetables and bacon on skewers in alternating patterns. This step may be completed up to 24 hours in advance. *□ illustration- when placing (any) items on a skewer, do not place them too close together. If the pieces on the skewer touch one another, they will not cook properly*

Turn the oven to *Broil* or *preheat a grill*

Season the skewers lightly with salt and smoked paprika. Cook underneath the broiler or on a hot grill until hot through and browned at the edges. Arrange the hot skewers on a platter or plates and sprinkle with vinegar, basil and crumbled feta cheese.

‘Scalloped’ Parsnips

Inspiration Ingredients- parsnip, lemon, dairy

Compound- terpinolene- parsnip, citrus, cheese, milk

Aroma- pine, sweet, plastic

Potatoes, cream and cheese is never going to be bad. But we wanted to transform that classic to something more suited for the grown ups table. Parsnips are an under-appreciated root vegetable with a sweet earthy flavor. Lemon is a great match for parsnips and dairy, but the acid in lemon juice would cause the creamy sauce to curdle, so we brighten things up with just the zest.

2 lbs parsnips (umami)
1 lb Yukon gold potato (umami)
2 cups chicken/vegetable stock (umami)
2 cups half and half (fat)
2 teaspoons kosher salt (salt)
1 garlic clove, smashed
1 bay leaf
Zest of 1 lemon
Pinch nutmeg
¼ grated parmesan (salt, umami)
½ cup grated sharp cheddar

Peel the parsnips and cut into 1/8 – ¼ inch slices using a mandoline or sharp knife. Peel and cut the potatoes to the same thickness and reserve separately.

Combine the parsnips, stock, half and half, salt, garlic and bay in a large pot and bring to a boil. When the parsnips reach a boil, stir in the potatoes and allow the pot to return to a boil once again. Remove the pot from the heat and add the lemon zest and nutmeg. □ *nutmeg is always*

best when scraped from whole nut. 2-3 scrapes down the length of a microplane will add the perfect amount. Whole nutmeg will last much longer in your spice cabinet. Pre-ground nutmeg can lose it's potent aroma very quickly, so it is extremely important to give nutmeg the 'sniff test' before using.

Cover the pot and let the mixture stand 5 minutes then remove the bay leaf and transfer to a 9X9 baking dish. Spread the vegetables into a an even layer and scatter the cheeses over the top. Cover the baking with aluminum foil and bake 20 minutes.

Remove the foil from the pan and bake 20-25 minutes longer, until the cheese is bubbling and browned. Cool 10 minutes before serving.

Summer Squash Gratin

Inspiration Ingredients- squash, tomato, mint, cheese

Compound- toluene

Aroma- glue, paint, solvent

We call this a gratin, but it's more like a light, fresh summer vegetable lasagna. Lightly roasting the zucchini and tomatoes before assembling the dish help to eliminate excess moisture and gives the vegetables a meatier texture so they stand up to the second baking. We love the pairing of zucchini and mint but basil is equally delicious in this recipe. We think recipe is a great base for all seasons- we've been to know to make it with asparagus in spring or roasted squash and caramelized onions in fall.

3 roma tomatoes (umami)

2 yellow zucchini

2 green zucchini

1 tablespoon minced garlic

2 teaspoons kosher salt (salt)

½ teaspoon ground black pepper (spicy)

1 tablespoon chopped mint

1 cup ricotta cheese (fat)

1 egg (umami)

¼ cup milk (fat)

¼ cup parmesan (umami, salt)

Preheat oven to 425°F

Slice the roma tomatoes and zucchini into ¾ inch slices. Place in a bowl and toss with the garlic, salt and pepper to coat evenly.

Spread the sliced vegetables out on baking pans, making sure that they are in a single layer and not over-crowded on the pan. *□the point of this step is to eliminate excess moisture from the vegetables. If they are too close to one another on the pan, they will steam and remain soggy.*

Roast in the oven for 6 minutes, then set aside to cool.

While the vegetables cook, combine the ricotta, egg and milk in a bowl and whisk until smooth.

When the vegetables are cool enough to handle, arrange half of the zucchini and tomato into a 9X9 or oval baking dish in overlapping rows. Scatter the mint over the vegetables, then spread 2/3 of the ricotta over the vegetables. Tile the remaining vegetables on top of the ricotta, then spread the remaining ricotta over the second layer of vegetables and scatter the parmesan cheese over top.

Bake again until bubbling on the edges, about 12 minutes. Cool slightly before serving.

Grains & Pasta

All grains are made up of three basic parts. The hull and bran are the edible outer seed covering. Bran adds a significant amount of nutrition to grains in the form of fiber. The endosperm makes up the bulk of the grain and contains mostly stored starches and small amounts of protein. The small kernel known as the germ is the nutritional powerhouse of grains. Like the yolk of an egg it contains nutrients, protein and trace amounts of unsaturated fat.

ILLUSTRATION: Grain Structure

Layers (in order): Hull/Bran, Endosperm, Germ

For better nutrition, shop for whole grains, since refined grains—such as white rice and products made with white flour (white bread, pasta, crackers, etc.)—are made by removing the bran and germ from the grain, thus stripping it of all nutritional value.

Cooking Grains

The simplest way to cook grains is to boil them in an excess of water, as you would when cooking pasta. Bring a large pot of lightly salted water to a boil and stir in the dry grain. If cooking quinoa, be sure to rinse it thoroughly before cooking. Reduce the heat to a simmer and cook until the grain is tender. Then strain the cooked grains through a fine strainer and serve as desired.

Dry grains or pasta can also be added to a soup or stew as it simmers or cooked risotto-style by gradually adding liquid and frequently stirring as they simmer. Cooking risotto-style is a particularly underrated technique for pasta, as this method captures all starches released during the cooking process and holds them in the sauce. We employ this technique in our ‘Baked Rigatoni,’ p. xx.

However you choose to cook your grains or pasta, the one thing that does not change is the need for salt. Grains must begin cooking in a salted liquid. Throughout the cooking the process, the dried starches found inside of grains or pasta begin to soften by bonding with water molecules. This process is your one opportunity to get salt to the interior of the grain which creates a properly seasoned bite. If salt is not introduced in the beginning, then you can only season the exterior of the grain, meaning each bite will always taste under seasoned.

General cooking times for grains:

Fresh Pastas 3-4 minutes

Dried Pastas 8-10 minutes

Quinoa 12-15 minutes

Teff 12-15 minutes

Brown Rice 40 minutes

Freekeh 20 minutes, cracked; 40 minutes, whole

Farro 45 minutes

Spelt 45 minutes

Spring Vegetable, Brown Butter and Lemon Pasta

Inspiration Ingredients- brown butter, peas, lemon

Compound acetaldehyde

Aroma- ether, floral, green apple, pungent

Many of the recipes in this section explore the extremely strong pairing between green vegetal aromas and deep roasted aromas. This can be a bit of a challenging pairing because green aromas are typically lost in the roasting process, meaning you get one, or the other. But we'll endeavor to highlight both. The best part about this particular pasta is that once everything is prepped, the sauce can be made in the same amount of time that it takes the pasta to cook.

1 medium zucchini (about 1½ cups)

1 lb long pasta like bucatini, perciatelli or thick spaghetti

1 stick butter (fat)

4 oz pancetta, diced (umami, salt) □ *you could make this meatless by using diced portabella instead of pancetta. It's the same preparation we use in our rigatoni alla gricia p.xxx*

1 bunch scallions, thinly sliced (about 1 cup)

2 cups sweet green peas (sweet)

Zest and juice of 1 lemon (sour)

½ cup grated pecorino or parmesan, plus more for garnish (salt, umami)

Prepare the zucchini by slicing into a julienne using a mandoline, julienne peeler or spiralizer. Alternatively, grate the outer portion of the squash on the box grater. Lightly salt the cut vegetables and set aside in a colander to drain. □ *Whichever method of cutting you choose, make sure you cut or grate only the outer, colored and fleshy portions of the squash. Stop cutting once you reach the seeds in the center. This inner core has a looser texture that becomes mushy and wet when cooked. The outer portion of zucchini and squash are firmer and better suited for cooking.*

Bring a large pot of salted water to a boil. When the water comes to a rapid boil, add the pasta and stir immediately to avoid sticking or clumping. Continue stirring occasionally as the pasta cooks.

Melt the butter in a large saute pan over high heat. When the butter is full melted, reduce the heat to medium. Watch the pan carefully, as the butter solids begin to toast and the butter takes on an amber color. Immediately add the diced pancetta to the butter and stir well. Continue cooking over medium heat until the pancetta is crisp and browned.

Add the scallions and saute approximately 30 seconds, until aromatic and tender. Stir in the zucchini and green peas and saute 1 minute more. Season to taste with salt and pepper. Stir in 1 cup of the pasta cooking water and bring to a boil. *□add each of the ingredients quickly, and as needed to control the heat in the pan. The goal is to get each of the vegetables quickly cooked and tender, but not browned. If you see signs of browning on the vegetables, quickly add the next ingredient to help cool the pan and stop the browning.*

At this point, the spaghetti should be ready (about 8 minutes). Transfer the spaghetti to the pan with the vegetables and simmer 1 minute more. Remove the pan from the heat and stir in the lemon juice, zest and cheese. Stir well for 30 seconds to emulsify and serve immediately. Garnish with additional cheese.

Creamed Kale and Caramelized Onion Mac and Cheese

Inspiration Ingredients- pasta, kale, caramelized onion

Compound dimethyl trisulfide

Aroma- cabbage, fish, onion, sulfur, sweat

Deep roasted and green aromas strike again! This is a rich and satisfying dish to make for a crowd. We love it just the way it is, but experiment with seasonal variations- add cut cherry tomatoes or artichokes in summer or diced squash in fall/winter.

- 4 tablespoons vegetable oil, divided
- 2 yellow onions, thinly sliced (about 4 cups)
- 4 tablespoons flour (to thicken, use gluten free alternative)
- 1 tablespoon granulated garlic
- 4 cups milk
- 1 lb chopped kale (bitter)
- 1 teaspoon vinegar based hot sauce (like Tabasco or Crystal) (spicy)
- 2 cups grated cheddar or gruyere (umami)
- 1½ cups grated parmesan, divided Umami, salt)
- 1 lb short tube pasta, cooked (like rigatoni or penne)

Preheat oven to 425°F

Heat 2 tablespoons of the oil in a dutch oven over high heat until it begins to lightly smoke around the edges of the pan. Add all of the onions at once and stir well. Cook for 30 seconds, then reduce the heat to heat medium low and continue cooking, stirring occasionally until the onions are tender and brown. If the onions are still light in the center and dark at the edges, reduce the heat further. The goal is to cook them slow and low so that they brown evenly, becoming tender and sweet. □ *properly caramelizing onions is a process that takes time, typically 30-40 minutes. Be patient and do not try to rush it, if the onions are cooked too quickly*

they will burn on the outside and remain pungent in the center. When done right, caramelized onions should be creamy and sweet throughout.

Add the remaining fat to the pan and stir in the flour and garlic. Cook together for 1 minute, scraping the bottom of the pan and breaking up any lumps.

Turn the heat to high and add the milk, 1 cup at a time, stirring constantly and taking care scrape the bottom and the corners of the pot to prevent scorching. Allow each addition to come to a boil before adding the next. When all the milk has been added, reduce the heat to medium and simmer 5 minutes. *□adding the milk gradually insures you will not get lumps in your sauce.*

Stir in the kale and hot sauce and simmer 5 minutes more. Remove the pot from the heat and stir in the cheddar or gruyere, plus 1 cup of the parmesan.

Add the cooked pasta and stir well to coat. Transfer the pasta to a large baking dish. At this point you may cool refrigerate the dish for up to 2 days, tightly covered. Just before baking, top with the remaining parmesan cheese.

Bake in a 425°F oven until golden brown on top and bubbling, approximately 15 minutes. Cool 5 minutes before serving.

Rigatoni alla Gricia with Mushroom ‘Pancetta’

Inspiration Ingredients- mushroom, maple syrup, soy sauce, vinegar

Compound- 2,3,5- trimethylpyrazine

Aroma cocoa, earth, must, potato, roast

Here we have a line-up of perfectly flavor-matched ingredients as well as sweet, sour, salty and umami working in harmony. The result is shitake mushrooms that act and taste just like crisp bacon in our vegetarian take on the classic Roman pasta, Rigatoni alla Gricia.

Serves 4

2 tablespoons soy sauce (salty, umami)

2 teaspoons maple syrup (sweet)

1 teaspoon red wine vinegar (sour)

1 lb shitake mushrooms, sliced (umami)

2 tablespoons extra virgin olive oil (fat)

8 ounces rigatoni pasta, cooked, plus 1 cup reserved cooking water

¼ cup grated pecorino cheese (salty, umami, fat)

Combine the soy sauce, maple syrup, and red wine vinegar in a bowl and whisk to combine. Add the mushrooms and toss well to coat. Leave to marinate 15 minutes, up to overnight. Drain well and pat dry before cooking. *the maple syrup and soy sauce will mimic the salt and sugar balance of cured pancetta, while the vinegar will soften the mushrooms and keep them from tasting too sweet. These ingredients will also help the cooked mushroom to develop a brown and crispy exterior like cooked pancetta.*

Heat the olive oil in a large sauté pan over high heat. When the oil begins to shimmer, add the mushrooms to the pan, reserving any excess marinade. Quickly spread the mushrooms into an

even layer and allow to sear for 1 to 2 minutes, until the mushrooms are crisp on one side. Stir well and cook 1 to 2 minutes more, then add the reserved marinade and continue cooking until all of the liquid is completely absorbed.

Add the cooked pasta and reserved pasta cooking water to the pan and toss well. Cook one minute, then remove the pan from the heat, add the cheese and stir well. Divide among bowl and top with more cheese.

Quinoa Salad

Inspiration Ingredients- hazelnuts, vinegar, sweet potatoes, honey

Compound- acetylpropionyl

Aroma- butter, caramel, cream, fruit, sweet

Honey has a particular affinity for grains, while the vinegar creates balance. We strongly recommend making this vinaigrette in big batches and keep a jar of it in the fridge. It is great for a simple green salad but this dressing can really take a fresh fruit salad to the next level.

Watermelon, grapes, cantaloupe and apples are all excellent match for this dressing.

Honey Vinaigrette

¼ cup apple cider vinegar (acid)

1 tablespoon honey (sweet)

1 tablespoon minced shallots

1 teaspoon Dijon mustard (acid)

½ teaspoon fresh thyme

¼ cup extra virgin olive oil (fat)

2 cups quinoa

2 cloves garlic, cut in half

½ cup hazelnuts, chopped

2 tablespoons extra virgin olive oil

2 medium sweet potatoes, sliced into rounds (sweet)

Optional garnishes:

Small greens like arugula, watercress or baby kale (bitter)

Cheese- shaved parmesan or pecorino or crumbled feta or goat cheese (umami, salt)

For the honey vinaigrette, combine all ingredients in a small bowl and stir well to combine. Set aside until ready to use. Vinaigrette may be made x3 or x4 and stored in the refrigerator for up to 2 weeks.

Bring a pot of salted water to a boil. While the water heats, rinse the quinoa thoroughly in a fine mesh strainer under running water. Add the quinoa to the boiling water and stir well. Reduce the heat to cook at a strong simmer. When the quinoa is cooked, strain through a fine mesh strainer and spread the cooked quinoa out on a sheet pan to steam dry. *□ a thorough rinsing will prevent the cooked quinoa from tasting bitter*

Preheat an oven to 425°F or a grill to medium high heat

Brush the slices of sweet potato on both sides with the honey vinaigrette. Then season lightly with salt and pepper. *□ Season the potatoes immediately before placing them in the oven or on the grill. If the salt sits on the sweet potatoes too long it will draw water to the surface and they'll end up steaming, not browning*

To cook the sweet potatoes in the oven, arrange them on a sheet pan in a single layer without touching and roast in the oven until tender, 12-16 minutes. This may require two sheet pans or cooking in batches. Make sure to not over crowd the pan or the potatoes or they will not brown properly.

To cook the sweet potatoes on the grill, arrange them cut side down and cook 2 minutes, then rotate 90° and cook 2 minutes more. Flip and repeat on the second side, continuing to rotate until the potatoes are tender through.

Set the cooked sweet potatoes aside to cool slightly before serving. Sweet potatoes may be prepared in advance and reheated to serve OR served at room temperature.

For to serve, combine the oil, garlic, and hazelnuts in a large non-stick sauté pan. Place the pan over medium heat and cook until the garlic begins to sizzle and brown around the edges. Immediately add the quinoa and stir once to coat. Turn the heat to high and cook undisturbed until the quinoa starts to 'pop.' Remove the pan from the heat. Arrange the roasted sweet potatoes slices on plates and top with the sautéed quinoa. Drizzle each plate with some of the honey vinaigrette and greens and cheese if using.

Faro and Crispy Kale Salad

Inspiration Ingredients-wheat, bell pepper, tomato, feta

Compound- 4- hydroxypentanoic acid lactone

Aroma- herb, sweet, floral

Lactones are a group of aromatics that have floral and tropical fruit aromas. It may very surprising to find them in tomatoes and pepper, but grains and cheese contain a surprising amount of lactones as well. Smoky aromas are another big factor here. They are naturally present in cooked grains and are developed further in the recipe by roasting the peppers and with the addition of smoked paprika.

1 cup faro

3 cups water

1 teaspoon kosher salt (salt)

Zest and juice of 1 lemon (sour)

2 tablespoons extra virgin olive oil + more for sautéing kale (fat)

1/4th teaspoon smoked Spanish paprika

Freshly ground black pepper (spicy)

10 oz bag chopped kale or 1 large bunch, stems removed and chopped (bitter)

6 cloves garlic, peeled, crushed

1 red pepper, roasted, skin removed, seeded, chopped (about 1 cup)

¼ cup almonds, lightly toasted, chopped (bitter)

½ cup crumbled feta (salt, umami)

Combine the faro, water and salt in a saucepot. Bring the pot to a boil, then reduce the heat to simmer. Cover the pot and cook 25 minutes or until tender. When tender, drain any remaining water and transfer to a large bowl.

□grains like faro will never get 'soft,' they always a little bite or 'al dente' feel. When cooking faro, you want to insure that there is no more gritty or chalky raw starches left in the center of the grain.

Pour the lemon juice and zest, 2 tablespoon of olive oil, smoked paprika over the faro and toss to combine. Allow to cool to room temperature.

In a large sauté pan over high heat, add 2 tablespoons of olive oil and 3 crushed garlic cloves. When the garlic begins to brown at the edge, add half of the kale. *□Add the kale carefully, it will sizzle and pop immediately, but that's a good thing it gives the kales crisp, crunchy edges.*

Allow the kale to cook undisturbed for 30 seconds, so that the edges begin to get a little crispy. Then gently toss or stir and cook 30 seconds more. Immediately transfer the kale to the bowl with the faro.

Wipe out saute pan with a paper towel and add fresh olive oil and garlic. Repeat the sautéing process with the second batch.

Add the roasted pepper, almonds and feta. Toss well to combine and adjust seasoning to taste with salt and pepper.

Flavor Flip *try with buckwheat instead. Thoroughly rinse the buckwheat in a fine mesh strainer under running water and drain well. Heat a tablespoon of oil in a medium saucepot and add the buckwheat. Saute over high heat for 2 minutes. Add water at a ratio of 1:1.5 (1 cup of buckwheat, add 1.5 cups water; 2 cups of buckwheat, add 3 cups water) and simmer until all the liquid is absorbed.*

Summer Squash 'Vongole '

Inspiration Ingredients- squash, clams, corn tomato, basil

Compound- dimethylamine

Aroma- fish

This a pasta that we make at Angelena's to celebrate all the glorious summer bounty of the Gulf Coast. It's based on the idea that clams and summer squash are flavor best friends. We also use this pairing (minus the spaghetti) occasionally on pizzas!

Serves 4

4 medium-sized yellow summer squash or zucchini

2 tablespoons olive oil, divided (fat)

2 tablespoon butter, divided (fat)

1 tablespoon minced garlic, divided

½ cup cherry tomatoes, cut in half (umami)

Juice of ½ lemon (acid)

50 Little Neck Clams, well rinsed

1 tablespoon chopped basil

Cut the squash into julienne strips using a mandoline, julienne peeler or spiralizer. Alternatively, grate the outer portion of the squash on the box grater. Lightly salt the cut vegetables and set aside in a colander to drain. *□ Whichever method of cutting you choose, make sure you cut or grate only the outer, colored and fleshy portions of the squash. Stop cutting once you reach the seeds in the center. This inner core has a looser texture that becomes mushy and wet when cooked. The outer portion of zucchini and squash are firmer and better suited for cooking.*

In a large sauté pan, combine 1 tablespoon of olive oil, 1 tablespoon of butter and ½ of the garlic over medium high heat. When the garlic begins to sizzle, gently squeeze the squash dry

with paper towels and add to the pan. Spread them quickly into an even layer and cook undisturbed for 1 minute. Gently toss and cook 1-2 minutes more, until the squash is tender but still retains a little bite. Transfer the cooked squash to a bowl and return the pan to the heat.

Add the remaining oil, butter, garlic, and the cherry tomatoes to the pan. When the garlic begins to sizzle, add the clams to pan and immediately cover with a tight fitting lid. Simmer 4 minutes then check to see that all the clams have opened. If they have not, stir well and replace the lid for 1 minute before checking again.

When all the clams have opened, uncover the pan and simmer for 1 minute to reduce the clam liquor. Stir in the lemon juice, par-cooked squash and fresh basil. Toss well to combine and season to taste with salt.

Divide among bowl and garnish with grated pecorino or parmesan cheese. *□ Cheese will add a little salt and fat that compliment the briny clams perfectly. Pecorino (which is made from sheep's milk) is ideal because it adds a little tang, but parmesan if fine substitute.*

Sweet Pea and Mushroom Rigatoni Carbonara

Inspiration Ingredients eggs, sweet peas, mushroom, wheat

Compounds- cumene

Aroma- solvent

Eggs, peas are an first ballot Hall of Fame pairing. Once again we are looking to pull in roasted/toasted aromas along with green aromas. At Angelena's we will occasionally toast the flour we are using to make pasta to give it a deeper, roasted, nutty flavor. You can achieve this same effect at home by toasting your pasta in a pan before you cook it, as we do in this recipe.

4 egg yolks (fat, umami)

1 cup grated parmesan cheese, divided (salt, umami)

1 tablespoon butter (fat)<-use two different fats for the right blend of browning and heat stability

1 tablespoon olive oil (fat)

1 lb dry rigatoni

2 garlic cloves, crushed

8 oz cremini mushrooms, cleaned and quartered (umami) *the best way to clean mushrooms is to quickly dunk them in bowl of room temperature water, swish them around and immediately transfer to paper towels and pat dry.*

Room temperature water, as needed

2 cups frozen petit peas

Combine the parmesan cheese and egg yolks in a bowl and whisk until smooth and set aside.

Place a large, wide, heavy bottomed over high heat. Add the butter and olive oil. When the butter begins to foam, stir in the dry pasta and reduce the heat to medium-high. Cook the pasta, stirring constantly until it is toasted and browned in spots, 3-4 minutes.

Add the garlic and mushrooms to the pot with the pasta and saute to soften the mushrooms slightly, another 3-4 minutes.

Deglaze the pot with 2 cups of water and stir well. Adjust the heat so that the liquid is at a strong simmer. Season well with salt. *□seasoning as you cook is essential. This dish will build a creamy well-seasoned sauce, but if you do not add salt early on, the pasta will not absorb it and you will be left with a beautiful sauce and bland pasta.*

Continue to cook stirring occasionally until nearly all of the water is absorbed. Repeat the process adding 1 cup of water at a time until the pasta is tender, typically about 5 cups of water total is required. Make sure to continue tasting and seasoning as the pasta cooks.

Once the pasta is cooked, reduce the heat to low and ladle approximately ½ cup of the hot liquid in pasta pot into the bowl of egg yolks. Whisk vigorously to combine.

Stir the frozen peas into the pasta. When the peas are warmed through, remove the pot from the heat.

Transfer the egg yolk mixture back into the pot and stir well so that egg yolks evenly coat and begin to thicken slightly. Continue stirring one minute to insure that the eggs yolks do not curdle. *□this is a basic tempering method, meant to introduce the egg yolks gently so that they create a rich, creamy sauce. But you must go through the process carefully to ensure the egg yolks do not curdle or become scrambled. Have a small amount of room temperature water on standby so that if the eggs yolks get too hot or too thick, you can add the water to stop them from curdling.*

Serve immediately. Divide among bowls and top with more cheese.

Vegetable Bolognese with Olive Gremolata

Inspiration Ingredients- fennel, olive, sage, orange, almond

Compound- myrcene

Aroma- balsamic, flower, fruit, herb, must

This dish is all about the subtle surprises. Olives and sage are a classic match for tomato sauce, but almonds and orange are rarely asked to join in. For a vegan twist, roasted and salted almonds stand in for cheese providing the same salty and nutty punctuation that you would get from parmesan or pecorino. Nuts are a great choice because olive and almond have such a strong pairing, and orange zest matches with the strong fruit aromas in the olive. But of course, it never hurts to finish it all off with a sprinkle of cheese if you desire.

Serves 4

¼ cup diced carrot (sweet)

¼ cup diced celery

½ cup diced onion (sweet)

1 tablespoon chopped garlic

½ teaspoon red pepper flakes (spicy)

1 tablespoon tomato paste (umami)

1 tablespoon soy sauce (umami)

1 cup white wine (acid)

28 fl oz can chopped tomato (acid)

2 branches fresh sage

2 branches fresh basil

2 teaspoons dried oregano

1 lb fresh pappardelle (or other pasta shape)

¼ cup toasted almonds, finely chopped

¼ teaspoon orange zest (approx. 1 full scrape on a rasp grater)
1 tablespoon chopped flat leaf parsley
1 tablespoon chopped celery leaves (bitter)
2 tablespoons niçoise olives, sliced lengthwise (salt, umami)

Heat the olive oil, garlic, and red pepper flakes in large saucepan over medium-high heat. Add the carrot, celery, and onions. Season the vegetables with salt and sauté until tender but not browned.

While the vegetables cook, combine the tomato paste, soy sauce and wine and whisk until smooth. Deglaze the vegetables with the wine mixture and reduce until nearly dry. *□mixing the tomato paste with other liquids help to loosen the paste and allows it to more even coat the vegetables, providing more depth and flavor in the dish.*

Add the canned tomatoes and herbs. Bring the mixture to a boil and season again to taste with salt and pepper. Reduce the heat to a simmer and cook 10 minutes.

Taste the sauce a final time, adjust the seasoning, and add a pinch of sugar if it is too acidic. The sauce may be pureed if desired or used as is. Keep sauce refrigerated for up to 2 weeks or freeze to keep for up to 6 months.

Bring a large pot of salted water to a rapid boil. Drop the pasta into the boiling water and stir immediately. Leave the pasta to cook, stirring occasionally, until al dente, approximately 3 minutes if fresh or 8 minutes if using dried.

Transfer the cooked pasta along with ½-1 cup of the pasta water to the pan with the sauce. Bring the pan to a simmer and cook 1 minute more and stir in the almonds. Add more of the pasta water if the sauce becomes too thick. Divide the pasta between bowls and top with garnishes.

Cheese

Good cheese provides texture, salt, fat, umami, and complex aromas all in a single bite. Cheese is a near perfect food on it own, yet when combined with other it only gets better. There are few foods that a little cheese cannot help to improve, later in this book you will even find cheese incorporated into a sweet dessert. The most amazing thing about this incredibly versatile ingredient which can be found in thousands of different forms around the world is that all cheese begins it's life as the same thing: Milk.

The flavor of a finished cheese will be completely determined by the type of milk that is used and the steps that milk is taken through in the cheese making process. We could easily fill volumes on this topic alone, but we won't. Let's take a broad, sweeping look at the basics. The first determining factor in the flavor of a cheese is the type of milk that is used.

The three most common used in cheese making and their flavor:

Cow- mild and milky

Sheep- tangy and salty

Goat- tart

Other factors that will affect the ultimate flavor of a finished cheese are:

Temperature Heat can destroy delicate compounds that give milk flavor. This is why some people forgo (sometimes) overly cautious food safety rules and prefer unpasteurized cheeses, or cheeses made with milk that has never been heated. Even if a cheese is made with pasteurized milk, the temperature that the milk is heated to during processing will still effect the texture of the curd and the flavor of the cheese.

Coagulating Agent Cheese curds are made up of coagulated fat and protein and whey. During processing, enzymes and/or acids are added to warm milk to initiate coagulation. These coagulating agents can impart their own flavor to the cheese making process.

Inclusions Any number of additional ingredients may be incorporated into the cheese during the production process. This could range from peppercorns or truffle peelings to additional bacteria or mold intentionally added for flavor, like in blue cheese

Rinsing Once the cheese is formed it may be washed or rinsed to impart unique flavors in the aging process. Beer, wine and salt brines are the most common rinses.

Aging The final factor effecting the flavor and texture of the cheese is time. Over months or sometimes years of aging, moisture is lost—making the taste of the cheese more intense—and complex aromas develop imparting a richer flavor.

From a flavor perspective, the dairy aromas found in all cheeses have particular affinity for floral, tropical fruit, sour and nutty aromas. The following recipes explores those pairings in a number of different forms.

Buffalo Bake

Inspiration Ingredients- cauliflower, blue cheese

Compound- methylamine

Aroma- fish, pungent

Capsicums, alliums and cheese in general are great matches for cauliflower, but blue cheese in particular is a very strong match. This recipe is kind of like buffalo, ranch and blue cheese all in one, but with vegetables! It's the ultimate party dish.

1 head cauliflower

2 tablespoons oil (fat)

2 teaspoons worchestershire (umami)

1 tablespoon Sriracha (spicy)

1 teaspoon Dried dill

1 teaspoon Onion powder

1 teaspoon Garlic powder (umami)

1 teaspoon kosher salt (salt)

8 oz Cream cheese, room temperature (fat)

½ cup Parmesan (salty, umami)

½ cup Milk

4 ea Scallion

½ cup Roma tomato (umami)

¾ cup Blue cheese crumbles, divided (1/2 mixed; ¼ top) (salty, fat)

Cut the head of cauliflower into florets.

In a large bowl, toss together the cauliflower florets, oil, Worcestershire sauce, sriracha, dill, onion powder, garlic powder, and salt. Mix well to coat and transfer to an 8X8 baking dish. □*you can put it somewhere else if you want but this way you'll have less dishes to wash.*

In the now empty bowl, combine the cream cheese, parmesan and milk. Beat until smooth. Stir in the scallions and tomato then return the seasoned cauliflower to the bowl and mix well.

Transfer the entire mixture back to the baking dish and spread into an even layer. Scatter the crumbles over the top of the cauliflower and bake 30 minutes or until the cauliflower is tender.

Cool 5-10 minutes before serving. Serve with crackers or toasted bread and celery sticks.

Cranberry Stuffed Baked Brie

Ingredient Inspiration- dairy, cranberry, orange

Compound methyl 2- methylbutanoate

Aroma- apple, fruit, green apple, strawberry, sweet

Cranberry, citrus, and Dijon mustard are all great matches for dairy. These ingredients are naturally bitter and acidic, putting them at odds with delicate milk... unless it first becomes cheese. This recipe can be prepared up to the point of baking, as much as a day in advance and stored in the fridge. It can go directly from the refrigerator to hot oven. Serve with additional Cranberry Chutney on the side and nuts, dried fruit and/or crackers.

1 whole round Brie, about 12 ounces (fat, umami)

1/2 cup Cranberry Chutney* store bought or see following recipe (bitter, sweet)

1 sheet puff pastry (from a 17.3-ounce package), thawed if frozen

1½ teaspoons Dijon mustard (spicy)

All-purpose flour, for dusting

1 large egg, beaten

Cranberry Chutney:

One 12-ounce bag fresh or frozen cranberries

1 Granny Smith apple, grated

1/3 cup brown sugar

1/2 teaspoon ground cinnamon

1/2 teaspoon ground ginger

Pinch ground cloves

Zest and juice of 1 orange

Kosher salt and freshly ground black pepper

Preheat the oven to 400 degrees F. Line a baking sheet with parchment paper.

Cut the round of Brie in half horizontally, as if splitting a hamburger bun. Lift the top off the Brie and spread the Cranberry Chutney over the bottom half. Replace the top.

Lay the sheet of puff pastry on a lightly floured work surface. Gently roll the pastry with a rolling pin so the dough become slightly thinner.

Spoon the Dijon mustard into the center of the pastry. Using a pastry brush, spread into a thin, even layer slightly larger than the brie.

Place the filled Brie in the center of the puff pastry.

Cut an approximately 1 inch piece from each corner of the pastry so that the shape resembles a stop sign. □ **illustrate cut and folding**

Fold in the 4 corners of the puff pastry so they meet in the center of the Brie. Lightly brush the dough where the pieces meet with the egg. Fold the remaining edges into the center so that the Brie is tightly wrapped.

Flip the Brie onto the prepared baking sheet seam-side down. Decorate the top of the round with the remaining puff pastry, if desired. Refrigerate 1 hour. This step can be completed up to 1 day in advance.

Bake until golden brown, about 35 minutes. Cool on the pan at least 10 minutes, then use a large spatula to transfer to a serving platter. S

For the cranberry chutney, combine the cranberries, apple, sugar, cinnamon, ginger, cloves, orange zest and juice and 1/4 cup water in a saucepan and bring to a boil. Reduce the heat and simmer until thickened, about 20 minutes. Season to taste with salt and pepper. Serve warm or chilled. Keeps refrigerated in an airtight container for up to 2 weeks.

Cherry Olive Jam

Inspiration Ingredients- cherry, olive, cheese, wine

Compound- ethanol

Aroma- alcohol, floral, ripe apple, sweet

They might look alike, but cherries and olives are two ingredients that couldn't be any different. It's not only the salty-sweet combo that makes this a great pairing, the way the flavors of these two ingredients come together is indescribable. Jamie first stumbled upon this combination as part of a dish he was developing for IBM's Chef Watson. While his version wound up accompanying duck, Brooke could imagine nothing better than to spread this on crackers and enjoy with mild, creamy cheeses. It is truly the GREATEST condiment for a cheese plate we have ever tasted.

Black Olive and Dried Cherry Coulis

½ cup dried cherries (sweet)

1 cup red wine

2 sprigs fresh sage or ¼ teaspoon dried sage

1 cup pitted nicoise olives, drained and brine reserved (salty, umami)

Combine the cherries, wine and sage in a saucepot. Bring the mixture to a simmer and cook until approximately 2 tablespoons of liquid remains in the pot. Remove from the heat and rest in the pot for 5 minutes.

Discard the sage (if using fresh). Combine the cherries and drained olives in a blender and process until smooth, gradually adding small amounts of the brine as needed until the mixture is smooth.

Grilled Cheese

Inspiration Ingredients- cheese, bread

Compound- 4-methylpentanoic acid

Aroma- floral

This ain't kids stuff. Grilled cheese is a beautiful sandwich that is all too often relegated to the kids table. What we're going to give you here are three incredible, grown up variations on the humble grilled cheese sandwich.

It all starts with the base cheese mixture. Trust us, you're going to want to make a bunch of it. When you have a container of this mixture sitting in your fridge, it means you're just a few minutes away from crusty, melty and perfectly balanced grilled cheese sandwiches at any time.

Ultimate Grilled Cheese Sandwich Base

2 tablespoons mayonnaise (fat)

2 oz cream cheese, softened to room temperature (mayo)

1 teaspoon garlic powder (umami)

½ teaspoon paprika (smoked if you prefer)

2 cup grated cheddar cheese (salt, fat)

1 cup grated provolone cheese (fat)

½ cup grated parmesan cheese (salty, umami)

Blend together mayonnaise, cream cheese, garlic powder, and paprika in a medium bowl. Fold in grated cheeses until well mixed. If not using immediately, store mixture in refrigerator in an airtight container. Keeps for up to 10 days.

For a plain grilled cheese, spread the cheese mixture between two slices of bread. Spread

softened butter or mayonnaise on the outside of the bread. Place a cast iron skillet or non-stick pan over medium heat. Add the sandwich to the pan, press lightly with a spatula and leave to cook until golden brown on one side, about 3 minutes. Flip the sandwich and brown on the second side, another 2 to 3 minutes. Remove the sandwich from the pan and cool slightly before cutting.

□ Cooking a perfect grilled cheese requires the perfect heat. Too high a flame will mean extra crispy (read: burnt) bread before the cheese inside reaches full, melty goodness. Too low of a flame means bread that's dry and pale rather than buttery, crisp and golden brown. In this process, preheating the pan is essential. Allow the pan to heat 3+ minutes before adding the sandwich to insure even heat distribution.

Spicy Kiwi and Bacon Grilled Cheese

Makes 4 sandwiches

Inspiration Ingredients- kiwi, jalapeno

Compound- 3-carene

Aroma- bell pepper, lemon, resin, rubber

This sandwich is all about the balance of sweet, spicy and salty so proportions matter. Make sure the cheese is spread into an even layer and arrange the kiwi slices so that they are not overlapping at all. They will not cover all of the cheese, but they don't need to.

1 cup grilled cheese base (p. xx)

2 ripe kiwi, peeled and thinly sliced (sweet)

½ jalapeno sliced into rings (spicy)

8 strips bacon, cooked crisp (salty, umami)

Softened butter, mayonnaise or reserved bacon fat as needed (fat)

Place 4 slices of bread on a cutting board. Spread approximately 2 tablespoons of the cheese mixture on each slice of bread. Arrange 3-4 slices of kiwi on top of the cheese.

Scatter a few jalapeno slices over the kiwi, then top with 2 slices of bacon on each sandwich. Break the bacon into pieces if it overhangs the sandwich too much. Set the sandwich bottoms aside.

Take 4 more slices of bread for the tops. Spread 1 tablespoon of the cheese mixture on each slice of bread then place on the bottom with the cheese side down.

Spread both sides of the sandwich with softened butter or mayonnaise. Alternatively, heat one tablespoon of reserved bacon fat in a cast iron skillet or non-stick pan over medium heat. Allow

the pan 3-4 minutes to preheat before cooking. Add the sandwich to the pan, press lightly with a spatula and leave to cook until golden brown on one side, about 3 minutes. Flip the sandwich and brown on the second side, another 2 to 3 minutes. Remove the sandwich from the pan and cool slightly before cutting.

Grilled Cheese Sandwich *kimchi, broccoli, cheddar*

Makes 4 sandwiches

Inspiration Ingredients- broccoli, kimchi

Compound- dimethyl trisulfide

Aroma- cabbage, fish, onion, sulphur

Sulphur compounds are known for creating deep, rich and somewhat funky aromas in foods- boiled eggs being one most the pure examples of sulphur compounds in food. But don't let that scare you off, the sulphur compounds shared between broccoli and kimchi are delicious in the best possible. In fact, if you want to take this sandwich to the next level, we strongly encourage you top it off with a fried egg.

¼ cup kimchi (spicy)

2 cups chopped broccoli (about 1 head, including stem)

1 cup grilled cheese base (p. xx) (salty)

Combine the kim chi and broccoli in the bowl of a food processor. Pulse in 3 seconds bursts to chop and mix well. Transfer to a strainer and set aside to drain.

Place 4 slices of bread on a cutting board. Spread approximately 2 tablespoons of the cheese mixture on each slice of bread. Divide the broccoli mixture evenly among the sandwiches.

Take 4 more slices of bread for the tops. Spread 1 tablespoon of the cheese mixture on each slice of bread then place on the bottom with the cheese side down.

Spread both sides of the sandwich with softened butter or mayonnaise. Place cast iron skillet or non-stick pan over medium heat. Allow the pan 3-4 minutes to preheat before cooking. Add the sandwich to the pan, press lightly with a spatula and leave to cook until golden brown on one side, about 3 minutes. Flip the sandwich and brown on the second side, another 2 to 3 minutes. Remove the sandwich from the pan and cool slightly before cutting.

Grilled Cheese Sandwich *olive, apple, pork*

Makes 4 sandwiches

Inspiration Ingredients- olive, apple, pork

Compound 1-heptanol

Aroma- chemical, green wood

Pork and apple are an all-time classic pairing. 1-heptanol plays a role in creating that and just happens to be prominent in olives and mustard as well, adding a little spicy and saltiness to an already great combo.

8 slices olive sourdough

2 tablespoons Dijon mustard (spicy)

4 slices smoked ham (umami)

½ apple, thinly sliced (sweet)

1 cup grilled cheese base (p. xx) (salty)

2-3 tablespoons extra virgin olive oil (fat)

Spread ½ tablespoon of mustard on one side of 4 slices of olive sourdough.

Fold the ham so that it is the same size as the bread. Place on top of the mustard. Top the ham with apple slices.

Divide the cheese mixture among the other 4 slices of bread and spread into an even layer.

Close the sandwiches by pressing the cheese side onto the apples.

Place cast iron skillet or non-stick pan over medium heat. Allow the pan 3-4 minutes to preheat before cooking. Add just enough olive oil to coat the bottom of the pan, then carefully place each sandwich into the pan, press lightly with a spatula and leave to cook until golden brown on one side, about 3 minutes. Flip the sandwich and brown on the second side, another 2 to 3

minutes. Remove the sandwich from the pan and cool slightly before cutting. Repeat as needed adding fresh olive oil with each sandwich.

‘To-Mayo’ Sandwich **bacon, lettuce, tomato**

Inspiration Ingredients- tomato, bacon

Compound- geranylacetone

Aroma- green, hay, magnolia

We’re aware that there is no cheese in this sandwich. But after you taste it, I can promise you forget about whatever nit-picky complaints you had about this being miscategorized under the Cheese section of this book. This is really all about the mayo, which is a bit of a game changer.

If you’re not hip to the homemade mayo game, it’s a popular misconception that emulsification requires egg yolks. The egg white mayonnaise conversation reminded me of a discussion I had with Hervé This when hosting him at The Institute of Culinary Education. In short, his visit culminated in him telling our students that he could (though, he insisted, he never would) make an emulsion from his spit! He explained that all that is required to create an emulsion, such as mayonnaise, is water and protein—both readily available in human saliva.

The next day, I went into the kitchen at ICE and made carrot “mayonnaise” with my students by emulsifying oil into a mixture of egg white powder, carrot juice, fish sauce and lime juice. It worked, and truth be told, it was delicious.

As we tasted and discussed, one of my students suggested making it again with tomato—to-mayo if you will. I immediately liked the idea and knew where this to-mayo should go: on a BLT! After all, a great BLT begins with good bread and mayo, so why not make that mayo out of tomato? Here’s a recipe for to-mayo for your next BLT.

1 tablespoon tomato paste (umami)

½ teaspoon sherry vinegar (sour)

1½ teaspoon soy sauce (umami)

2 tablespoons aquafaba OR 1 teaspoon egg white powder

8 fl oz canola oil (fat)

Combine the tomato paste, sherry vinegar, soy sauce and aquafaba or egg white powder in a small bowl and whisk until dissolved—the mixture should become a bit foamy.

Gradually add the canola oil, pouring in a steady stream while whisking vigorously until the oil is emulsified.

Ultimate BLT

Per sandwich

4-6 slices thick cut bacon

2 slices good quality bread

2 slices beefsteak tomato, room temperature

Salt and pepper as needed

2 leaves crisp lettuce

Cook the bacon slices to desired doneness (chewy or extra crispy, we won't judge) in a large saute pan or cast iron skillet.

Remove the bacon to paper towels to rest. Leave about 1 tablespoon of the bacon in the pan.

Add the bread to the pan and toast each slice on one side in the bacon fat.

Remove the slices of bread to a plate. Spread the non-toasted side of one bread slice with to-mayo. Arrange the bacon on top of the to-mayo, add a layer of tomato slices and season lightly with salt and pepper. Place the lettuce on top of the tomatoes, then spread the non-toasted side of the other piece of bread with to-mayo and place on top. Serve immediately.

Frico aka Crispy Cheese Cake *broccoli rabe, onion, potato*

Inspiration Ingredients- broccoli rabe, cheese, onion

Compound- 2-butenal

Aroma- pungent

We love making roasted broccoli sprinkled with parmesan cheese. Inevitably, when it comes out of the oven, we are fighting over the crispy bits of parmesan that are left baked onto the pan, known in Italian as frico. Then it hit us, why fight over those little bits when we can make an entire dish that is just one giant crispy piece of cheese with delicious vegetables tucked inside.

Serves 4 as a side or appetizer

1 russet potato

Kosher salt, as needed (salt)

1 bunch broccoli rabe (bitter)

1 tablespoon olive oil (fat)

1 cup sliced onion

3 garlic cloves, thinly sliced (umami)

8 ounces grated white cheddar cheese, about 3 cups (fat, salt)

4 ounces grated parmesan cheese, about 1 cup (salt, umami)

Peel the potato and slice into rounds about ½ inch thick.

Combine the cheddar and parmesan cheese in a bowl. Mix well and set aside.

Place the potato slices in a pot and add enough cold water to cover by 3-4 inches. Season the water well with kosher salt.

Bring the pot to a boil and reduce the heat to simmer. Cook until the potato is just tender through, about 6 minutes after coming to a boil. *□ be careful not to overcook the potatoes here, if the water is boiling too rapidly or the if the cook for too long, the potatoes will turn into a crumbly mess.*

Carefully lift the potatoes from the simmering water and place on a plate to steam dry, leaving the simmer water on the stove.

Trim the bottom inch-and-a-half of stem from the bunch of broccoli rabe. Drop the broccoli rabe into the simmer and cook 3 minutes. Remove and spread into a single layer on a separate plate to steam dry. When cooled enough to handle, chop into ½ inch pieces.

Heat the olive oil in a 10 inch non-stick sauté pan, over medium high heat. When hot add the onions and sauté for 2 minutes to soften, add the garlic cloves and cook 2 minutes more. Slid the onions and garlic onto the plate with the potatoes and return the pan to the heat.

Scatter one-half of the cheese mixture into the pan and spread it into an even layer with a heat proof spatula. Reduce the heat to medium and let the cheese cook without touching it. *□just give the pan an occasional gentle swirl to insure the cheese is not sticking.*

When the cheese is completely melted, begins to look a little greasy and has lightly browned at the edges, scatter half of the potato and onions over the cheese, then add half of the chopped broccoli rabe on top of that. Gently spread and press the vegetables into the cheese. At this point, the cheese should be one solid disk that is well browned on the pan side and soft and creamy on the vegetable side.

Remove the pan from the heat. Carefully slide a heat proof spatula under one side of the cheese and in one quick motion, fold it in half, like a omelet. Slide the folded frico onto a plate to cool before slicing.

Repeat the process with the remaining ingredients.

Serve sliced into wedges, slightly warm or room temperature.

Chicken

Introduction

Chicken makes frequent and recurring appearances on our dinner table, and our lunch plates too for that matter. A generally lean protein with mild flavor, it is exceptional at taking on flavors, blending seamlessly into nearly any cuisine or cooking style. We don't like to tie specific cuts of chicken to any one recipe or preparation. There are cuts we prefer for some recipes, but truly any type of chicken (or other variety of poultry) can be used in any recipe. Cooking chicken is all about the time/temperature relationship.

Lean cuts from the breast should be cooked just to the point of doneness and no longer. These lean cuts lack the intramuscular fat and connective tissue that help keep dark meat (wings and legs) juicy throughout the cooking process. When any piece of meat is cooked, proteins begin to denature as a result of being exposed to heat. Think of the denaturing process as a net unraveling. When done properly, this net holds in the majority of the water that muscle fibers naturally contain. The longer the proteins are exposed to heat the more the net tightens, eventually to a point that all of the water is squeezed out of the net and you are left with dry, tasteless chicken. The problem with chicken is that this 'net' starts to squeeze tight at temperatures that are just beyond what is regarded as safe for consumption (165°F). The best approach when cooking lean cuts of chicken is to cook them to approximately 155°F then remove them from the pan, grill, oven or wherever they may be cooking and place them on a rack to rest. As the meat sits on the rack it will continue to cook (its internal temperature will continue to rise) for at least 5 minutes; by the time the temperature stops rising, it should have reached the 165°F mark. The chicken can then be served directly or returned to the pan or sauce for holding or serving. Cooked chicken will begin to rapidly lose moisture at 175°F (79°C), which means there is a very short grace period between cooked and dried out. This temperature is less of a concern with chicken legs and wings as the fat and connective tissue

naturally present in the meat will help keep the meat more tender and moist than lean white meat cuts.

Monitoring temperature during cooking and learning how to anticipate temperature rises after cooking are essential to properly cooked chicken. It may all sound a bit complex and technical, but the rules are simple and become quite intuitive with practice.

Brines can provide an extra layer of protection from dry, overcooked meats by promoting water retention or actually adding moisture to the meat before it is cooked. For a comprehensive look at brining techniques and methods, see pages XX-XX in the Tools & Techniques chapter.

Light or Dark?

Lean cuts of poultry lend themselves well to high heat cooking methods where cook times are kept short. Methods like grilling, searing/sauting, and roasting cook efficiently and provide for flavor meat via Maillard browning (see p. XX in the Intro for a review of Maillard flavors).

As previously mentioned, fat and connective tissue naturally present in poultry legs and wings help the meat to become tender and remain moist through longer cooking times, where low temperatures are preferred. BBQ, smoking, slow roasting, braising, and slow cooking are ideal cooking methods for these cuts. Any of the following recipes may adapted to use light or dark meat cuts.

Recipes that call for light meat cuts. To substitute dark meat, remove the bone from the meat and lightly pound the meat to create an even thickness. Reduce the cooking rate by approximately 20% (i.e. brown over medium heat rather than medium-high) and extend the cooking time by 2½ minutes per side.

Recipes that call for dark meat cuts. To substitute light meat, increase the cooking rate by approximately 20% (i.e. if the recipe calls to roast at 325°F, increase the heat to 390°F) and reduce the time, cooking just until the chicken reaches 165°F.

The Myth of Cooking on the Bone

For ages we've heard the mantra that cooking meats on the bone means more flavor. Here's the problem: the bone imparts no flavor whatsoever to the meat while it cooks. Rather the bones, which contain approximately 15% water, are very poor conductors of heat and slow the cooking process. Because of this, the meat surrounding the bone cooks more slowly and will likely retain more moisture. The slower cooking also means it's less likely to overcook and become dry.

Bottom line—bones make meats cook more slowly. Even if you monitor cooking temperatures and are careful not to overcook the meat, bones provide no added benefit in the cooking process.

Why do we cook chicken to 165°F?

165°F is the temperature at which the bacteria salmonella is killed instantly. If the chicken's internal temperature reaches 165°F, there is no possible chance for salmonella infection. This is an important consideration when dealing with ground meats, where bacteria like salmonella could potentially be found anywhere within the meat. However, when dealing with whole cuts like a whole chicken breast or thigh, any bacterial contamination would be most likely to occur on the surface of the meat. In these cases, surface bacteria will be eliminated in the normal cooking process (grilling, searing, roasting, etc).

Therefore, in the following recipes you will see that we advise you cook all chicken to 165°F because we don't want to take any responsibility if you get sick. However, at home when we know that we have good, wholesome chicken, we routinely cook chicken breasts to 145-150°F. This is particularly true if we are cooking sous vide. Though the USDA does not regard this temperature as universally safe, the proteins in chicken have gone through the denaturation process (aka it's cooked) by the time the internal temperature reaches 145°F (62.8°C).

Chicken Cooking Temperature Line (graphic design TBD)

40°F Storage Raw. Chicken should be kept at or below 40°F before cooking.

135°F* Connective tissue begins to break down and tougher cuts like legs, wings and thighs will become tender over time. *Not safe to eat at this temperature

145°F Proteins are denatured (aka 'cooked'). At this temperature, the proteins in chicken have nearly completed the denaturation process and have a firm, cooked texture.

165°F Safe Temperature. Salmonella is instantly killed at this temperature

175°F Overcooked. Lean proteins begin to rapidly lose moisture and become dry.

One-Pan Roasts

Pan Roast Chicken with Creamed Greens and Potatoes

Ingredient Inspiration- chicken, mustard greens, cream, lemon

Compound- leaf aldehyde

Aroma- almond, fat, green apple, grass

Once your ingredients are prepared this recipe is ready in 30 minutes. Or make it with chicken breasts instead and cut the cooking time in half. The flavors focus on pungency and their balancing tastes- cumin and mustard seeds are pungent, lemon and cream create balance.

Serves 4

- 1 tablespoon vegetable oil (fat)
- 1.5 lbs boneless chicken (approximately 6 thighs or 4 breasts) (umami)
- 8 fingerling potatoes, cut in half lengthwise (umami)
- 4 garlic cloves, crushed
- 2 branches fresh thyme
- 1 teaspoon cumin seeds or ½ teaspoon ground cumin
- 1 tablespoon butter (fat)
- 2 cups mustard greens or kale, chopped (bitter)
- 12 fl oz chicken stock (1½ cups)
- 2 tablespoon sour cream (sour)
- Juice of ½ lemon

Preheat oven to 375°F

Heat the oil in a large sauté pan over medium high heat.

Lay the chicken out on a plate and season well on both sides with salt and pepper. Pat the skin side of the chicken dry. Transfer the chicken to the hot pan with tongs; placing each piece skin side down.

Reduce the heat to medium and leave to chicken to cook 3-4 minutes undisturbed. Check to see that the chicken is well browned on the one side, if it has not continue cooking until it has.

Remove the chicken back to the plate they were on. *□this is fine, because one side of the chicken is still raw and will be cook further.*

Drain any remaining fat from the pan, and return to the heat. Immediately add the butter, garlic, thyme and cumin. When the butter is melted, add the potatoes and toss to coat. Season the potatoes with salt and pepper and return the chicken to the pan.

Transfer the pan to the oven and roast for 12-14 minutes for boneless breasts or 25 minutes for boneless thighs.

When the chicken is finished cooking, remove it from the pan to rest and set the saute pan back on the stove over medium-low heat. Cook for 2-3 minutes, stirring often, so that the fat remaining in the pan begins to sizzle and turns clear (not colorless). *□ this is process of 'clarifying the fat' is basic technique for many pan gravy. By simmering the pan drippings, the juices that came out of the chicken during cooking dehydrate and turn into fond that clings to the bottom of the pan and is then released in the sauce.*

Add the mustard green to the pan and saute until just wilted, about 1 minute. Deglaze the pan with chicken stock and reduce by three-quarters.

Remove the pan from the heat, and stir in the sour cream and lemon juice, then return the chicken to the pan and toss to coat. Serve immediately.

Pan Roast Chicken with Herbed Carrots and Mustard

Ingredient Inspiration- dill, butter, lemon, carrot

Compound- terpinolene

Aroma- pine, plastic, sweet

Think of this as Chicken Picatta's sweet, saucy cousin. Carrots, dill, and lemon pair incredibly well thanks to the underlying woody and resinous aromas that all three contain (α-pinene and α-terpinolene). They are very similar to (and friendly with) citrus aromas. It's all the flavors of Picatta without the pucker—carrots lend sweetness and crunch which the original lacks.

Serves 4

4 chicken breast cutlets (umami)
kosher salt, as needed (salt)
freshly ground black pepper, as needed
2 tablespoons olive oil (fat)
2 tablespoons scallion, thinly sliced
2 cups of carrots, peeled and sliced (sweet)
juice and zest of 1 lemon (acid)
4 tablespoons butter, cut into pieces (fat)
1 teaspoon Dijon mustard (spicy)
1 tablespoon chopped dill or 1 teaspoon dried dill

Season the chicken with salt and pepper and set aside on a plate.

Heat the olive oil in a large sauté pan over high heat. When the oil begins to shimmer and lightly smoke at the edge, pat the chicken dry and add to the pan. Reduce the heat to medium and cook until well browned on one side, about 3 minutes. Flip the chicken and brown well on the second side, about 3 minutes more. Remove the chicken from the pan and cool on a rack.

Drain any excess oil from the pan and add the scallions. Sauté briefly until aromatic, but not browned. Stir in the carrots and continue cooking until just tender, 2-3 minutes.

Add the lemon juice to the pan and bring to a bare simmer. Immediately remove the pan from the heat and stir in the butter. Stir constantly until all the butter is melted and incorporated into the sauce. Stir in the mustard and dill and return the chicken to the pan. Turn the chicken to coat with the sauce and serve immediately.

Ultimate Honey Mustard Chicken

Ingredient Inspiration- yogurt, mustard, basil, honey

Compound- 2-phenylethanol

Aroma- fruit, lilac, rose, wine

This is honey mustard chicken with the flavor turned up to 11. Short and simple, this is the best grilling or broiling marinade you'll ever use. But don't stop at the grill, this marinade makes amazing fried chicken too!

Serves 4

¼ cup plain yogurt (sour)

2 tablespoons fresh basil leaves, chopped

1 tablespoon Dijon mustard (spicy)

1 tablespoon honey (sweet)

1 teaspoon kosher salt, plus more as needed (salt)

Few dashes worchestershire sauce (umami)

1.5 lbs boneless chicken (approximately 6 thighs or 4 breasts) (umami)

2 teaspoon smoked paprika

Combine the yogurt, basil, mustard, honey, salt and Worcestershire sauce in a bowl or container suitable for marinating. Whisk until it forms a smooth mixture.

Add the chicken to the marinade and turn to coat evenly. Place a lid on the container or transfer the chicken and marinade to a zip top bag, squeeze out all the air and seal.

Refrigerate the chicken in the marinade over night, up to 48 hours.

To cook, remove the chicken and wipe off any excess marinade. Season the pieces lightly on both sides with salt and smoked paprika. *□ if too much marinade is left on the chicken it will tend to cause it to burn or stick to the grill.*

Transfer the chicken immediately to a well-oiled grill (medium heat) or place on a rack and roast underneath the broiler, turning once half way through the cooking. For boneless breasts cook approximately 6 minutes per side, for boneless thighs approximately 8 minutes per side (though they will need to be turned more often to avoid charring).

Southwestern Grilled Chicken Salad

Ingredient Inspiration- grapes, grilled corn, tomato, cilantro

Compound- 4- hydroxynonanoic acid

Aroma- apricot, cocoa, coconut, peach, sweet

Chicken and fruit aromas pair well, this recipe gives you fruit aromas from all corners of the produce section, without being 'fruity.' Use any type of cooked chicken you like. If you're firing up the grill for the corn, grilled chicken is ideal, but a store-bought rotisserie chicken will do just fine.

Serves 4

- 1 tablespoon cider vinegar (sour)
- 1 teaspoon soy sauce (umami, salt)
- 1 teaspoon sriracha hot sauce (spicy)
- 1 bunch scallions, thinly sliced
- 1 cup grape tomatoes cut in half (umami)
- 1 cup green or red grapes, cut in half (sweet)
- 2 ears corn, grilled or roasted, kernels cut from the cob (sweet)
- 2 tablespoons cilantro leaves, chopped
- 1 cup feta cheese, crumbled (umami, salt)
- 2 tablespoon olive oil (fat)
- 2 lbs cooked chicken (umami)

In a large bowl, combine the vinegar, soy sauce, sriracha and scallions. Mix well and set aside while you prepare the remaining ingredients.

Add the tomatoes, grapes, corn kernels, cilantro, feta cheese, olive oil and cooked chicken to the bowl and toss well to combine. *□ if you're not a fan of cilantro- it's ok if you're not, it your parents fault not yours. Most people who strongly dislike cilantro genetically have a different aroma receptor that gives cilantro a soapy flavor- substitute fresh basil*

Taste for seasoning, adding salt and more hot sauce as needed. The flavor of the salad will improve for a few hours after mixing, but will begin to after 24-36 hours as the grapes and tomatoes begin to break down and become watery. If you need to prepare this recipe in advance it is best to keep all the ingredients separate and mix together 3-4 hours before serving.

Creamy Tomato Chicken Ragu

Inspiration Ingredients- tomato, cream, onion, vodka

Compound- 2-heptanone

Aroma- bell pepper, blue cheese, fruit, green

Again with the fruit aromas that pair with chicken... but in a very different way. Here we are accessing the fruity aromas in tomatoes and complementing them with the floral aromas from dairy. Which land this recipe in a delicious spot that is square between Bolognese and classic tomato vodka sauce. Of course this sauce is perfect for pasta, but no noodles required. It is equally good with cooked grains, roasted potatoes, or all on its own. Heck, we've even been known to throw it on a bun and dig in sloppy joe style... or Sloppy Giuseppe?

Serves 4

2 tablespoons vegetable oil (fat)

1 lb ground chicken (or turkey) or 1 lb boneless chicken thighs cut into ½ inch cubes (umami)

Kosher salt (salt)

Freshly ground pepper (spicy)

1 yellow onion, grated, juice reserved *□grating the onion is way easier than mincing, but also will help the onion to break down completely in the recipe releasing maximum flavor into the dish. The best way to do this is to place your grater inside a bowl, then grate the onion, to insure you capture all the juice*

1 tablespoon minced garlic

2 tablespoons fresh basil, divided

½ cup vodka

28 oz can chopped tomatoes (umami)

8 fl oz heavy whipping cream (fat)

Grated parmesan cheese for garnish

Heat the oil in a heavy-bottomed, wide pot.

When the oil is hot, add the ground chicken season well with salt and pepper. Cook stirring occasionally until the meat is well browned, 8-10 minutes. *□take your time with step, it is where all of the flavor is developed. First the meat will turn grey, meaning that it is cooked, and will release some liquid. Continue cooking, stirring occasionally until all of that liquid that was released, has evaporated and turned flavor rich fond on the bottom of the pot.*

When all of the liquid has evaporated from the chicken, stir in the grated onion and minced garlic.. If the pot seems dry before you add the onion, add a little extra oil. Cook until the onions and garlic are completely tender, 4-5 minutes. Stir often as the onions cooks and scrape the bottom of the pot with a wooden spoon or heat proof spatula to release the fond before it burns.

Slide the pot off of the heat and carefully pour in the vodka. Stir well and slide the pan on the heat. Stand back, in case the vapors ignite. If they do, don't panic, simply stand back and let the alcohol burn off. The flames will last no more than 45 seconds and nothing in the pot will burn or be damaged in that time.

Stir in the tomatoes and 1 tablespoon of basil. Mix well and place a lid on the pot and reduce the heat to a simmer. Cook 15 minutes, stirring occasionally. Remove the lid and cook 10 minutes more, still stirring occasionally. Stir in the cream and simmer another 5 minutes or until thickened. Remove from the heat and stir in the basil.

To serve with pasta, cook one pound of desired shape of pasta to al dente in a heavily salted water. Transfer the pasta directly to the pot of sauce incorporating about ½ cup of pasta water as well. Simmer the pasta and sauce together for 2 minutes stirring constantly. Serve immediately. Garnish with cheese

Ginger Beer Battered Chicken Fingers

Ingredient Inspiration- ginger beer, lime, chili powder

Compound- 1,8-cineol

Aroma- camphor, cool, eucalyptus, mint

Ginger, lime and chili are outstanding pairings for chicken, but not very surprising, so here we take these perfectly paired ingredients and put unique twist on them. Classic tempura batter with ginger beer.

1¼ cups ap flour, plus more as needed

¾ cup corn starch

1 tablespoon baking powder

1 tablespoon plus 1 teaspoon salt, divided (salt)

12 fl oz ginger beer, chilled □ *look for a low sugar ginger beer. If it is too sweet, the batter will burn before the chicken is finished cooking through if you can find a low sugar option cut the ginger beer with a bit of club soda.*

ice cold water, as needed

2 lbs boneless, skinless chicken, cut into strips (umami)

1 tablespoon chili powder (spicy)

Lime wedges for garnish (sour)

Combine the flour, corn starch, baking powder, and 1 teaspoon salt in a bowl. Whisk well to combine.

In a separate bowl, mix the remaining tablespoon of salt with the chili powder. Set aside.

Whisk the ginger beer into the flour mixture. It should have the consistency of pancake batter. If the mixture is too thick, add cold water 1 tablespoon at a time to reach desired consistency.

Heat a pot of oil to 350°F. In a small bowl mix the salt and chili powder.

Lay the strips of chicken out on a platter or baking sheet and season lightly with the chili salt mixture. Sprinkle with flour over the chicken and toss to lightly coat. □ *coating the chicken with small amount of flour will help the batter to stick to the meat. Otherwise the crisp coating will tend to slip or flake off.*

Take the strips of chicken, one at a time and dip them into the batter. Then gently lower into the hot oil. Cook until golden brown and crisp, about 6 minutes a piece. When the chicken is cooked remove and rest on a rack. Season immediately with the chili-salt mixture.

Serve with lime wedges to squeeze on top. Or make a dipping sauce by mixing 1 cup mayonnaise with the zest and juice of lime.

Indian Chicken and Green Vegetable Stir Fry

Ingredient Inspiration- Green Pea, Cucumber, and Hazelnut

Compound- E (2)-nonenal

Aroma- cucumber, fat, cut grass, watermelon

Serves 4

Sauvignon blanc typically brings thoughts of citrus and melon. However, green vegetable—and more specifically green pepper—are one of the most important aromas found in the wine. This recipe brings green vegetable and spice in bunches to an unexpected pairing. Peas, cucumber and jalapeno all hits different notes of “green” that compliment the wine perfectly while crisp acidity in each sip refreshes between spicy bites.

2 lbs boneless, skinless chicken (umami)

2 teaspoons garam masala (spicy)

1 teaspoon kosher salt Salt)

1 tablespoon ghee or vegetable oil (fat)

3 tablespoons butter (fat)

1 cup onion, thinly sliced

3 garlic cloves, crushed

3 slices fresh ginger

1 teaspoon cumin seeds

½ teaspoon ground coriander

1 cup jasmine or basmati rice

1 cup petite green peas, thawed

2 kirby cucumbers, thinly sliced

¼ cup hazelnuts or peanuts, toasted and chopped

½ green jalapeno, sliced and seeds removed (spicy)

1 tablespoon lime juice (sour)

2 tablespoon extra virgin olive oil (fat)

2 tablespoons cilantro leaves for garnish

Season the chicken on both sides with the garam masala and salt.

Place a large saute pan over high heat and add the oil. When the pan and oil are very hot, add the chicken to the pan, make sure they remain in a single layer without touching. Cook in batches if necessary. Cook the chicken until well browned on both sides. When all the chicken pieces are browned, remove them from the heat and rest on a rack.

Drain any remaining fat from the pan and add the butter. Place the pan back over high heat and add the onions, garlic and ginger. Cook until tender and lightly browned. Stir in the ground cumin, coriander and rice. Sauté 1 minute longer, until very fragrant. Stir the chicken stock into the pan and return the seared chicken pieces to the pan. Cover with a lid and bring to a simmer. Reduce the heat to low and cook 15 minutes or until the rice has absorbed all the liquid. Then remove the pan from the heat, gently stir in the peas then replace the cover and let stand covered 5 minutes.

While the chicken is simmering, combine the sliced cucumber, hazelnuts, jalapeno, lime juice and olive oil in a bowl. Mix well and season to taste with salt.

To serve, spoon the chicken and rice onto plates, top with the cucumber salad and garnish with cilantro.

Spicy Pomegranate and Chicken Lettuce Wraps

Ingredient Inspiration- chicken, pomegranate, chili, lemongrass

Compound- 2-ethyl-1-hexanol

Aroma- citrus, green, oil, rose

Sometimes dinner just need to be quick and easy, and that exactly what this recipe is. It's ready in about 15 minutes, but the best part is that it is as good the day you make it as it is the next day for lunch. This craveable combination is based on the strong citrus aromas that link together pomegranates, chili and lemongrass. It's a strong match and chicken is happy to come along for the ride.

Serves 4-6

2 tablespoons sesame oil (fat)

1 tablespoon minced garlic

1 tablespoon minced ginger

2 tablespoons thinly sliced scallion

1 jalapeno, seeds removed and minced (spicy)

2 3-inch pieces lemon grass stalk, crushed (optional)

1 lb ground chicken (umami)

4 kaffir lime leaves (optional)

1 teaspoon corn starch (or rice flour)

2 tablespoons soy sauce (umami, salt)

8 fl ounces pomegranate juice (sweet, bitter)

1 tablespoon toasted sesame seeds

1 tablespoon mint leaves

1 tablespoon cilantro leaves

½ head iceberg lettuce, cut in half, core removed for garnish

Heat the oil in a large saute pan or wok over high heat, add the lemongrass if using. When the lemongrass begins to sizzle, stir in the garlic, ginger, scallion and jalapeno. Stir fry, until aromatic, about 20 seconds.

Add the ground chicken and lime leaves. Immediately break the chicken up into pieces. Continue cooking over high heat, until the chicken is fully cooked any juices that accumulated in the pan are completely dry, 6-8 minutes more. Sprinkle the cornstarch over the chicken and stir until absorbed.

Deglaze the pan with the soy sauce and stir until nearly dry. Add the pomegranate juice and simmer, stirring constantly until the juice is thickened and slightly syrupy. Transfer to a plate and garnish the chicken with sesame, mint and cilantro. Serve with lettuce wedges for making wraps.

Beef, Pork and Lamb

These meats fall into three distinct categories in our minds, yet they are really far more similar than they are different. All three animals have the same bone and muscle structure, and have very similar diets. Further, the ultimate flavor developed in a cut of meat, is largely influenced by it's cooking method, as raw meat generally has very little aroma. The flavor in cooked meats mostly comes from the pyrazines and pyridines generated by Maillard reactions also known as searing. Sometimes incorrectly referred to as 'caramelization,' Maillard reactions require amino acids as well as reducing sugars to creating pleasing roasted aromas. Though there are three main fatty acids that play a role in the flavor of these and they are influenced by genetics, but more importantly by diet.

- Grass-fed animals have higher concentrations of linolenic acid (clean fatty flavor)
- Grain-fed animals develop more oleic and linoleic acid (fatty, fried flavor)
- Corn-finished animals develop less caproic acid (cheesy-fatty flavor)

Aging also plays a surprising important role in the flavor of meat. Beef, in particular, has very little flavor before aging. A key aromatic in beef and pork flavor, *1-octen-3-ol*, can increase 10x in the first 14 days of beef aging. The compound (E,E)-2,4- decadienal (aroma: cilantro, deep-fried, fat, oil, oxidized) is created by the oxidation of linoleic acid and has been identified as contributing significantly to the unique of lamb.

Retail cuts of meat undergo a minimum of 7 days aging. The vast majority of meat that you find at your local grocery is wet aged. Despite its name, no liquid is added to the meat during this process. In wet aging, the meat is simply vacuum sealed to prevent oxidation and moisture loss as it ages under standard refrigeration. In dry aging, the meat is held in a highly precise environment where temperature, humidity, and even light exposure are carefully controlled. During the dry aging process, the meat will lose moisture giving it more concentrated flavor.

During this process, enzymes also develop in the meat that begin to naturally break down connective tissue within the muscles, making the meat more tender.

Like chicken, and most other proteins, for that matter, there is no single perfect answer for which cut of meat should be used in a particular recipe. With beef, the proper cut is sometimes best chosen by budget and time. I can assure that if you were to substitute ribeye or lamb chops in any of the following recipes, and adjust the cooking time accordingly, the results would be delicious. However, that would be an extraordinarily expensive way to put dinner on the table night after night.

As we mentioned before, fat is the key to flavorful meat. It also helps to keep the meat moist throughout the cooking process. This holds true for other types of meat as well. Specifically, pork and lamb which we will explore in the following sections. Highly marbled cuts of meat are the most forgiving to cook and they simply taste better. They also tend to be the most expensive. Leaner cuts, or cuts with more connective tissue can be a bit more temperamental, but can be equally delicious if cooked properly. Though, when properly handled, these cuts from beef stew meat (typically cubes from the shoulder) to pork shoulder to lamb shanks can yield deeply flavorful and highly satisfying meals which some prefer to the more expensive cuts.

How you chose to cook your meat should be largely determined by the type of cut you select. Those that are naturally low in connective tissue (CT) should be cooked quickly, over high heat. Cuts with lower amounts of connective tissue come from the center part of the animal (between the shoulder blade and the hips and are your typical 'steak' cuts. They are naturally tender and will be the cuts are cooked to desired temperature in beef and lamb or to 145°F in the case of pork.

Cuts that come from the shoulder and hind leg have higher amounts of connective tissue and will require longer cooking and lower heat. As the connective tissue is exposed to heat, it slowly begins to break down and transforms from dense and chewy to tender and creamy. The elastin

and collagen in CT actually becomes gelatin, giving slow cooked meat their distinctive, rich mouthfeel. It is this process that allows these leaner cuts to become tender and moist. Though it must be done slowly, if cooked at high heat, the connective tissue will toughen and the meat will dry out.

No matter what cut of beef, lamb or pork you are cooking, salt is the most essential part of the cooking process. As soon as you get your meat home, you should take it out of whatever packing it is in and pat it dry. Lightly salt the meat on all sides and then wrap in paper or loosely cover in a glass or plastic pan. This light salting will do two things- first it will help to dry the surface of the meat so that when you do eventually cook it, the meat will more readily develop a brown, seared crust on the exterior. Secondly it will give the salt time to penetrate the meat resulting in fuller flavor throughout.

When it comes time to cook, there is no need to remove the meat from the refrigerator for any set time before cooking. It's a myth that withstood the test of time, but the truth is setting a steak on plate on your countertop for 30 minutes before taking it to the grill will have zero net effect on the eventual doneness or browning of said steak.

In the following recipes you will see that we use a variety of different cooking techniques for a variety of cuts of meat. Don't feel bound to replicate ours exactly. Try the beef recipes with lamb or the pork recipes with beef. There are no rules! Cook what you like, keeping mind the rules that we mentioned above, most importantly employing the right methods for tender and tough cuts.

Slow Cooker Sweet Potato and Ale Beef Stew

Inspiration Ingredients- sage, sweet potato, beer

Compound- limonene

Aroma- balsamic, citrus, fragrant, fruit, greenery, herb

Limonene, as its name would suggest, is the most prominent compound in the aroma of all citrus fruits. In this recipe, we take a look at the other side of the compound; pulling together citrus-adjacent ingredients without using any actually citrus. Beer and sage both have particular strong pairings with citrus which add a bright flavors to this hearty and crowd pleasing stew.

Serves 4-6

2 lbs beef chuck cubes (umami)

1 tablespoon kosher salt (salt)

1 teaspoon ground black pepper (spicy)

1 teaspoon smoked paprika

¼ teaspoon ground cumin, plus more as needed

2 tablespoons flour (to thicken, use gluten free alternative)

1 head garlic, with top cut  *yup, the whole thing, as you would for preparing roasted garlic (illustration)*

1 (12 oz) ale style beer (bitter)

4 branches fresh sage or 1 tablespoon dry sage

2 sweet potatoes, peeled and cut into 2 inch cubes (sweet)

1 tablespoon extra virgin olive oil (fat)

Toasted bread for garnish (see note in recipe)

½ cup sour cream (sour)

Place the beef cubes in a bowl, add the salt, pepper, paprika and cumin. Toss well to combine.

Add the flour and toss again to coat.

Heat the oil in a large cast iron or non-stick pan over high heat. When the oil is hot, add the beef cubes so that the pan is full, but none of the pieces are touching one another. Turn so that the beef is browned evenly on all sides. □ *overcrowding the pan will prevent the beef from browning properly, work in batches if necessary*

As the beef cubes brown, transfer them to a slow cooker. When all the beef has browned, add the onion and garlic to the pan and sauté for 2 minutes, then transfer the onions and garlic to slow cooker as well.

Pour in the beer and sage. Set slow cooker to low and cook for 8 hours or high 4 hours. Add the sweet potatoes halfway through the cooking process. □ *you can add them earlier if you must for convenience, but the sweet potatoes may turn out a little 'mushy'*

Remove the garlic from the pot and set aside to cool. When the garlic is cool enough to handle squeeze the whole head from the root end over a bowl. The creamy cooked cloves of garlic will pop out into the bowl. Add the olive oil to the garlic and mash with a fork until a rough paste forms. Spread on toasted bread and serve with the stew. □ *baguette slices would do, but toasted rye bread or pumpernickel cut into pieces would be ideal match here.*

To serve, scoop the stew into bowls and garnish each bowl with a spoonful of sour cream, a pinch of cumin and toasted bread with garlic.

Pan Seared Steak with Bourbon Raisin Sauce

Ingredient Inspiration- raisins, bourbon, mustard, Worcestershire

Compound- 2,4,5- trimethyloxazole

Aroma- nut, sweet

Serves 4

In this recipe Taste steps to the forefront, with the savory-sweet combo of roasted meat and bourbon soaked raising taking center stage. There are plenty of aromas that tie these ingredients together but success of this dish comes down to kitchen fundamentals. A deep brown sear on the meat creates abundant savory aromas via Maillard reactions, which are essential to balance the sweet notes coming from raisins and bourbon. If you find the sauce too sweet for your tastes, add a few drops of red wine or sherry vinegar at the end to balance.

1 (2-pound) piece beef tenderloin or strip loin, trimmed and tied (umami)

2 tablespoons canola oil (fat)

2 tablespoons butter (fat)

4 garlic cloves, crushed

4-6 branches fresh thyme

3-4 branches fresh rosemary

4 fl oz bourbon

8 fl oz chicken stock

¼ cup raisins (sweet)

2 tablespoons Dijon mustard (spicy)

2 tablespoons butter (fat)

3 dashes worchestershire sauce (umami)

Season the roast with a generous amount of salt and pepper, and let stand at room temperature for 30 minutes to 1 hour. □ *allowing the meat to come to room temperature before searing will*

result in more even cooking, this time all allows the salt to penetrate, making the meat it more flavorful.

Preheat oven to 425° F

Before cooking, pat any excess moisture off of the meat and season again with salt and pepper. Place a large sauté pan or cast iron skillet over high heat. Add the canola oil and heat until the oil begins to smoke. Add the roast and brown well on all four sides, 3 to 4 minutes per side. Drain the excess fat from the pan; add the butter, garlic, thyme, and rosemary. As the butter melts and the herbs sizzle, spoon the melted butter over the roast repeatedly, basting for 2-3 minutes.

Transfer the entire pan to the preheated oven and cook until an internal temperature of 130° F (for medium rare) is reached. Remove the meat from the pan and place on a rack to rest while you prepare the sauce. *□there are a number of factors that are going to effect how quickly the meat cooks including actual temperature of your pan and oven, diameter of the roast, the fat and moisture content of meat, and more. Therefore an exact time in the oven is difficult to predict, so it is best to err on the safe side and check the internal temperature after 6 minutes in the oven and calibrate timing from there.*

Drain the excess fat and aromatics from the pan. Place the back on the stove top with the burner off. Pour in the bourbon and raisins and stir well. The pan should sizzle immediately. When the bubbling subsides, turn the burner to medium and allow the bourbon to reduce by ¾. *□be careful, the bourbon may still ignite! If it does, just step back let it burn out- the flames should only last for 30 seconds at most.*

Add the chicken stock and reduce again by ¾. Remove the pan from the heat and add the mustard, butter and Worcestershire sauce. Stir until the butter is melted ad emulsified. Spoon the finished sauce over the steaks and serve immediately.

Seared Steak with Almond-Cherry Pesto

Ingredient Inspiration- almond, garlic, oregano, cherry

Compound- benzaldehyde

Aroma- bitter almond, burnt sugar, cherry, malt

Serves 4

As seen in the previous recipe with raisins, beef pairs well with fruity aromas. This is an Italian spin on a pretty unique combination of ingredients- making Sicilian style almond pesto from Trapani. Almonds and cherry (our unique addition) are an off the charts pairing- botanically, the plants that the two fruits are nearly identical. Benzaldehyde creates the distinctive aroma of almonds- in fact imitation almond flavoring is just a diluted benzaldehyde solution- it also carries strong aromas of cherry fruit.

2lbs hanger, flat iron or skirt steak, trimmed (umami)

Kosher salt, as needed (salt)

Freshly ground black pepper (spicy)

1 tablespoon vegetable oil (fat)

2 tablespoons butter (fat)

2 cloves garlic, smashed

Lay the trimmed steaks out on a tray and season on all sides with salt and pepper. Rest 15-30 minutes before cooking.

Heat the oil in a large cast iron or other heavy bottomed pan over high heat. When hot, pat the steaks dry and carefully add them to the pan, working in batches if necessary to avoid overcrowding.

If using skirt steak, these cooking times should be cut in half □ Sear the steaks on one side for 3-4 minutes or until well browned. Turn and brown on the second. After turning, add the butter and garlic clove and baste as the steak continues cooking. After 3 minutes on the second sides, turn the steak again back to its first side and baste 1 minute more. Remove the steaks from the pan and place on a rack to rest. □ *resting is an important step for cooked proteins. It allows the muscle fibers to relax and hold more of the juices inside. This is particularly important for the cuts in this recipe as they will be sliced before serving. If they are not rested properly, all the juices from the steak will be left behind on your cutting board.*

After resting cut the meat into thin slices, taking care to slice against the grain of the meat. Serve with almond-cherry pesto.

Almond-Cherry Pesto

Makes 1 pint

½ cup water

½ cup red wine vinegar (sour)

½ cup dried cherries (sweet)

1 teaspoon oregano

1 cup toasted almonds (fat)

1 clove garlic

¼ cup grated pecorino or parmesan cheese (umami, salt)

½ cup extra virgin olive oil (fat)

While the steaks are cooking, combine the water and vinegar in a small saucepot and bring to a boil. As soon as the liquid reaches a boil, add the cherries and oregano. Reduce the heat to simmer and cook until only 1-2 tablespoons of liquid remain. Remove the pot from heat and set aside to cool.

While the cherries cool, combine the almonds, garlic and pecorino in a food processor and blend until they form a paste.

Add the cherries and any juices that remain in the pot, and the olive oil. Blend again to form a thick sauce. Add more olive oil if needed. Season to taste with salt and pepper. Keep refrigerated up to 10 days.

Grilled Beef and Eggplant with Espresso Butter

Ingredient Inspiration- beef, eggplant, espresso

Compound- linalool

Aroma- bergamot, citrus, cilantro, lavender, lemon, rose

Serves 4

Beef and eggplant are two ingredients that absolutely LOVE the flavor of char. In this recipe, we use espresso powder like a megaphone to blast those 'roasted' and 'charred' aromas throughout the dish. Compound butters are a chef's secret weapon, they are the most effective way to add lots of big flavor to a dish with very little effort. You can make them, then wrap tightly and store in the freezer for months. Surprisingly, the same charred aromas that make this dish so craveable link to citrus-y aromas like linalool that also make your finished plate bright and fresh.

8 oz butter (fat)

Zest and juice of ½ lemon (sour)

2 tablespoons espresso powder or coffee grounds (bitter)

1 tablespoon chopped mint

2 teaspoons kosher salt (salt)

½ teaspoon ground black pepper (spicy)

4 steaks of your choice grilled to desired temperature (umami)

4 Japanese eggplant (bitter)

4 tablespoons extra virgin olive oil, divided (fat)

2 kirby cucumbers, sliced (about 1½ cups)

2 teaspoons lemon juice (sour)

Kosher salt (salt)

Freshly ground black pepper (spicy)

¼ cup mint leaves, for garnish

Season the steaks well with salt and pepper. Cook to desired temperature over high heat. As soon as the steaks come off the grill, place them on a rack to rest and top each with 1 tablespoon of the espresso butter. The butter should melt and run over the steaks as they sit.

For the eggplant, split the eggplant in half lengthwise. *□if they have large curve or bend, cut each eggplant into 2 pieces before splitting in half to make it easier.*

Turn the eggplant so that the flesh side faces up, and lightly score the flesh with a pairing knife to create a diamond pattern. Season the flesh side lightly with salt and let stand for 5 to 10 minutes. *□this will help to draw some of the excess moisture out of the eggplant before cooking.*

Pat the eggplant dry and brush the flesh of the eggplant with a generous amount of olive oil, about 2 tablespoons. Season again light with salt and pepper.

Place the eggplant on a hot grill flesh side down. Cook 3 minutes or until the edges begin to char and turn black. Flip the eggplant and cook 3 minutes more on the skin side. Turn again for a finish cooking 1 minute on the flesh side. Remove the eggplant to the rack with the steaks and immediately spread the flesh side with espresso butter. Let stand 1 minute.

To serve, toss the cucumbers with the lemon juice and remaining 2 tablespoons olive oil. Season lightly with salt.

If the steaks have cooled, quickly reheat them on the grill, then cut into slices. Place the eggplant pieces each plate and top with a teaspoon of espresso butter. Place the sliced steak directly on top of the butter. Garnish each plate with cucumbers and mint leaves.

Chianti Braised Beef with Grits

Inspiration Ingredients- red wine, corn, butter

Compound- 2- phenylethanol
Aroma- fruit, honey, lilac, rose, wine

Serves 6+

This is the signature dish at our restaurant Angelena's. The Flavor Matrix showed us that beef, red wine and corn were a great pairing, but exactly how to pull that combination eluded us for quite a while. It wasn't until we started thinking outside of the 'cob' that we found what we were looking for. Corn flavor can be found in places other than yellow kernels on the cob. Here we find it in the flavor of buttery grits.

4 lbs beef short ribs, trimmed
Kosher salt
Freshly ground black pepper
1 teaspoon fennel seeds
2 tablespoons vegetable oil
½ chopped carrots (1 inch pieces)
½ cup chopped celery (1 inch pieces)
6 garlic cloves
4 branches fresh thyme
4 branches fresh sage
3 cups Chianti wine
3 cups chicken stock
Butter, as needed
1 recipe stone ground grits, prepared

Cut the short ribs into portion-sized pieces- if the short ribs are bone in, cut them into sections with one bone each. If boneless, cut into pieces approximately 3"X4" (about 6 ounces each). Season the meat generously with salt and pepper and place in a pan in a single layer. Scatter the

fennel seeds over the meat, cover the pan and return to the refrigerator overnight. *□if you don't have the time, let the ribs stand one hour at room temperature.*

Preheat oven to 325°F

Place a large, heavy bottomed pan over high heat and add the oil. While the oil heats, pat the surface of the short ribs and season again lightly with salt. Carefully, plate the ribs into the hot pan and sear until well browned on all sides. Work in batches, if necessary, to avoid overcrowding. Set the seared ribs aside on a clean baking pan, and reduce the heat to medium and drain any excess fat. Add the carrots, onion and celery to the pot and season with 1 teaspoon of kosher salt. Sauté 5 minutes, then stir in the garlic, sage and thyme; cook 1 minute more.

Degalze the pan with wine and cook, stirring occasionally until reduced by $\frac{3}{4}$. Then return the short ribs to the pan along with any juices that may have collected. Pour over the chicken stock and bring to a boil.

Once the pot reaches a boil, cover with a lid, or aluminum foil, and transfer to the oven to cook for 2½ hours.

After 2½ hours remove the ribs from the oven, uncover and rest on the stove top for 30 minutes. Remove the ribs to a clean sheet pan or platter, strain the cooking liquid into small sauce pot. Bring the cooking liquid to a bare simmer and cook for 15 minutes to reduce. While the liquid is simmering, skim any fat from the surface and discard. If not serving immediately, transfer the ribs to a storage container. Cool the reduced cooking liquid and pour over the ribs. Store in the refrigerator for up to 5 days.

To serve the short ribs, place the desired number of portions in a saute pan and add 4 oz ounces of cooking liquid per portion. Cover the pan and bring to a boil. Remove the cover and continue

boiling until the liquid is reduced by half. *□as the liquid is reducing, turn the ribs so that they heat evenly and occasionally baste the ribs with the cooking liquid*

When reduced, remove the pan from the heat and add 1 teaspoon of butter per portion. Swirl the butter into the pan until emulsified. Place the ribs on top of the prepared grits and spoon over 2 ounces of braising liquid.

Creamy Stone Ground Grits (or polenta)

5 cups water

1 teaspoon salt

3 tablespoons butter, divided

1 cup stone ground grits

1 ear of corn, kernels removed

½ cup grated parmesan cheese

Bring the water, salt and 1tablespoon butter to a boil in a medium saucepot. When the water reaches a boil, remove the pot from the heat and add the grits, whisking vigorously for 10 seconds.

Let the pot stand pieces of husk will rise to the top of the water, while the grits sink. Using a small, fine-meshed strainer, scoop the husk from the surface of the water and discard. Whisk the pot again, braking up any clumps that may have formed at the bottom. Return the pot to the heat and adjust the temperature so that the grits cook at simmer.

Cook, until the grits are thick, creamy and tender, typically 30-40 minutes. *□be sure to taste the grits. Do not cook only until they are thick, sometimes the grits will thicken quickly, yet remain tough or “gritty.” If this happens add more water and continue cooking until tender.*

When the grits are done, remove from the heat and whisk in the parmesan and remaining butter.

When cooking grits, you should always work in a proportion of 4:1. Four parts water to one part DRY grits. Lightly salt the water, add a pat of butter and bring it to a rapid boil. Add the grits in a steady stream, whisking the whole time to avoid lumps. Continue stirring and return the pot to a boil. Reduce heat to a simmer and continue cooking, stirring occasionally until the grits are tender – 30 to 45 minutes.

Pork

Heritage breeds of pork, like Berkshire, Duroc, Spotted and Mangalitsa grow to maturity more slowly than commercially raised breeds. These pigs require more feed and space to roam and therefore are more expensive. Though higher priced, heritage breeds of pork yield meat much more flavorful meat, with a higher ration of fat to lean, a richer coloring (more red than white). Like grass-fed beef pasture raised pigs develop high quantities of omega 3-fatty acids and linoleic acid.

When cooking pork, the goal should be to reach an internal temperature of 145°F. At this temperature the interior of the pork should still have a rosy blush. It is not necessary to cook pork to well done so that the interior is completely white and firm. The concerns with undercooked have been nearly completely eliminated with better farming practices and higher food safety standards. By cooking all of the color out of your pork the only thing you're eliminating is flavor.

Maillard flavors make the best pairings for pork, especially darker aromas like caramel or roasted. Pork also goes very well with fruity aromas, whether they are found in tropical fruits, chilies, or tomatoes.

Pan Seared Pork Medallions with Apples, Balsamic and Bourbon

Inspiration Ingredients- coffee, balsamic, apple, bourbon

Compound- 4,5- dimethylthiazole

Aroma- green, nut, roasted, smoke

This is our seasonal variation on one of our favorite recipes from THE FLAVOR MATRIX. We originally devised this recipe with peaches and it became a staple in our house. But making in cool Fall and Winter weather just didn't feel right, so we decided with switch it up with apples and add Balsamic vinegar to for a little tartness. This remains a ridiculously simple recipe, one that you can pull out whenever you are in a hurry but looking to impress; it packs in unexpected flavors at every step, and it can be on the table in about 20 minutes.

Serves 4

2 teaspoons kosher salt, plus more as needed (salt)

1 teaspoon ground coffee bean (bitter)

1 teaspoon ground coriander

½ teaspoon freshly ground black pepper, plus more as needed (spicy)

2 pounds pork tenderloin, trimmed and cut crosswise into 12 medallions (umami)

2 golden delicious apples (sweet)

2 tablespoons butter (fat)

4 tablespoons extra virgin olive oil, divided (fat)

2 fl oz bourbon

1 fl oz soy sauce (salt, umami)

8 fl oz chicken stock

2 tablespoons scallion, thinly sliced

Zest and juice of 1 lemon (sour)

2 tablespoons balsamic vinegar (sour)

Basil leaves for garnish

Combine the salt, coffee, coriander and pepper in a small bowl and mix well.

Arrange the sliced pork medallions on a plate or baking pan and sprinkle evenly on both sides with the spice mixture. Set aside to rest while you prepare the apples.

Peel one apple and cut into quarters. Cut away the core and cut each piece in half lengthwise to create 8 wedges. Cut the second apple in half, leaving the skin in, and remove the core. Cut both halves into slices and toss with the lemon juice and zest, 2 tablespoons of olive oil and the scallions. Mix well to coat and season with a pinch of kosher salt and ground black pepper

Heat a large cast-iron skillet over medium-high heat. Add 1 tablespoon each of the butter and olive oil. Pat the pork medallions dry with a paper towel.

Cook the pork medallions in batches, browning them well on both sides before removing to a rack to rest. Halfway through the cooking, discard the fat in the pan, wipe the pan out and add fresh butter and oil.

When all the pork has been cooked, add the apples and thyme to the pan and cook 2 minutes, stirring well.

Deglaze the pan with the bourbon and reduce until nearly dry. Return the pork to the pan along with the soy sauce and chicken stock. Simmer 2 minutes, then remove from the heat. Divide the pork and apples among plates. Top with the fresh apples and drizzle each plate with balsamic vinegar.

Slow Cooker Pork with Mustard, Orange and Sage

Ingredient Inspiration- mustard, orange, sage

Compound- 1-heptanol

Aroma- chemical, green, wood

The slower cooker is awesome tool to have in your kitchen, but things that come out of it can sometimes be a little 'one note' in the flavor department. Not this time... pork, citrus and sage is a phenomenal pairing. It creates deep and sophisticated flavors that are both comfortable and unexpected. Best of all, it can all happen while you are at work or running weekend errands.

Serves 4

1 tablespoon, plus 1 teaspoon kosher salt (salt)

2 tablespoons grainy mustard (spicy)

Zest of 1 orange

2 teaspoons dried sage

3 to 4 lb pork shoulder roast, bone out (umami)

1 yellow onion, thinly sliced

1 tablespoon olive oil (fat)

1 tablespoon soy sauce (salt, umami)

2 cups chicken stock

Combine the salt, mustard, orange zest and sage in a small bowl and mix well.

Pat the pork shoulder dry and rub on all sides with the mustard mixture. Making sure to coat the should completely and use the entire mixture. Place the shoulder in a baking dish or zip top bag and refrigerate overnight.

To cook, place the sliced onions, olive oil and soy sauce in a bowl and toss well to coat. Place the onions in the bottom of a slow cooker insert. Set the pork on top of the onions and pour in the chicken stock.

Cook for 8 hours on low or 6 hours on high. When finished, remove the pork to a cutting board to rest and transfer the onions and juices to a large saute pan. Bring the mixture to a boil and cook to reduce by half.

To serve, slice the pork shoulder and spoon the onions and reduced cooking liquid over the top.

Apple-Brined Pork with Caramelized Onion and Bourbon Gravy

Ingredient Inspiration- apple, pork, bourbon, onion

Compound- 2-acetylfuran

Aroma- balsamic, cocoa, coffee, smoke tobacco

*This recipes leans on the ‘fruitier’ side of the pork and apple pairing, while our Pan Seared Pork Medallions (p. xx) takes the same pairing in a more robust setting. Or as Jamie would say, in true food nerd fashion, ‘this is all about the esters.’ *Brooke rolls her eyes* Whatever, brining does magical things to pork chops, keeping them tender and juicy throughout the cooking process. Meanwhile bourbon creamed onions may very well become one of your favorite toppings for any type of roasted meat. Or potatoes. Ever.*

- 1 cup apple juice (sweet)
- 1 cup water
- 2 tablespoons kosher salt (salt)
- 1 tablespoon brown sugar (sweet)
- ½ lemon, thinly sliced (sour)
- 4 pork rib chops, bone in or pork tenderloin (umami)
- 1 tablespoon vegetable oil, plus more as needed (fat)
- 1 yellow onion, thinly sliced
- 2 ounces whiskey
- 1 cup heavy cream (fat)
- 2 tablespoons thinly sliced chives

Combine the apple juice, water, salt and sugar in a bowl and whisk until dissolved. Pour the liquid into a large zip top bag, add the pork chops and seal. Refrigerate overnight, up to 36 hours.

Before grilling the pork chops, place a large saute pan over high heat. Add just enough oil to coat the bottom of the pan. When the oil is hot and begins to shimmer, add the onions all at once and stir well. Allow the onions to settle and cook 1 minute without stirring, then mix again. Reduce the heat to medium-low and continue to cook, stirring every 2-3 minutes until they are tender and caramelized, about 30 minutes. *<--caramelizing onions is a process that takes some time to do properly. The key is maintaining proper heat throughout the cooking process. Start the onions on very high heat, then reduce the heat and cook slowly for the remainder. If you notice the onion slices are getting brown at the edges but remain white in the center, the heat is too high and needs to be reduced.*

When the onions are caramelized- tender through and evenly browned- remove the pan from the heat and add the whiskey. Carefully return the pan to the burner and turn the heat to high. Stir well to scrape up any brown bits from the base of the pan. When the whiskey is reduced by half, add the cream and cooked until reduced and thickened, 3-4 minutes. Remove from the heat and stir in the chives. Reserve warm and serve with the pork chops.

Set up a grill with one side on high heat and the other at medium. *□setting different temperatures on the grill allows you to cook more evenly, creating hotter areas for brown the exterior and cooler areas for evenly cooking though*

Remove the pork chops from the brine and pat dry. Use 1 tablespoon of the oil to coat the surface of the pork chops. Place the chops down on the hot side of the grill and cook 2 to 3 minutes. Lift each chop, rotate it 90° and place back on the grill on the same side. Cook an additional 2 to 3 minutes.

Flip the pork chops to the second side and place over medium heat. Cook 5 minutes, then rotate 90° and cook 5 minutes more. At this point, the pork should be cooked to medium/medium well, about 140°F degrees. Remove to a rack and rest.

Grilled Pork with Spicy Pineapple and Basil Relish

Ingredient Inspiration- pork, pineapple, basil

Compound- (Z)- beta- ocimene

Aroma- citrus, flower, herb, mold, warm spice

This recipe is designed to complement the smoky, rich flavor of grilled pork. Try it with your favorite cut- grilled tenderloin, chops, sausages or smoked pork shoulder. It works really great with chicken and fish too. Pineapple and basil are the star pairing here. All other ingredients play a supporting role.

Serves 4

4 ea Choice of pork cuts, grilled (umami)

½ jalapeno, seeds removed, thinly sliced (spicy)

2 tablespoons white wine vinegar (sour)

2 ears corn, shucked (sweet)

4 pineapple 'rings' (fresh pineapple, sliced ½-1 inch thick, core removed) (sweet)

3 tablespoons Vegetable oil (fat)

kosher salt (salt)

2 tablespoons mayonnaise (fat)

1 tablespoon soy sauce (umami, salt)

1 tablespoon tomato paste (umami)

2 tablespoon basil leaves, chopped

Place the sliced jalapeno in a medium bowl, pour over the vinegar and set aside.

Brush or lightly coat the corn and pineapple rings with oil and sprinkle with salt. Cook on a grill over high heat until lightly charred on all sides. Remove to a pan and cool.

When cool enough to handle, cut the kernels off of the cob and chop the pineapple rings into ½ inch pieces.

Add the mayonnaise into the jalapenos and mix well. When smooth, stir in the soy sauce, tomato paste, corn and pineapple. Mix well. Reserve until ready to serve. May be made up to 2 days in advance.

To serve, stir in the basil and spoon the mixture over the fully cooked protein.

Brown Butter, Maple and Pear Pork Roast

Inspiration Ingredients- pear, pork, maple

Compound- 2-pentanone

Aroma- burnt plastic, ether, fruit

Pork chops and apples are classic. One of the driving concepts behind The Flavor Matrix, was that when two ingredients have similar aromatic profiles that can—and should—be substituted for one another in recipes. Anywhere apples are used pears are an easy swap. In this recipe, since pears will tend to become soft with heavy roasting, brown butter and maple syrup are added used to boost roasted aromas without over cooking the fruit.

Serves 4

2 teaspoons kosher salt, plus more as needed (salt)

1 teaspoon ground coriander

2 pounds pork tenderloin or pork loin trimmed and cut crosswise into medallions (umami)

2 branches rosemary

2 bosc or d'anjou pears, slightly firm (sweet)

6 tablespoons butter, divided (fat)

2 tablespoons extra virgin olive oil, divided (fat)

Zest and juice of 1 lemon (sour)

Arrange the sliced pork medallions on a plate or baking pan and sprinkle evenly on both sides with salt and coriander. Set aside to rest while you prepare the pears.

Slice the pears in half through the core, then cut each half again lengthwise through the core to make 4 equal wedges. Trim away the core and cut each piece in half lengthwise again to create 8 wedges.

Heat a large cast-iron skillet over medium-high heat. Add 1 tablespoon each of the butter and olive oil along with one of the branches of rosemary. Pat the pork medallions dry with a paper towel.

Cook the pork medallions in batches, browning them well on both sides before removing to a rack to rest. Halfway through the cooking, discard the fat in the pan, wipe the pan out and add fresh butter, oil and rosemary.

When all the pork has been cooked, wipe out the pan and remove from the heat. Add the remaining 4 tablespoons of butter. Stir constantly as the butter melts. Return the pan to low heat and continue stirring until the butter stops foaming and takes an amber color. Remove the pan from the heat and continue stirring, watching the color carefully. As soon as the butter turns brown, add the pears and stir well. They may splatter a bit when first added. *□browning butter can a bit tricky- the key is to remove the pan from the heat before the butter is completely browned. Anticipate that the heat in the pan will cause the butter will turn one to two shades darker after being removed from the heat. Adding the pears will stop the browning immediately.*

When the pears stop sizzling, return the pan to low heat and cook until the pears are just softened. Stir in the maple syrup, lemon juice and zest. Stir well and return the pork to the pan to coat. Serve immediately.

Lamb

There are two main styles of lamb that you will find on the market- domestic and imported.

Domestic lamb, raised in the U.S. tends to be larger and have a more mild, sweet flavor.

Imported lamb, mostly raised in Australia and New Zealand will do not grow quite as big have a slightly stronger, though not 'gamey,' flavor. The main reason for this difference is the diet of the two animals. Most domestic lamb is grain fed, meaning it will grow more rapidly, however, grain fed animals develop less linoleic (fatty) acid than grass fed lamb. Linoleic acid is the precursor to (E,E)-2,4- decadienal, the compound most responsible for lamb's distinctive flavor. In short, grass fed lamb tastes 'lamb-ier'.

The process for cooking lamb is identical to that of cooking beef. The main difference being that lamb should not be prepared rare. Lamb is perfectly safe to eat at this temperature, it simply does not taste as good. At 'rare' temperatures (around 120°F) lamb fat will not be completely softened, meaning that it has not released it's full flavor and it tend to leave a slick, greasy mouthfeel. Otherwise, feel free to swap beef and lamb in any recipe you find in this book.

Braised Lamb with Blue Cheese and Mushroom Potato Gratin

Ingredient Inspiration- lamb, blue cheese, mushroom, potato, brandy

Compound- heptanoic acid

Aroma- apricot, floral, rancid, sour, sweet

Shepard's Pie was a dinner of our childhood- ground beef, mashed potatoes, a few veggies and cheese. It was simple and satisfying- every busy mother's dream. It wasn't until we traveled to Ireland, a 'babymoon' before our son August was born that we fell in love with authentic Shepard's pie with tender chunks of lamb that shredded at the touch of a fork. We've taken that memory and kicked it up with some seriously bold flavors from brandy and blue cheese.

For the lamb:

2 tablespoons vegetable oil (fat)
3 lbs lamb stew meat, cut into cubes (umami)
2 teaspoons kosher salt (salt)
½ teaspoon ground black pepper (spicy)
2 cups yellow onion, thinly sliced
4 garlic cloves, smashed
6 fl oz brandy
16 fl oz chicken stock
4 branches fresh thyme or ½ teaspoon dry thyme
1½ cup crumbled blue cheese (umami)

For the potatoes:

24 fl oz milk (fat)
4 russet potatoes, peeled and cut into thin slices (umami)
1 tablespoon minced garlic
1 tablespoon kosher salt
½ cup grated parmesan cheese (umami, salt)

Preheat oven to 325°F

Heat the oil in large, heavy bottomed sauté pan or dutch oven.

Season the lamb with salt and pepper. Add the lamb to hot pan and cook until well browned on all sides. When the meat is browned, remove to a bowl and set aside to rest. *□give the lamb adequate space to brown. Smaller pieces of meat will release more liquid; if you over crowd the pan, the meat will not brown properly. Cook in batches if necessary.*

Add the sliced onions and garlic to the pan the lamb was cooked in. Sauté over medium heat, until tender and lightly browned, 3-4 minutes. Return the lamb and any collected juices to the pan and turn the heat to high. Cook until all the juices have evaporated. Remove the pan from the heat and pour in the brandy. Stir well and carefully return the pan to the heat. Cook until reduced by half.

Add the chicken stock and thyme and bring to a boil. Taste the stock for seasoning (it should taste fully seasoned). Cover the pan and transfer to the oven and cook 1 hour.

While the lamb is in the oven, place the milk, potatoes, garlic and salt in a large sauce pot and bring to a boil. Reduce the heat to a simmer and cook 10 minutes. Remove from the heat and stir in the parmesan cheese. Transfer the potatoes to large oval or 13x9 baking dish and spread into an even layer. Set aside until the lamb is finished cooking.

After 1 hour, remove the lamb from the oven. Turn the oven to 450°F Uncover the pan and place on the stove top over medium heat. Simmer, stirring occasionally until the liquid in the pan is reduce to a glaze, only 4-6 ounces should remain.

Spoon the lamb, onions and sauce over the potatoes. Scatter the blue cheese over the lamb and transfer the baking pan to the oven. Bake until bubbling and browned, 10-12 minutes. Remove from the oven and cool at least 5 minutes before serving.

Braised Lamb with Spicy Peach Chutney

Ingredient Inspiration- corn, peaches, Grenache

Compound- (E)-2- nonenal

Aroma- cucumber, cut grass, fat, paper, watermelon

Most people pass off lamb as simply ‘gamey’, but fruity aromas (esters and lactones) are surprisingly strong in lamb. In this recipe we bring those flavors to the forefront. Braising is such a great way to cook lamb, the cuts for braising are typically less expensive and the meats transforms to tender and succulent. We love the bright flavors of Grenache and peach that make this dish comforting, yet bright and fresh tasting at the same time.

Serves 4

4 lamb shanks, trimmed (umami)

Kosher salt (salt)

Ground black pepper (spicy)

2 tablespoons vegetable oil (fat)

1 cup yellow onion, chopped

½ cup celery chopped

½ cup carrots

1 teaspoon fennel seeds

3 garlic cloves

1 ripe peach, pit removed and chopped (sweet)

16 fl oz Grenache wine (sour, butter)

16 fl oz chicken stock

2 tablespoon butter (fat)

For the peaches,

1 tablespoon yellow mustard seed

1 teaspoon cumin seed

1 fl oz water
2 fl oz red wine vinegar (sour)
½ teaspoon kosher salt (salt)
1 tablespoon sugar (sweet)
1 teaspoon finely grated fresh ginger
1 jalapeno, sliced (spicy)
2 firm/under ripe peaches, pit removed and diced (sweet)
4 fl oz Grenache wine (sour, bitter)

Season the lamb shanks with a generous amount of salt and pepper.

Heat the vegetable oil in a dutch oven. When hot, add the lamb shanks and sear to brown thoroughly on all sides. When all of the lamb shanks are well browned, remove to a pan and set aside. Drain the fat from the pan and return to the stove on medium heat. Add the onion, carrots and celery. Saute the vegetables until the onions are slightly softened, about 5 minutes. Add the fennel seeds, garlic and peaches and cook 1 minute more.

Deglaze the pan with the Grenache wine and return the lamb shanks. Bring the wine to a boil and cook for 1 minute before adding the chicken stock and basil. Cover the pot and transfer to the oven to cook for 2 hours.

While the lamb cooks, place a small sauce pot on the stove top over medium heat. Add the mustard seeds and cumin seeds and toast. Swirl the spices in the dry pot; when they begin to crackle and become fragrant, they are ready. *□be careful of over toasting (aka burning) the spices, they will become acrid and bitter.*

Add the water, vinegar, sugar, salt, ginger and jalapeno to the pot and bring to a boil.

Stir in the peach and cook at a simmer until the mixtures thickens and the peaches are tender, but not mushy. Adjust season in to taste, adding salt and sugar as necessary. Remove from the heat and cool slightly before stirring in the wine. Reserve until ready to serve the lamb.

After two hours, remove the lamb from the oven. Take the shanks out of the pot and set aside on a pan to rest. Pour the cooking liquid into a bowl, then strain the liquid back into the pot. Press lightly on the vegetables and peaches in the strainer to extract any remaining liquid, then discard what ever remains in the strainer.

Bring the liquid to a light simmer on the stove top and skim any fat that has collected on the surface. Simmer to reduce the liquid by half. When the braising liquid is reduced, remove from the heat, whisk in the butter and return the lamb shanks to the pot. Turn the meat to coat well with the sauce and allow them to sit at least 5 minutes before serving. If left to cool overnight in the sauce and reheated to serve the next day, they will improve even more.

Serve with peach chutney.

Roast Lamb and Carrots with Mint and Lemon

Ingredient Inspiration- carrot, mint, lemon, garlic

Compound- heptanoic acid

Aroma- apricot, floral, sour

We live for recipes like this- one simple seasoning that make two wildly different ingredients taste delicious. Thanks to the shared compounds between lamb, carrots and mint we can make prep and cooking a snap while producing a meal that has some serious 'wow' factor.

2 tablespoons mint, chopped, plus whole leaves reserved for garnish

3 tablespoons olive oil, divided (fat)

1 tablespoon kosher salt (salt)

Zest of 1 lemon, juice reserved (sour)

2 garlic clove, finely grated

1 teaspoon smoked paprika

1 lamb rack or 3-4 lb lamb leg roast, tied (umami)

½ lb carrots cut into pieces 1 inch wide and 2-3 inches long (sweet)

1 tablespoon vegetable oil (fat)

Preheat oven to 425°F.

In a small bowl, combine the mint, 2 tablespoons olive oil, salt, lemon zest, garlic and paprika. Mix well to form a smooth paste.

Place the lamb on a roasting pan and spoon 3 tablespoons of the spice mixture over the lamb rub to evenly coat on all sides. Set the lamb aside 30 minutes to come to room temperature. The spice mixture may be applied to the lamb up to 24 hours in advance.

Place the cut carrots in a bowl and add 2 tablespoons of the spice mixture. Toss to coat. Reserve any remaining spice mixture.

Heat the oil in a large saute pan. When hot, add the lamb and quickly sear to brown on all sides.

□though often repeated, searing meat before roasting does not lock in the juices. What it does is provide coloring to outside of the meat that might not otherwise for cooking times of 1 hour or less. It also allows more roasted aromas to develop in the meat as it cooks in the oven.

After searing, transfer the lamb to a pan with a rack set over it and rub any remaining spice mixture over the lamb. place in the oven to roast. For lamb rack, cook 20-25 minutes and for leg cook 45-50 minutes for medium. Check the temperature with an instant read thermometer before removing from the oven. The center of the roast should read between 125°F and 130°F before resting. Rest the lamb 10 minutes before slicing.

Spread the seasoned carrots into a single layer on a separate baking pan. Add them to oven while the lamb cooks. Roast until just tender, 15-20 minutes. Remove the carrots from the oven and tent with aluminum foil to keep warm while the lamb rests.

To serve, toss the carrots with the reserved lemon juice, remaining 1 tablespoon of olive oil and mint leaves. Slice the lamb and serve with the dressed carrots.

Horseradish and Citrus Crusted Lamb Roast

Ingredient Inspiration- lamb, orange, basil, horseradish

Compound- nonanal

Aroma- citrus, fat, floral, green, soap

Say it with us- basil = mint, mint = basil. Ok, so not literally but their compounds are so similar that they can be used interchangeably. Here are shaking up the tried and true mint and lamb combo in favor of basil. But things really get interesting when we pull in some of basil favorite matches- like orange and horseradish. It all comes together to create one of those 'OMG, this is so good, what's in it? But don't worry, it can be our little secret.

1 lamb rack or 3-4 lb lamb leg roast, tied (umami)

Kosher salt, as needed (salt)

1 tablespoon fresh grated or prepared horseradish, drained (spicy)

1 tablespoon fresh basil leaves, chopped

1 tablespoon extra virgin olive oil (fat)

2 teaspoon Dijon mustard (spicy)

1 teaspoon freshly grated orange zest

2 tablespoons vegetable oil (fat)

Pat the lamb dry and season with salt on all sides. Set aside for 10 minutes.

In a small bowl, combine the horseradish, basil, olive oil, mustard, and orange zest; mix well.

Dry any moisture that has collected on the surface of the lamb after salting. Heat the vegetable oil in a large sauté pan. When hot, add the lamb and quickly sear to brown on all sides. Remove to a rack to rest.

When cool enough to handle, rub the horseradish mixture on all sides of the meat to evenly coat. Return the roast to the rack and place in the oven. *□if the rub is placed on the meat before searing, the roast would not brown properly, and the rub would burn- adding afterward maximizes flavor.*

For lamb rack, cook 20-25 minutes and for leg cook 45-50 minutes for medium. Check the temperature with an instant read thermometer before removing from the oven. The center of the roast should read between 125°F and 130°F before resting. Rest the lamb 10 minutes before slicing.

Grilled Lamb with Anchovy Vinaigrette

Inspiration Ingredients- lamb, radish, anchovy, lettuce

Compound- pyrrolidine

Aroma- ammonia, fishy

This recipe is not for the timid. We are talking about some seriously big flavors here, taking the some of the stronger aromas in lamb and amping them up with even more intense anchovy. The beauty in this is the way it all comes together. You make a single mixture that doubles as marinade and salad dressing.

2 fl oz red wine vinegar (sour)

3 cloves garlic, finely grated

2 anchovy filets, minced *Or 1 teaspoon anchovy paste* (umami)

4 fl oz vegetable oil (fat)

2 fl oz extra virgin olive oil (fat)

Boneless lamb leg or lamb chops, trimmed (umami)

Freshly ground pepper (spicy)

2 heads romaine or 4 heads gem lettuce, chopped

½ cup grated parmesan cheese (umami)

1 cup sliced radishes (spicy, bitter)

Optional garnish:

Thinly sliced chives

Toasted breadcrumbs or croutons

Combine the red wine vinegar, garlic and anchovy in a bowl and mix well. Set aside to rest 5 minutes, then stir in the oils.

Place the lamb in a zip top bag and add ¼ cup of the dressing. Press as much air as possible out of the bag and seal. Gently massage to get the lamb evenly coated. Marinate one hour at room temperature, or up to 48 hours refrigerated.

When ready to cook, wipe any excess marinade off of the lamb and season the meat lightly with ground pepper.

Prepare a grill with one side on high heat, and the other medium. Place the lamb down on the hot side of the grill and cook 2 to 3 minutes. Lift each piece, rotate it 90° and place back on the grill on the same side. Cook an additional 2 to 3 minutes. Flip the meat and place on the medium side of the, continue to cook, rotating the meat every 3-4 minutes until a thermometer inserted into the center of each cut reads between 125°F and 130°F. □ (approximate) total cook time for:

lamb rib chops- 6 minutes

lamb loin chops- 8 minutes

boneless lamb leg pieces- 8-15 minutes (depending on size)

After cooking, remove the lamb to a rack to rest. Rest at least 5 minutes before slicing.

Place the chopped lettuce, radish and parmesan in large bowl. Add 3-4 tablespoons of the dressing and toss well. Season with freshly ground black pepper. Serve with sliced lamb.

Fish

Now we will take a look at fish in sweeping terms, because like beef, pork and lamb, all types of fish are really more alike than they are different. The specific species of fish that you cook in a given recipe really doesn't matter. The best fish that you can buy for any given recipe is the one that is most fresh. So let that be your guide and your inspiration as you cook, rather than trying to hunt a specific type and compromising on quality, simply because the recipe said so.

In this section we will discuss the best methods for cooking fish and break fish into two main categories: mild and full flavored. The distinction between those two groups has mostly to do with fat content. The higher in fat (with fish we are generally talking about high quality fats like omega-3) a fish the stronger it's flavor will be. These are also known as 'high activity' fish, meaning they swim faster and travel greater distances throughout their lifetime. They employ a music analogy, the flavor of lean and fatty fish is made up of the same notes, fatty fish just plays those notes at a higher volume.

No matter the type of fish you are cooking, you want to take care in the cooking process. When fish is fully cooked, it becomes flaky and will tend to fall apart on the grill or in the pan. That is why it is most important to account for 'carry over' cooking with fish. Carry over cooking means that a food will continue to cook, even after it is removed from the heat source. Generally carry over cooking will cause the internal temperature of a protein to rise about 5°F after it comes off the heat. So with fish, you must try to remove it before it is fully cooked so that carry over cooking can do the job of bringing the delicate fish to it's perfect doneness.

Whether grilling, searing or broiling, fish should be cooked over high heat. There are, of course, exceptions (i.e. poaching), but those are for another time. Cooking fish over high, direct heat helps to create a textured and flavorful exterior while leaving the interior, tender and moist.

Properly cooking fish is not quite as simple as just throwing a filet on a hot grill or saute pan. However, there are steps you can take to help ensure success. When preparing to sear a piece

of fish, the first decision to be made regards the skin – will it be left on or removed? Fish skin, if not cooked properly, can turn into an unappetizing, slimy mess. Skin on or off also determines which side of the filet will be cooked first. If the skin is left on, then the skin side must be cooked first.

However the fish is prepared, it is essential to begin with a very hot and well-oiled pan or grill. Fish cookery most often goes wrong because the fish sticks to either the pan or the grill. Fish sticks typically stick because moisture on the surface of the flesh. Making sure that the fish is dry and that your cooking surface is properly oiled and as hot as possible will help prevent sticking.

To begin cooking, season the skin side with salt and pepper. Dusting with flour is helpful in drying the skin and making it crisp, but too much flour will result in black, burned pieces so be sure to pat off any excess flour before cooking. Also, do not salt the fish too far in advance because the salt will draw moisture to the surface of the filet. If this happens, be sure to pat filets dry before cooking.

If cooking in a pan, carefully lay the fish in the pan letting it fall away from you so hot oil does not splash toward you. Immediately press the fish with the back of a flexible spatula, to insure that it is in full contact with the pan and the filet remains flat. Hold the fish flat with firm pressure for 10 seconds. You will feel the skin tightening and trying to curl as soon as it hits the pan; use just enough pressure on the spatula to force the skin flat, but not crush the fish. If searing skinless fish, lay the filet down with the skin side facing up; only a light press with the spatula will be required to insure even contact with the pan.

Cook over a high to medium-high flame for three to four minutes. When the edges of the fish begin to brown, turn the fish and cook approximately 2 minutes more (depending on thickness and desired doneness).

The fish should be slightly translucent in the center when you remove it from the pan.

Remember, the fish will continue to cook after it is removed from the heat. Allow the fish to rest for 2 minutes before serving.

Mild (Bass, Halibut, Snapper, Grouper, Flounder)

Pan Seared Fish with Broccoli, Balsamic and Garlic Croutons

Ingredient Inspiration- breadcrumbs, broccoli, balsamic

Compound- (E,E)-2,4- heptadienal

Aroma- fat fish, nut, plastic

This recipe is all about creating roasted aromas in different forms. First the pan seared fish, then roasting broccoli in large pieces allows you to get browning and texture on the outside, while staying tender with fresh aromas in the interior. Toasted breadcrumbs and balsamic add depth and texture that make this taste like a special occasion meal that easy enough for a busy weeknight supper.

4 mild white fish filets, seared or grilled (see page XX) (umami)

1 head broccoli, cut in ¼'s through the stem

2 tablespoons butter (fat)

1 tablespoon extra virgin olive oil, plus more as needed (fat)

2 cloves garlic, smashed

2 cups breadcrumbs **see note*

Basil leaves for garnish

Lemon for garnish (sour)

2 tablespoons good quality balsamic vinegar (sour)

**Tender, coarse breadcrumbs are ideal for this recipe. These can be made by cutting day old bread into cubes and pulsing in a food processor or by using a bread knife to cut fresh bread into ½ inch cubes.*

Preheat the oven to broil.

Bring a large pot of salted water to a boil. When at a rolling boil, add the broccoli florets and cook for 2 minutes. Remove from the water with a slotted spoon and set on a plate to drain and steam dry.

Arrange the pieces of broccoli on a baking pan with the cut sides facing up. Brush or spray the exposed face of the broccoli pieces with oil and season with salt. Place in the oven underneath the broiler and cook until lightly charred.

Saute or grill the fish according to instructions on p. xx.

When the fish is finished cooking, drain any excess oil from the pan and add the fresh butter, olive oil and garlic. Saute until the garlic becomes golden on the edges.

Add the add the coarse breadcrumbs and saute until golden brown and crisped. Remove from the pan with a slotted spoon and set on paper towels to drain.

To serve, place one piece of fish on plate with the a quarter of broccoli next to it. Grate a few scrapes of fresh lemon zest directly over the fish and broccoli, scatter a few basil leaves (chop them if they are large) and drizzle each plate with ½ tablespoon balsamic vinegar.

Pan Seared Fish with Sweet Pea and Ham Quinoa Pilaf

Ingredient Inspiration- peas, mushroom, ham

Compound- acetophenone

Aroma- almond, flower, meat, must, plastic

This recipe based of the strong pairing between fish and toasted nuts. Rather than adding nuts to the dish we let quinoa stand in with to toasted nuts are great pairing so the nuttiness of grains stands in perfectly.

4 ounces sliced smoked ham (or bacon) (umami)
3 tablespoons butter (fat)
2 cups cooked quinoa or brown rice (bitter)
8 ounces cremini mushrooms , finely chopped (umami)
1 tablespoon chopped garlic
16 oz frozen green peas
12 fl ounces chicken stock or water
1 tablespoon chopped mint
2 tablespoons olive oil (fat)
Kosher salt (salt)
Ground black pepper (spicy)
4 pieces sautéed or grill fish (see page xx) (umami)

Stack the ham slices up on a cutting board and cut them in half horizontally, then cut cross-wise into thin strips so that the pieces of ham are no more than 3 inches long and ¼ inch wide.

Melt the butter in a large sauté pan over high heat. When the butter begins to foam, add the ham and saute briefly until lightly browned at the edges, 1-2 minutes. *□at this point it may seem like you've got way too much butter in the pan, but you don't; so don't worry*

Add the chopped mushrooms and garlic. Saute, stirring often until no liquid remains in the pan and the mushrooms are lightly browned, 4-5 minutes more.

Reduce the heat to medium and add quinoa or rice. Stir well and continue cooking until the grains begin to sizzle or pop. Add the peas and chicken stock and bring to a simmer. Cook until the peas are warmed through and tender. Remove from the heat and stir in the mint, olive oil and vinegar. Season to taste with salt and pepper.

Spoon onto plates and top with cooked fish.

Pan seared Fish with Bacon and Creamy Tomato Farro

Ingredient Inspiration- tomato, faro, bacon cream

Compound- 2- heptanone

Aroma- bell pepper, blue cheese, green, nut, spice

This is another variation on putting fish together with big, toasted aromas. Grains, this time farro provide the nuttiness, while bacon, tomato and cream enhance the flavors of the grains and fish. Have fun with this recipe and try it out in different forms- you can omit the bacon and substitute sautéed mushrooms. The farro makes a fantastic side to just about any protein or can be a pretty satisfying meal all on it's own.

15 oz can diced tomato (umami)

16 fl ounces chicken stock

4 oz thick cut bacon, diced (umami)

2 garlic cloves, smashed

2 cups faro, rinsed (bitter)

2 teaspoons kosher salt

½ cup heavy cream (fat)

Freshly ground pepper (spicy)

1 teaspoon red wine vinegar (sour)

1 tablespoon chopped basil, plus extra leaves for garnish

4 pieces sautéed or grill fish (see page xx) (umami)

Optional: halved cherry tomatoes for garnish

Combine the tomatoes and chicken stock in a medium sauce pot and bring to a boil. Once the pot reached a boil, turn the heat and leave on the stove.

Place a large sauté pan over medium heat, add the bacon and 1 tablespoon of water. Cook until the water evaporates and the bacon is browned and crisp. *□adding water to the pan will actually make the bacon crispier- water steams the bacon pieces, allowing them to begin*

cooking, and rendering fat, before it begins browning, mean that more fat can be rendered from the bacon with burning it

Add the garlic and rinsed faro to the pan. Sauté 1 minute, then stir in the tomato mixture and salt. Cover the pan and reduce the heat to medium low. Simmer 20 minutes, or until all of the liquid is absorbed.

Remove the lid and stir in the heavy cream. Simmer in minute more and season to taste with salt and pepper. Keep warm while you cook the fish. Just before serving, reheat the faro mixture if necessary and stir in the vinegar and basil.

Spoon the faro into bowls and place the fish on top.

Pan Seared Fish with Lemon and Dill Lettuce Saute

Ingredient Inspiration- fish, lettuce, lemon, dill

Compound- 2,6- nonadienal

Aroma- cucumber, green

Sautéed lettuce may sound like a crazy idea, but it is a delicious and unexpected side that just happens to be an idea pairing for fish. Green aromas are a strong pairing for fish and sautéing lettuce really brings those green aromas forward.

1 lb new potatoes or fingerling potatoes (umami)

Kosher salt as needed (salt)

2 tablespoons olive oil, plus more as needed (fat)

3 garlic cloves, smashed

Pinch red pepper flakes (spicy)

1 teaspoon poppy seeds optional (bitter)

1 head romaine lettuce chopped

1 lemon, cut in half (sour)

1 tablespoon dill

4 pieces sautéed or grill fish (see page xx) (umami)

Cut the potatoes in half or into quarters, depending on their size (they should be bite sized).

Place the cut potatoes in a large sauce pot and add enough water to cover them by 1-2 inches.

Season the eater with a generous amount of salt. Place the pot on the stove over high heat and bring to a boil. When the water comes to a boil, cook 2 minutes, then drain the potatoes and spread them into a single layer on a sheet pan to steam dry. *□the potatoes will likely not be cooked through completely at this stage, they will cook a little more in the next step, but if they are on the larger side they may need to boil for up to 4 minutes. If the start to turn white at the edges, remove them immediately, as the potatoes are beginning to over cook.*

Heat the oil, garlic, red pepper flakes, and poppy seeds (if using) together in a large sauté pan over medium high heat. When the garlic begins to sizzle and lightly brown at the edges, add the warm potatoes and stir well. Allow the potatoes to settle in the pan and brown on one side.

Add the chopped romaine and cook 1 minute more, or until the lettuce is partially wilted.

Remove the pan from the heat and squeeze in fresh lemon juice to taste. Season with salt and stir in the dill.

Serve with cooked fish and drizzle each plate with olive oil.

Better-than-it-sounds Milk-Poached Fish with Basil

Inspiration Ingredients- fish, milk, onion, basil

Compound- 2-heptanone

Aroma- bell pepper, blue cheese, fruit, green, nut, spice

Fat = flavor, it's an age old saying; and it's true, but not quite in the way that you think it is. You see, fat brings flavor, but it's more of a transport vessel than it is a creator. Fat is the ultimate way to capture to capture aromas and deliver them from one food to another. In this recipe, milk takes all the lovely aromas from butter, lemon, herbs and alliums gives deposits them in your fish filets.

4 meaty white fish filets, skinless (grouper, halibut, tile fish, cod, sea bass) (umami)

2 teaspoons kosher salt, plus more as needed (salt)

2 ounces butter (fat)

1 tablespoon olive oil (fat)

2 shallots, thinly sliced

4 cloves garlic, thinly sliced

2 teaspoons all-purpose flour *(to thicken, use gluten free alternative)*

2 cups whole or 2% milk, plus more as needed (fat)

Zest of 1 lemon, removed with a vegetable peeler

4 branches fresh basil

½ teaspoon freshly ground black pepper, plus more as needed (spicy)

2 scrapes fresh grated nutmeg

Optional garnish: toasted almonds or breadcrumbs

Season the fish filets with salt. And set aside on a plate.

Melt the butter in a wide bottomed pan over medium-high heat. Cook until the butter is lightly browned and has a roasted, nutty aroma.

Pat the fish filets dry. Add the oil to the butter and place the fish in the pan, skin side facing up. When well browned on the first side, remove the filets from the pan and place back to the plate.

Add the shallots and garlic to the pan and sauté for 2 -3 minutes, until lightly browned. Add the flour the pan and stir until dissolved. Cook, stirring constantly, for 1 minute more. Gradually whisk the milk into the pan and bring the mixture to simmer. Add the lemon zest, basil, 2 teaspoons salt, pepper, and nutmeg.

Return the fish to the pan and bring the milk to a simmer, then reduce the heat slightly. For proper poaching you're looking to maintain the temperature between 160°F and 180°F. Allow the fish to poach 12-15, or until they are just cooked through. *□no need to pull out a thermometer here, if the milk is completely still and barely steaming the temp is a bit too low, if it is bubbling consistently all over the pan, the heat is a bit too high. Partially covering the pan can help keep the temperature more consistent*

Remove the fish from the pan rest on a rack. Serve with your choice of vegetables or the poaching liquid may be reduced to create a sauce if desired. Reduce until thickened, then remove the basil and take the pan off of the heat. Stir in 1 tablespoon of capers, lemon juice to taste, and more fresh basil. Garnish with chopped toasted nuts or breadcrumbs.

Miso and Maple Glazed Fish

Inspiration Ingredients- maple syrup, miso, vinegar

Compound- phenol

Aroma- medicinal, phenol, sharp, smoke, spice

This is an amped up version of a recipe that built an empire. Nobu's miso glazed is one of the dishes that propelled that restaurant to icon status. Big bold flavors and layered Maillard aromas that come from miso, maple syrup and smoked paprika make this a real crowd pleaser. The glaze mixture can be made in bigger batches and held for up to 10 days in the refrigerator, so this recipe could easily wind up in heavy rotation at your house.

2 tablespoons white miso (umami)

1 tablespoon, plus 1 teaspoon rice wine vinegar (sour)

1 tablespoon maple syrup (sweet)

Zest of 1 lime, finely grated

1 teaspoon smoked paprika

Pinch cayenne (spicy)

4 fish filets (salmon, cod, black cod, sable) (umami)

Combine the miso, rice wine vinegar, maple syrup, lime zest, smoked paprika, and cayenne in a large zip top bag. Mix thoroughly.

Add the filets to bag and gently turn to coat completely.

Place the bag in the refrigerator to marinate 2 hours, up to overnight. Remove the fish from the bag and wipe off any excess marinade.

Preheat the oven to broil.

Set a rack over a sheet pan and spray the grate of the rack with pan release. Arrange the fish on the rack and place in the oven on low oven rack underneath the broiler.

Cook, until the fish is well browned on top and cooked through, about 12 minutes.

Serve with Coco-slaw (p. xx) or A Different Green Salad (p. xx).

Pan Seared Fish with Brown Butter and Grape Vinaigrette

Inspiration Ingredients- brown butter, fennel, grapes

Compound- 2,3-butanediol

Aroma- fruit, herb, onion

Once again we are digging into the unexpected to pull out some delicious flavor pairings. Grapes and brown butter are not your standard accompaniment to fish, but their flavors are an incredible match. Browning the butter in a separate allows it to add a deep, rich roasted flavor that is the perfect bridge between the fish and the grapes.

1 tablespoon kosher salt (salt)

1 teaspoon fennel seeds

½ teaspoon ground black pepper (spicy)

1 tablespoon vegetable oil (fat)

4 fish filets (tuna, swordfish or salmon) (umami)

1 shallot, thinly sliced (or 2 tablespoons sliced scallions)

3-4 branches fresh thyme

1 stick unsalted butter (fat)

1 cup seedless grapes, cut in half (sweet)

¼ cup white wine vinegar (sour)

Heat the oil in a large sauté pan over high heat.

In a small bowl, mix together the salt, fennel seeds, and pepper. Season the fish well on both sides with the mixture.

Carefully add the fish to the pan and reduce the heat to medium-high. When well browned on one side, flip and brown on the second side. □*For tuna, cook 1-2 minutes per side for*

rare/medium rare; salmon cook 3-4 minutes per side for medium; swordfish should be cooked medium well- cook 4-5 minutes per side

While the fish is cooking, melt the butter in a small sauce over medium heat. The butter will melt, then boil. Continue cooking until the butter takes a light amber color, then remove from the heat and reserve. □ *remember that butter will turn 1-2 shades darker after removing from the heat.*

When the fish is finished cooking, remove to a rack to rest. Drain any excess oil from the pan, leaving behind approximately 1 teaspoon of oil to sauté the shallots. When the shallots are tender, add the thyme and grapes and continue cooking until the grapes begin to soften, 2-3 minutes more. Pour in the vinegar and remove the pan from the heat. Stir the butter into the grapes and mix well.

Discard the thyme from the pan and spoon the grapes and butter over the cooked on plates.

Fennel Roasted Fish with Chilled Green Beans, Sesame, and Dill

Ingredient Inspiration- sesame, dill, green bean

Compound- (Z)-3- hexen-1-ol

Aroma- bell pepper, grass, green, herb, unripe banana

Dill and fennel work very well together. But this dish is all about getting those aromas to work together in different forms—the toasted aromas from the fennel seeds and the freshness from the dill are then layered with more fresh and green aromas in the green beans and more toasted ones with the sesame seeds.

Serves 4

½ cup mayonnaise

1 tablespoon soy sauce (salt, umami)

2 teaspoon tomato paste (umami)

1 teaspoon sesame oil (optional) (fat)

1 teaspoon sriracha, or more to taste (spicy)

1 lb green beans, trimmed

4 (5 oz) portions firm white fish such as halibut, grouper, or bass

kosher salt, as needed (salt)

½ teaspoon ground fennel seed

¼ cup olive oil, divided (fat)

2 tablespoons toasted sesame seeds

1 tablespoon fresh dill, chopped

juice of 1 lemon (sour)

jarred crispy shallots, as needed

Bring a large pot of salted water to a boil.

In a small bowl, whisk together the mayonnaise, soy sauce, tomato paste, sesame oil, and sriracha. Combine and season to taste.

Add the green beans and cook until just tender, about 2.5 minutes. Remove the green beans and plunge into a bowl of ice water to chill quickly. When cooled, drain and set aside.

Season the fish with salt on both sides and season the fish on the bone side with the ground fennel seed. Set aside with the oil heats and pat the fish dry just before adding to the pan.

Heat 2 tablespoons of olive oil in a large sauté pan over medium high heat. When the pan is hot and the oil begins to shimmer, add the fish to the pan, skin side up. Gently shake the pan and leave the fish to sear undisturbed for 2 minutes.

Reduce the heat to medium and scatter the sliced fennel over the fish. Cook 2 minutes more, then turn the fish and check doneness. Cook 2-4 minutes until the fish is cooked through. Remove the fish to plates and keep warm until ready to serve.

In a bowl, combine the green beans, soy sauce, dill, and sesame seeds. Toss well to combine and add the remaining olive oil and lemon juice. Mix again and adjust the seasoning if needed.

To serve, place a spoonful of the mayonnaise sauce on the plate. Top with the fish and arrange the green beans next to the fish. Garnish the green beans and fish with the crispy shallots.

Pan Seared Fish with Sweet Potato, Mushroom and Poblano Hash

Inspiration Ingredients- sweet potato, mushroom, basil, honey, mustard

Compound-

Aroma-

This is probably the biggest and boldest of fish recipe you will find in this book. It is somehow comfort food plus light and fresh at the same time. Jamie says this dish reminds him of Brooke- a little bit spicy and a little bit sweet. We think you'll fall in love the sweet potato hash, which is an incredible side dish for just about anything- matched with fish as in this recipe or find it a spot on your holiday table matched with turkey or roast beef.

2 large sweet potatoes, cut into 1 inch cubes (4 to 5 cups) (sweet)

8 oz cremini mushrooms cut into quarters (umami)

4 cloves garlic, smashed

4 branches fresh thyme

1 teaspoon salt (salt)

4 (5 oz) portions firm white fish such as halibut, grouper, or bass

3 tablespoons olive oil, divided plus more for garnish (fat)

1 poblano pepper, diced (spicy)

1 tablespoon minced garlic

2 tablespoons cider vinegar (sour)

1 tablespoon honey (sweet)

2 teaspoons Dijon mustard (spicy)

1 tablespoon chopped basil leaves, plus more small leaves for garnish

Optional ¼ cup toasted pumpkin or sunflower seeds

Preheat oven to 400° F

Combine the chopped sweet potatoes, mushrooms, garlic, thyme salt and 2 tablespoons olive oil in a bowl. Toss well to season and spread the squash into a single layer on a baking pan. Place on the center rack of the oven and cook until tender and lightly browned, about 12 minutes.

Pan sear the fish and set aside, covered with aluminum foil to stay warm.

Heat the remaining 1 tablespoon olive oil in a large sauté over medium heat. Add the poblano and garlic to the pan and cook, stirring often until the peppers are tender. *□you can (and should) do this in the same pan you use to cook the fish.*

While the pepper cooks, mix together the vinegar, honey, mustard and basil.

Add the roasted sweet potatoes and mushrooms to the sauté pan and toss well to coat. Adjust the seasoning with salt and pepper.

Remove the pan from the heat and toss in the honey mustard mixture. Mix well. Divide the sweet potatoes among plates and top with the cooked fish. Or pass separately. Garnish with a drizzle of fresh olive oil, basil leaves and optional toasted seeds

Pan Roast Fish with Spiced Potatoes and Avocado Cream

Ingredient Inspiration- potato, tomato, avocado

Compound- diacetyl

Aroma- butter, caramel, fruit, pastry

This recipe is really all about the all-too-interesting-to-ignore pairing of avocado and potatoes. With one pan you'll roast potatoes and cherry tomatoes together. Then make a thick, creamy dressing from avocado to serve alongside the potatoes and fish. It can make an elegant presentation of a plate or wrap that whole thing up in a tortilla for some killer fish tacos!

1 lb new potatoes or fingerling potatoes (umami)
2 teaspoons cumin seeds
2 teaspoon garlic powder
1 teaspoon ground turmeric (bitter)
¼ cup extra virgin olive oil (fat)
1 cup cherry tomatoes, cut in half (umami)
2 teaspoons kosher salt (salt)
¼ red onion, sliced as thinly as possible
¼ cup red wine vinegar (sour)
½ teaspoon sugar (sweet)
1 avocado, cut into cubes (fat)
½ cup sour cream (sour)
1 tablespoon minced cilantro
1 teaspoon sriracha or other hot sauce (spicy)
½ teaspoon ground cumin
Juice of 1 lime (sour)
1 cup baby arugula leaves (bitter)
4 portions of grilled or pan seared fish (grouper, halibut, bass)
Optional garnish: sliced radishes, cilantro leaves

Preheat oven to 425°F

Cut the potatoes in half or into quarters, depending on their size (they should be bite sized).

Place the potatoes in a bowl and add the cumin seeds, garlic powder, turmeric and olive oil.

Toss well to coat.

Spread the potatoes out in a single layer on a baking sheet. They should not touch one another, if there is not enough room, divide among two pans. Make sure that each of the potatoes are arranged so that the flat, cut side of the potato is facing down on the pan. Sprinkle the salt evenly over the potatoes and immediately place them in the oven. *□Salt draws water out of food. Adding the salt at the last moment will help insure that the potatoes get crisp and brown in the oven by making sure that the roasting pan does not fill up with water before roasting*

Let the potatoes cook 8 minutes, then add the cut cherry tomatoes to the pan and return to the oven for 8 minutes more, or until the potatoes are tender. When they are finished roasting, use a spatula to stir the vegetables on the pan, then set aside to cool slightly.

While the potatoes cook, combine the vinegar, sugar and a pinch of salt in a small bowl; stir until dissolved. Add the onions, mix well and set aside until ready to serve.

Combine the avocado, sour cream, cilantro, hot sauce, cumin, and lime juice and mash with a fork or whisk until smooth, or blend in a food processor.

Transfer the roasted potatoes and tomatoes to a large bowl. Drain the onions from the vinegar mixture and add the onions to the potatoes and toss well to coat. Divide the mixture among plates, and garnish with a large dollop of the avocado dressing and arugula leaves. Serve with cooked fish.

Shrimp, Crab and Lobster

The flavor of crustaceans comes mainly from nitrogen containing compounds known as amines and thiazines. Amines are derived from ammonia and thiazines also contain an sulphur atom. These compounds are responsible for the 'fishy' and ammonia odors associated with seafood that is past it's prime. Fresh fish and shellfish should have little to no smell or smell like the ocean. Often if they have been stored in a closed plastic the odor can become more pronounced even if it had not gone bad. To check, pat the items dry and give them another sniff in a clean air space. The strong smell remains, it is probably best not to eat them.

There is one simple rule for cooking crustaceans: less is more. All crustaceans- shrimp, lobster, crab, crawfish, langoustine- have a little fat or connective tissue, meaning that they can go from perfectly to tough and rubbery in a matter of moments. The quicker they can be cooked, the better they will be. This also means that if being served hot, they should not be cooked until moments before being served. Keeping any crustaceans hot for an extended period of time will cause them to become dry and tough.

When boiling crustaceans, it is best to cook them with their shells on, then peel and devein after cooking. It's not the easiest way, but when cooking shrimp without their shells, the water can dry out the meat of the shrimp leaving them with a mealy texture. When using dry heat cooking methods like sauté or grill, it is fine to cook them with the shells already removed. Though with all dry heat methods make sure they are well coated with fat (oil or butter) to protect the flesh from the direct heat which can toughen the exterior.

Poached Shrimp Salad with Chilled Carrot Soup

Inspiration Ingredients- carrot, ginger, red wine

Compound- terpinolene

Aroma- pine, plastic, sweet

Carrots ginger and lime are an outstanding combination all on their own. In this recipe, they make an awesome soup that is delicious served hot or cold. The additional of fresh poached shrimp takes this pairing to the next level for both flavor and presentation. The sharp, 'pine-y' aromas in carrots, lime and ginger are all a great match for shrimp flavor-wise. The soup and the shrimp can both be prepared a day in advance .

Lime Dressing:

2 tablespoons extra virgin olive oil (fat)

2 teaspoons lime juice (sour)

1 teaspoon cilantro, chopped

Pinch salt (salt)

Freshly ground pepper (spicy)

2 cups chicken stock

20 fresh shrimp (21-25 count), shell on (umami)

2 tablespoons olive oil (fat)

4 cloves garlic, chopped

1 teaspoon ground coriander

1 cup onion, minced

2 inch piece of fresh ginger, peeled and thinly sliced

4 cups carrots, thinly sliced (sweet)

2 teaspoons kosher salt (salt)

1 cup heavy cream (fat)

2 limes, juiced (sour)

1 avocado, sliced (fat)

Prepare the lime dressing, in a medium bowl, combine the olive oil, lime juice, cilantro, salt and pepper. Mix well and set aside.

Rinse the shrimp, with their shells on, under cold water. Bring the chicken stock to boil in a small saucepot, add the shrimp and stir once. Put a tight fitting lid on the pot, then turn the heat off and let the pot stand covered for 5 minutes.

Remove the shrimp from the pot with a slotted spoon and place on a large plate or baking pan to cool- do not submerge in water. Remove the shrimp with a slotted Reserve the chicken stock.

When the shrimp are cool enough to handle, peel and de-vein the shrimp and toss with the lime dressing.

Heat the olive oil, garlic and coriander in a wide sauce pot over medium-high heat. When the spices begin to sizzle and become aromatic, add the onions and sauté until translucent. Reduce the heat as needed to prevent the onions from browning.

Add the ginger, carrots, and salt to the pot and stir well. Continue cooking for 5 minutes, stirring often. If the pot begins to brown reduce the heat or a tablespoon of water at a time to prevent browning.

When the carrots are slightly softened, add the red wine and cook until nearly dry. Then add the reserved chicken and simmer with carrots until reduced by half, about 6 minutes more. *□The carrots are going to be pureed so they need to be completely soft. To check doneness, remove one piece of carrot with a spoon and try smash with the back of the spoon on a cutting board. If it does not crush easily, continue cooking until it does.*

When the carrots are tender, transfer the mixture to a blender and puree until smooth. Return the puree to the pot and stir in the cream. Bring the soup to a boil and stir in the lime juice, to taste. Adjust seasoning with salt and pepper. Transfer to a large bowl to chill.

Divide the soup between four café au lait sized bowls garnish with 5 shrimp each, a couple slices of avocado and a drizzle of good quality olive oil.

Creamy Shrimp Pasta with Charred Tomato and Pumpkin

Inspiration Ingredients- shrimp, pumpkin, cream, bread, pasta

Compound- 2,3- pentanedione

Aroma- bitter, butter, caramel, cream, fruit, sweet

Shrimp and pumpkin are one of the more intriguing pairings we found in the Flavor Matrix. We just couldn't resist turning it into delicious fall pasta... hearty, yet light.

- 1 beef steak tomato (umami)
- 2 cloves garlic
- 4-6 fresh basil leaves
- 2 cups chicken stock
- 3 tablespoons butter, divided (fat)
- 1 cup yellow onion, finely chopped
- 2 cups diced pumpkin or butternut squash
- 1 teaspoon dried sage
- 8 oz dried mezze rigatoni, or rigatoni
- 2 teaspoons kosher salt (salt)
- 2 cups water
- 1 lb shrimp, peeled and deveined (umami)
- ½ cup toasted breadcrumbs

Cut the tomato in half and char the cut side on a hot grill, underneath a broiler or in a lightly oiled cast iron pan. □*The goal here is to burn the tomato, so don't hold back. You want to cook the tomato until the cut sides are completely black.*

Transfer the tomato to a medium sauce pot, add the garlic, basil and chicken stock. Bring the pot to a boil, then cover and remove from the heat. Set aside for 15 minutes to infuse. After infusing, strain the stock into a bowl and press the tomatoes with the back of spoon to crush

them and extract as much liquid as possible. Reserve the liquid and discard the basil, garlic and remaining tomato solids.

While the broth is infusing, prepare the risotto. Melt the 2 tablespoons butter in a wide bottomed sauce pot. Add the onions and sauté until translucent and tender, without browning. *□Adding a pinch of salt to the onions as they cook will draw a little water out of the onions and help them to soften with browning.*

When the onions are translucent, add the remaining tablespoon of butter, the pumpkin and sage. Sauté 2 minutes. Deglaze the pan with the white wine and cook until nearly dry.

Stir in the dried pasta and mix well. Add the reserved chicken stock, 2 cups of water and 2 teaspoons salt. Bring the pot to a simmer then cover with a lid. Cook, stirring occasionally, until the pasta is nearly tender, about 8 minutes.

Remove the lid from the pot and turn the heat to high. Add the shrimp and cream, stir constantly until the shrimp are cooked though, 3-4 minutes. *□at this point the sauce should be thick and creamy, if it is not, continue boiling until the sauce thickens.*

Divide among bowls and garnish with toasted breadcrumbs.

Brazilian Shrimp and Coconut ‘Stew’

Ingredient Inspiration- Shrimp, tomato, coconut, pepper

Compound- hexanal

Aroma- fresh, fruit, grass, green, oil

Tomato + coconut + shrimp are a stunningly delicious combination. Together they can make a simple meal feel like an escape to the tropics. This recipe is made in the style of ‘Moqueca’ Bahia’s famed seafood stew, which we learned to cook while visiting the idyllic UXUA Hotel and Casas in Trancoso, Brazil.

8 oz shrimp, peeled and deveined (umami)

1 teaspoon kosher salt (salt)

2 tablespoons vegetable oil (fat)

1 teaspoon paprika

1 garlic cloves, smashed

1 lemon (sour)

15 oz can coconut milk (fat)

½ cup diced red bell pepper (sweet)

½ cup diced onion

½ jalapeno, diced (spicy)

½ cup diced tomato (umami)

1 tablespoon chopped parsley

1 tablespoon fresh coriander

Optional garnish:

2 cups cooked rice or toasted/grilled bread

¼ cup toasted quinoa or breadcrumbs

Season the shrimp and set aside.

Combine the oil, paprika, and garlic in a large saute pan over medium heat. When the garlic sizzles and begins to brown at the edges, pat the shrimp dry and add to the pan. Cook on one side for 30 seconds then flip and cook 30 seconds longer on the second side. Remove the shrimp to a clean plate next to the stove. *□The shrimp will not be fully cooked at this point, they will finish cooking later.*

Add the bell peppers, onion and jalapeno to the hot pan and sauté 2-3 minutes, until tender. Stir in the tomatoes and mix well.

Add the coconut milk and bring the mixture to a simmer. Cook the coconut milk at a low simmer for 2-3 minutes.

Return the shrimp to the pan along with the parsley and cilantro. Simmer until the shrimp are just cooked through, about 4 minutes. Remove from the heat and season to taste with lemon juice, salt and pepper.

Serve over cooked rice or toasted/grilled bread and garnish with toasted quinoa or breadcrumbs.

Lump Crab Bruschetta

Ingredient Inspiration- tomato, sesame, dill, toast

Compound- 2- ethyl-1- hexanol

Aroma- citrus, green, oil, rose

Pan con tomat, is one of the essential foods of Spain- crisp slices of bread, still tender on the side and saturated with garlic, tomato and olive oil. We really didn't it could get any better. Then, it did... looking at some ways to have some fun with the strong pairing between crab and tomato we decided to top pan con tomat with a creamy, herby crab salad. It is summer-eating perfection.

Grated zest and juice of 1/2 lemon (sour)

2 teaspoons chopped fresh dill

2 tablespoons mayonnaise (fat)

Kosher salt, as needed (salt)

Freshly ground black pepper (spicy)

1 pound lump crabmeat, picked over for shells and cartilage (umami)

4 large slices country-style sourdough bread, or semolina bread

2 tablespoons extra virgin olive oil (fat)

1 garlic cloves, whole

1 ripe beefsteak tomato, cut in half (umami)

2 tablespoons toasted sesame seeds

Combine the lemon zest and juice, dill and mayonnaise in a large bowl and mix well. Add the crabmeat and gently fold to combine. Season to taste with salt and pepper. Cover and refrigerate until ready to serve, up to 1 day in advance.

When ready to serve, brush the bread with a generous amount of olive oil and both side. Toast on a grill or under a broiler until charred at the edges. Remove the bread to a rack and rub one side of the bread with the garlic clove. Then, rub each slice of bread with a cut the tomato, gently crushing the tomato in your hand as you scrape it across the bread. *□Imagine the crisp bread is like a cheese grater and you are trying to grate the tomato. The goal is to break down the tomato and get it to soak into the bread.*

Top the tomato soaked bread with the dressed crab. Sprinkle each toast with toasted sesame seeds. Serve immediately.

Crab and Crisp Apple Salad

Ingredient Inspiration apple, mayonnaise, chive, lime

Compound- caprylaldehyde

Aroma- citrus, fat, floral, green, pungent

*The tart crunch of the apple is essential in this dish. The crab can be dressed and seasoned up to a day in advance, but do not add the apples until immediately before serving. It can be served as an hors d'oeuvre or an appetizer. To serve as an hors d'oeuvre, finely dice the apple, mix it into the crab and place a spoonful of the mixture on top of toasted crostini. To serve as an appetizer, gently mound ¼ cup of the crab mixture in the center of a plate and tuck 4-5 thin slices of apple into each mound of crab. **Illustration of both serving options***

Grated zest and juice of 1/2 lime (sour)

2 teaspoons chopped fresh chives

¼ teaspoon celery seed (bitter)

2 tablespoons mayonnaise (fat)

Kosher salt, as needed (salt)

Freshly ground black pepper (spicy)

1 pound lump crabmeat, picked over for shells and cartilage (umami)

1 Granny Smith apple (sour, sweet)

Combine the lime zest and juice, chives, celery seed and mayonnaise in a large bowl and mix well. Add the crabmeat and gently fold to combine. Season to taste with salt and pepper. Cover and refrigerate until ready to serve, up to 1 day in advance.

Grilled Lobster and Butternut Squash with Tomato, Olive and Preserved Lemon Relish

Ingredient Inspiration- lobster, butternut squash, olive, tomato

Compound- 2,5- dimethylpyrazine

Aroma- burnt plastic, cocoa, medicinal, roast beef, toasted nut

Maillard (deep roasted) aromas are the common link between lobster, butternut squash, olives and tomato. This recipe amplifies those flavors by seasoning them both lobster and squash with smoked paprika and sage, then cooking both on the grill. The result is smoky, delicious and completely unexpected.

- 1 tablespoon salt (salt)
- 1 teaspoon smoked paprika
- 1 teaspoon dried sage
- 4 lobster tails, shell on, split lengthwise (umami)
- 1 medium butternut squash, peeled and sliced into rounds ½ inch thick
- ½ cup cherry tomato, cut in half (umami)
- ¼ cup Castelvetrano or picholine olives, pitted (salt, umami)
- ¼ cup extra virgin olive oil, plus more as needed (fat)
- 2 tablespoons red wine vinegar (sour)
- 2 teaspoons preserved lemon, minced *Or zest of 1 lemon* (sour)
- 1 cup baby arugula (bitter)
- Optional garnish: toasted pinenuts, pickled onions (p. xx)

Prepare a grill on high heat

Combine the salt, smoked paprika and sage in a small bowl and mix well. Brush the lobster tails and butternut squash with olive oil and season with the salt mixture.

Place the lobster tails, cut side down on the grill. Cook 4 minutes, then turn and cook 5-6 minutes more, until the meat is no longer translucent in the center. Remove and place on a rack to cool slightly.

Grill the butternut squash at the same time as the lobster tails, turning every two minutes until they are easily pierced through the center with the tip of a knife, 6-8 minutes total.

Arrange the grilled lobster and squash together on a platter.

Combine the tomatoes, olives, olive oil, vinegar and lemon in a bowl and mix well. Spoon the mixture over the platter of lobster and squash. Garnish with baby arugula, pinenuts and pickled onions.

Drinks

Crafting drinks had long been a hobby of ours, but after having two children it became a necessity. There is nothing that we enjoy more than sitting down with a simple cocktail at the end of the day.

Drinks, more so than any other food item, rely on aromas to convey flavor. The Flavor Matrix showed how aromatic compounds (those things that create flavor) are part of a larger group called volatile compounds. Volatile compounds are just that... volatile; meaning that they readily dissipate into the air. Aromas from herbs, spices and fresh produce can only be held onto for so long before their flavor vanishes into thin air. This is why most of our favorite sips rely on a good deal for fresh fruits, vegetables and herbs. These recipes are mere suggestions, you should feel encouraged to substitute fruits, vegetables and herbs based on the season or whatever you may have on hand.

Grapefruit Cucumber Collins

Ingredient Inspiration: grapefruit, gin, aperol

Compound- myrcene

Aroma- geranium, herbaceous

We don't mind digging into a little chemistry when it produces crisp, herbaceous, and refreshing cocktails like this. Myrcene is a compound that is present in all citrus and has the aroma of geranium and herb. However, myrcene is more prominent in grapefruit, making it the perfect match for the herbaceous and floral aromas in gin, Aperol, and cucumber.

**Myrcene also contributes to the pairing of grapefruit and sage, making this drink the perfect pairing for slow cooker pork recipe on page XX.*

Makes 1 cocktail

1 oz Gin

1 oz Aperol

3-4 leaves fresh rosemary

½ slice pink grapefruit

2 oz Pink grapefruit juice

Splash soda water

Garnish: sprig rosemary, ¼ slice grapefruit

Combine the gin, aperol, rosemary and sliced grapefruit in a cocktail shaker. Muddle thoroughly. Fill the shaker with ice and add the grapefruit juice. Shake vigorously to chill. Strain into a wine glass filled halfway with ice. Garnish with rosemary and grapefruit, then top with soda water.

These next two recipes were designed specifically around the aromas naturally found in each wine. They take advantage of the fact that vodka is an excellent solvent, it can strip aromas from ingredients, hold on to them and help carry them to other places. As the raw aromatic ingredients sit in the vodka the alcohol will help pull both and aroma from the ingredients. You can make these infusions in larger batches, but do not let the fruits sit in the vodka for more than a week or they will begin to break down. After 5 to 7 days maximum, strain the fruit and aromatics from the vodka and keep the vodka in an airtight jar or bottle in the refrigerator or freezer. The strained fruits can also be frozen to reserve for garnish.

Cucumber, Apple, Sauvignon Blanc Sangria

Inspiration Ingredients- cucumber, sauvignon blanc, grape, apple

Compound- linalool oxide D

Aroma- citrus, floral, fragrant, green

16 fl oz vodka

1 kirby cucumber thinly sliced

1 cup green grapes, cut in half

1 granny smith apple, core removed and diced

Sauvignon blanc, as needed

Club soda, as needed

Combine the vodka, cucumber, grapes and apple in a glass pitcher or jar and refrigerate overnight, up to 5 days.

To serve, mix the vodka with the wine in a 1:3 ratio. For a single glass: 1 oz infused vodka, 3 oz wine. For a pitcher mix 1 cup of the infused vodka with 1 bottle of white wine.

Garnish drinks with some of the infused fruit and top each glass with a splash of soda water.

Peach, Basil, Rosé Sangria

Inspiration Ingredients- basil, peach, lime, rosé

Compound- ethyl benzoate

Aroma- camomile, celery, fat, flower, fruit

16 fl oz vodka

¼ cup basil leaves, loosely packed

1 peach, pitted and thinly sliced

1 lime, zested

provençal rosé

club soda, as needed

Combine the vodka, basil, peach and lime zest in a glass pitcher or jar and refrigerate overnight, up to 5 days.

To serve, mix the vodka with the wine in a 1:3 ratio. For a single glass: 1 oz infused vodka, 3 oz wine. For a pitcher mix 1 cup of the infused vodka with 1 bottle of rosé.

Garnish drinks with some of the infused fruit and top each glass with a splash of soda water.

Cocoa and Lemon Old Fashioned

Ingredient Inspiration- cocoa, whiskey, lemon

Compound- alpha- phellandrene

Aroma- citrus, mint, black pepper, turpentine, wood

This drink is all about bring in citrus aromas in two forms, the aromatic peel of the orange zest pair perfectly with the citrus notes in cocoa. Then, lemon brings brightness and acidity to the cocktail.

Cocoa syrup

½ cup sugar

2 tablespoons cocoa powder

1 cup sugar

1 orange

cocktail

1 teaspoon cocoa syrup

1½ ounces whiskey

1 wedge lemon

1 oz club soda

Lemon twist

Combine the sugar and cocoa powder in a saucepot. Whisk thoroughly to combine. Add the water while still stirring with a whisk. Place the pot on the stove and bring to a boil. Stir gently to insure all the sugar and cocoa have dissolved. Remove the pot from the heat. Remove the zest from ½ of the orange with a vegetable peeler. Add the strips of zest to the hot syrup and set aside to cool to room temperature. When cooled, strain and reserve in an air tight container in the refrigerator up to 3 months. □ *cocoa powder has a tendency to form clumps when added to liquid. Whisking it together with the dry sugar before adding the water will help prevent this.*

To make a cocktail, fill a rocks glass halfway with ice. Add the cocoa syrup and whisky. Squeeze in the juice from the lemon wedge (about 1 teaspoon according to taste) and stir in the glass to thoroughly chill. Top with club soda and garnish with a lemon twist.

Spiced Butternut Squash Old Fashioned

Inspiration Ingredients- bourbon, butternut squash

Compound- 2-acetylfuran

Aroma- balsamic, cocoa, coffee, smoke, tobacco

We and (The Flavor Matrix) love to take credit for coming up with this incredible pairing.

However, we first sampled a version of this cocktail at the bar of the restaurant Redd in Yountville, CA. In the years that have past since first tasting it, Jamie has tinkered with the recipe until finally settling on this spice laden version which remains a staple on our refrigerator shelf throughout the Fall.

Spiced Butternut Squash Coulis

1 teaspoon coriander seed

1 teaspoon clove

1 teaspoon black peppercorn

2 cups water

2 cinnamon sticks

Zest of ½ lemon

1 teaspoon salt

2 cups butternut squash, cubed

Tie the coriander, clove and peppercorn in a sachet. Add to the water with cinnamon, lemon zest and salt. Bring the mixture to a simmer and add the butternut squash. Continue cooking until tender.

Pour the mixture through a strainer, reserve the liquid. Add the butternut squash cubes to a blender and pour in 4 oz of the cooking liquid. puree until smooth. Reserve in the refrigerator up to 2 weeks.

Cocktail

1 ounce spiced butternut squash coulis

1½ ounces bourbon

Club soda, to taste

Lemon twist for garnish

Combine the coulis and bourbon in a shaker filled halfway with ice. Shake to chill thoroughly.

Strain into a rocks glass filled with ice. Top with club soda and garnish with a lemon twist.

Mezcal Michelada

Inspiration Ingredients- mezcal, beer, coriander

Compound- gamma-terpinene

Aroma- bitter, citrus, gasoline, resin

Freshly ground coriander or crushed coriander seeds really make all the difference in this recipe as they bring powerful citrus aromas which are seriously lacking in jars of ground coriander. It is a tasty match for the smokiness of Mezcal. If you are serving at a party, the lime, mezcal, hot sauce and Worcestershire can be mixed in pitcher, then set out for guests to add their drinks according to their own tastes.

2 tablespoons kosher salt

1 teaspoon chili powder

1 teaspoon ground coriander

1 ounce lime juice

½ ounce mezcal

2 teaspoons Mexican-style hot sauce

2 dashes worchestershire sauce

12 oz light Mexican-style beer

Combine the kosher salt, chili powder and coriander in small, shallow dish and mix well. Rub the rim of a pint glass with a lime wedge and dip the rim of the glass into the seasoning mixture.

Pour the lime juice, mezcal, hot sauce and worchestershire sauce into the glass. Fill the glass with ice and slowly pour in the beer. Serve immediately.

Non-alcoholic

Raspberry Vanilla Soda

Inspiration Ingredients- vanilla, raspberry, thyme

Compound- geraniol

Aroma- geranium, lemon zest, rose, peach

We love to have the option of sipping on something that is delicious, sophisticated and alcohol free. The pairing of raspberry and thyme is bright, fresh and totally surprising. When making this recipe be careful to not let the syrup cook at a strong boil or those fresh flavors will begin to fade.

2 cups fresh raspberries

½ cup sugar

1 teaspoon vanilla extract

Zest and juice of 1 lemon

1 branch fresh thyme, leaves picked

Combine the raspberries, sugar, vanilla extract, lemon juice and zest and thyme leaves in a sauce pot. Stir and mash with a wooden spoon to break up the berries. Place the pot over medium low heat and bring the mixture to a simmer. Cook at a gentle simmer until the berries completely break down, 3 to 4 minutes.

Remove the pot from the heat and transfer the mixture to a fine mesh strainer set over a bowl. Slowly stir the mixture in the strainer until all of the liquid has drained and only the solids remain. Reserve the strained syrup.

Place 1 oz (two tablespoons) of the raspberry syrup in a tall glass and fill with ice. Fill the glass halfway with soda water and stir to thoroughly combine. Fill the remainder of the glass with soda and serve. Garnish with fresh thyme sprig.

Grapefruit Shrub

Ingredient Inspiration- grapefruit, honey, vinegar

Compound- carvacrol

Aroma- caraway, spice, thyme

Shrubs, also known as drinking vinegars, are an essential tool in your non-alcoholic mixology game. They are able to remain bright and crisp because the mixture is never heated. Acetic acid in the vinegar helps to breakdown the fruits and strip aromatics to preserve them in the resulting syrup. Let this recipe guide for creating shrubs with fruit and herb combinations.

2 large grapefruit

$\frac{3}{4}$ cup sugar

$\frac{1}{4}$ cup wildflower honey

1 cup Apple cider vinegar

Peel the zest from the grapefruit in large strips using a vegetable peeler. Trim the white pith from the grapefruit, and cut the fruit into pieces. Place the cut fruit, zest and sugar in a bowl and gently mix. Cover the bowl and leave at room temperature for 8-24 hours.

Stir in the honey and vinegar and refrigerate another 24 hours. Strain and reserve. The shrub may be consumed as is or diluted with water or club soda.

Spicy Peach Soda

Ingredient Inspiration- peach, pepper, ginger

Compound- limonene

Aroma- balsamic, citrus, fragrant, fruit, green, herb

The pairing between peach and jalapeno is the star of this recipe. However, if you want to capitalize on the fruity aromas of peppers without the spice, substitute slices of bell pepper for the jalapeno.

2 ripe to slightly over-ripe peaches

$\frac{1}{2}$ cup sugar

2 slices fresh jalapeno

Small pinch kosher salt

2 inch piece fresh ginger, peeled and thinly sliced

Juice of 1 lime

Remove the pits from the peach and cut into pieces. Combine the peaches, sugar, jalapeno, salt and ginger in a blender. Blend on high for 1 minute. Strain and reserve the puree.

Place 2 oz ($\frac{1}{4}$ cup) of the peach puree syrup in a tall glass and fill with ice. Fill the glass halfway with soda water and stir to thoroughly combine. Fill the remainder of the glass with soda and serve.

Homemade Cola

Ingredient Inspiration- citrus, star anise, cinnamon, mint

Compound- valencene

Aroma- citrus, green oil, wood

The flavor of your favorite cola is truly a wonder of chemistry. The mingling of complex spice and citrus aromas are perfectly balanced to create something that is far greater than it's individual parts. Making your own cola is a bit of a process, but the results are incredible.

Zest of 2 oranges

Zest of 1 lemon

Zest of 2 limes

1/4 teaspoon ground cinnamon

5 scrapes fresh nutmeg

1 pod star anise pod, crushed

1 tablespoon minced ginger

1 teaspoon vanilla extract

1½ cups sugar

½ cup brown sugar, packed

2 cups water

1 tablespoon mint leaves, loosely packed

Combine all ingredients, except for the mint in a sauce pot and bring to a boil. Remove the pot from heat, add the mint and cover the pot with a lid. Set aside for 10 minutes to infuse.

Line a colander with cheese cloth or a clean kitchen towel. Slowly pour the syrup through the cloth. Transfer to a clean container and refrigerate until thoroughly chilled.

Place 3 to 4 tablespoons of the cola syrup in a tall glass and fill with ice. Fill the glass halfway with soda water and stir to thoroughly combine. Fill the remainder of the glass with soda and serve.

Baked

The browning reaction that happens with amino acids and sugar breakdown under heat, also known as a Maillard Reaction is one of the most sure fire ways make food highly craveable. It's the reason that most grocery stores vent their bakery facilities at the front door. The smell of fresh baked anything can instantly induce hunger and get salivary glands flowing. Whether your baking is savory or sweet the final result will be determined by three main factors: gluten, moisture and leavening. To consistently reach the best result it is important to understand these elements.

Gluten. Much has been made of gluten recently and we will not get into anything health-related about gluten, rather our interest is it's function in baked goods which is structure. Think of gluten as scaffolding, a rigid structure that forms around air bubbles in baked goods. Gluten is a protein naturally found in wheat and other grains. When the gluten protein bonds with water—a process called hydration—it becomes elastic. The more a dough is mixed, the more elastic it will be become. However, mixing is not required for gluten development. Water, on the other hand is. You could a bowl full of dry flour for days on end and gluten would never develop; until water is added. Water causes gluten to develop; mixing speeds gluten development. The more gluten in a mixture, the denser and chewier the resulting baked item will be. In breads, gluten is desirable, in cakes it is less so. In other words, mix bread a lot, mix cakes as little as possible.

Moisture. Moisture is one of the trickier elements in baked good. Moisture is essential to form gluten and allow the elements of a dough or batter to come together. However, an excess of moisture will prevent the structure of gluten from properly forming and the baked good will “collapse.” On the other hand too little moisture will produce a dry, pasty or mealy result. This is why accurate measuring is important in recipes. All quality recipes are carefully formulated to have the proper amount of moisture allow of mixing and gluten development as well as structure.

It is also important to note that there are certain ingredients that can help retain moisture in baked goods without interfering with structure. Chief among these is sugar, which will bond with water and prevent it from evaporating. This is the reason that you will find many commercial baked goods to have high amounts of sugar, it helps to insure that the item will not dry out even after days or weeks of sitting on a shelf. Fruit and vegetable purees can also help retain moisture in baked goods as fiber in these ingredients bonds and holds water in the same way that sugar can. However, these items can have a greater impact on the structure of baked goods, so they should be added carefully.

Leavening. Leavening is the term bakers use for describing how air gets into an item. The most common kitchen leaveners are yeast, baking soda and baking powder. Here's how they work:

Yeast are microorganisms that digest sugars naturally present in the dough and release carbon dioxide and ethanol. Yeast is slower acting than chemical leaveners, typically requiring 3 or more hours for optimal results. Yeast produces a unique flavor in baked goods while chemical leaveners are flavor neutral. Yeast is a living organism and is therefore requires moisture and is sensitive to pH, it cannot survive in an environment that is too acid or basic. Yeast does not have to be mixed with liquid water before using, though it is typically a good idea to do so to insure that the yeast is active. When mixed with water, active yeast should produce a foam on the surface within about 5 minutes.

Baking Soda, or sodium bicarbonate produces gas whenever it comes in contact with a liquid acid. Most people are familiar with this process from their grade school experiments in volcano making. The process is no different in baked goods a liquid acid- buttermilk, yogurt, lemon juice, etc will meet baking soda in a batter and immediately release gas to lighten the batter. Once all of the acid has reacted, no more gas will be released. This means that if a baking soda leavened batter sits too long before being cooked, it will fall flat.

Baking Powder, works in a similar way to baking soda. Baking powder is a mixture of baking soda and cream of tartar an acid in powder form. Since both elements are present in powdered form any liquid (not just an acid) will produce the leavening reaction. Double-acting baking powder produces a second reaction when exposed to heat.

-Savory

Walnut and Olive Focaccia

Ingredient Inspiration- olive, walnut, rosemary

Compound- pentanal

Aroma- almond, bitter, malt, oil, pungent

Focaccia is a staple in our restaurant, Angelena's. We bake about a dozen loaves of plain focaccia a day just for our bread service. Not to mention what we go through for sandwiches. This recipe has a few special additions in olive, walnut and rosemary; all three ingredients have particularly strong pairings with the aroma of baked wheat.

18 ounces "00" flour or bread flour

12 fl ounces water

1½ teaspoons dry active yeast

1 teaspoon kosher salt

3 tablespoons olive oil

½ cup nicoise or kalamata olives, pitted and corasley chopped

½ cup walnut pieces

Combine all of the ingredients in the bowl of an electric mixer fit with a paddle attachment. Mix until the ball of dough that forms is smooth and clings to the dough hook, about 5 minutes.

Add the olives and chopped walnuts and mix 1 minute more to incorporate.

Without an electric mixer, combine the ingredients in a bowl and mix with a wooden spoon or spatula until a ball forms. It will be a bit of a shaggy mess at first. Continue to knead the dough in the bowl by hand until a smooth ball of dough forms. Add the olives and chopped walnuts and continue kneading to incorporate.

Transfer the dough to a lightly oiled bowl and cover with plastic wrap. Set aside at room temperature for 1 hour if baking the bread the same day. If preparing in advance, place the dough in the refrigerator overnight.

Spread the 3 tablespoons olive oil evenly over a 13x9 baking pan. Gently turn the dough out of the oiled bowl onto the pan. Dip your finger tips into the oil in the pan and gently press the dough out towards the corners of the pan. Cover the pan and set aside at room temperature for a minimum of 1 hour, up to 2½ hours.

When ready to bake, preheat oven to 425°F. Uncover the pan and gently press the dough out using just your finger tips until it fills the pan completely. Sprinkle the top of the bread with salt and rosemary.

Bake in a preheated oven 20 minutes. Remove the pan and cool on a rack 5 minutes, then remove the bread from the pan using a large metal spatula and cool the bread directly on a rack.

Cherry Tomato and Nectarine Crostata

Ingredient Inspiration- cherry tomato, nectarine, balsamic, mint

Compound- beta- damascenone

Aroma- cooked apple, floral, fruit, honey, tea

Crostata is a typically Italian dessert, a free form tart made with what ever fruit is ripe and bountiful. We decided to give it a savory twist, however- this recipe's beauty is in it's versatility. Served with cheese it is the perfect app, topped with a dollop of sweetened whipped cream it can be a great dessert.

Crostata Dough

12 ounces ap flour (about 2½ cups)
¼ cup granulated sugar
1 teaspoon baking powder
½ teaspoon kosher salt
4 oz cold butter, cut into cubes
2 eggs + 1 yolk, beaten (white reserved)

Filling

2 nectarine or peaches, pit removed and sliced ½ inch thick
2 cups cherry tomatoes, cut in half
1 tablespoon balsamic vinegar, plus more for garnish
1 tablespoon fresh mint, torn, plus more for garnish
1 teaspoon sugar
1 teaspoon salt
2 tablespoons corn starch
2 tablespoons sliced almonds

Combine the flour, sugar, baking powder, salt, and butter in the bowl of food processor.

Run the machine in 3 second bursts until the butter is cut into small bits (pea-sized).

Add the beaten eggs to the flour and process again in short bursts until a ball begins to form. It should still be broken and uneven.

Turn the mixture out onto a table gently squeeze together to form a ball. Wrap tightly and press into a thin disk. Refrigerate at least 30 minutes before rolling.

For the filling, combine the cut nectarines or peaches, tomatoes, balsamic, mint, sugar and salt. Gently mix with spatula to combine.

When well mixed, sprinkle the cornstarch over the mixture and continue mixing until absorbed.

Clear a large area for rolling out the dough and lightly flour the surface. Place the round if chilled dough in the center of the of the area and roll the dough out into a large circle. The rolled dough should be $\frac{1}{4}$ to $\frac{1}{2}$ inch thick and 18 to 20 inches in diameter.

□always roll from the center of the dough out towards the edge. It is best to always roll in one spot and rotate the dough, rather than turning the pin at various to roll out. If the dough is sticking to the surface as you roll, add more flour in small increments or roll the dough out on lightly flour parchment paper.

If the dough is not yet on a piece of parchment paper, carefully transfer it to one. Brush the entire surface of the dough with the reserved egg white. Using a slotted spoon, pile the nectarine and tomato mixture in the center of the dough, making sure maintain a 3 to 4 inch border of dough around the fruit on all sides. Reserve the remaining juices in the bowl

When all of the mixture has been added, fold one side the uncovered dough over to partially cover the piled fruits, then rotate the crostata and fold another side. Repeat the process until all

of the dough has been folded up and there is a large area of exposed fruits in the center of the tart.

Transfer the tart, and the parchment paper, to a baking pan. Pour the reserved juices into the center of the tart. Brush the surface of the folded dough with the egg white and sprinkle with toasted almonds.

Bake in a 350°F oven for 45 minutes. Cool at least 20 minutes before cutting, though this tart is excellent slightly warm to room temperature.

Sweet

Nutty Better Cookies

Inspiration Ingredients- peanut, fish sauce, cayenne

Compound- guaiacol

Aroma- burnt, medicinal, phenol, smoke, wood

These are a simplified version of our these amazing cookies that were “accidentally” created from The Flavor Matrix. One day after a demo of Flavor Matrix recipes Jamie found himself with loads of left over Spicy Fish Sauce Peanut Brittle (p. 65 in FM) so he decided to grind it all up until it was basically sugar again. He then used that sugar to make the most mind-blowingly unique chocolate chip cookies we had ever tasted. The fish sauce was imperceptible, but lent this incredible umami and savory-ness that was the perfect counterpoint to the chocolate chip cookies, while the cooked sugar made the cookies extra chewy.

Makes approx. 4 dozen

1 cup brown sugar

½ cup granulated sugar

½ cup roasted salted peanuts

2 teaspoon fish sauce or soy sauce

1/16 teaspoon cayenne pepper *□you don’t need to actually measure the cayenne, just know that it is the tiniest possible pinch! (you can always add more)*

8 oz butter, softened

2 eggs

1 teaspoon vanilla extract

2¼ cups ap flour

2 cups bittersweet chocolate chips

Combine the brown sugar, sugar, peanuts, fish sauce, and cayenne in a food processor and pulse in short bursts until finely chopped and well combined.

Combine the brown sugar mixture and the butter in a bowl and beat with an electric mixer until light and fluffy.

Beat in the eggs one at a time, fully incorporating one before adding the next. Beat in the vanilla extract. Add the flour and beat gently until just incorporated.

Lay a large sheet of plastic wrap on a clean work surface. Transfer the cookie dough to the plastic wrap and wrap tightly, squeezing out as much air as possible. Refrigerate the dough overnight before baking.

Scoop the chilled dough to desired size and place on a parchment lined baking sheet. Flatten the scoops slightly before baking.

Bake at 375°F for approximately 10 minutes

Bourbon-Pecan and Coffee Sweet Rolls

Ingredient Inspiration- bourbon, pecan, coffee

Compound- diacetylformoin

Aroma- caramel, toasted almond, butter

There are ingredients that we 're just meant to go together. They are loads of rich roasted and toasted aromas, plus the triple threat taste combo of salty, sweet and buttery. They're like a morning on Bourbon Street, but without the bad smells.

1 cup milk

1 package quick-rise yeast (2¼ teaspoons)

¼ cup granulated sugar, divided

3½ cups all-purpose flour

1 teaspoon salt

1 large egg, lightly beaten

6 tablespoons melted butter, divided

1 cup brown sugar

½ cup chopped pecans

1 tablespoon ground cinnamon

2 teaspoons ground coffee

Icing:

½ cup sour cream

1 teaspoon vanilla extract

1 cup powdered sugar

Preheat oven to 350°F

In a small bowl, whisk together the milk, yeast and 1 tablespoon of granulated sugar. Set aside.

In a large mixing bowl, combine the flour, remaining sugar and salt. Mix well.

Whisk the egg and 4 tablespoons melted butter into the milk mixture.

Stir the milk mixture into the dry ingredients and mix with a dough hook in an electric mixer, or by hand, until a dough forms, about 5 minutes.

Turn the dough out onto a lightly floured surface and knead until smooth. Return the dough to a lightly oiled bowl, cover with plastic wrap and set aside for 30 minutes.

In a small bowl, whisk together the remaining 2 tablespoons melted butter, brown sugar cinnamon, coffee, chopped pecans and butter and stir thoroughly to combine.

Turn the risen dough out onto a floured surface and roll into a rectangle (about 18 x 11). Sprinkle the brown sugar mixture evenly over the dough, gently pressing to help it hold in place.

Roll the dough along the long side like a jelly roll and pinch the seam together. Slice the roll crosswise into 1 inch pieces (there should be 18 pieces). *□to make the dough easier to slice, transfer the roll to the freezer for 15 minutes before cutting.*

Arrange the cut pieces into a 13x9 baking dish. Cover the baking pan and let rise 30 minutes at room temperature. Or refrigerate, covered overnight and bake the next morning.

Bake the rolls in preheated 350°F oven for 20 minutes or until lightly browned. Remove and rest the pan on a cooling rack for 5 minutes. Turn the rolls out onto the cooling rack and cool five minutes more.

In a small bowl whisk together the sour cream, vanilla extract and confectioner's sugar. Whisk until smooth. Spoon the glaze over the warm cinnamon rolls and serve immediately. *▫the glaze may also be made a day in advance if desired.*

Tropical Caramel Upside Down Cake

Ingredient Inspiration- citrus, banana, pineapple, caramel

Compound- furaneol

Aroma- burnt, caramel, cotton candy, honey, sweet, toasted

Butter, brown sugar, rum and tropical fruits are really a no brainer. Pineapple up-side down cake is a classic, so are Bananas Foster. So we got to thinking that these two things would taste pretty darn good if we put them together. Boy, were we right!

2 ounces butter

½ cup packed brown sugar

2 tablespoons dark rum

1 cup fresh pineapple, cut into 1 inch pieces

1 banana sliced into rounds ¼ inch thick

3 eggs, separated

1 cup sugar, divided

1 teaspoon vanilla extract

Zest, finely grated and juice of 1 orange

1 cup all-purpose flour

1 teaspoon baking powder

1/4 teaspoon salt

Preheat oven to 375°.

Combine the butter, brown sugar and rum in a large, nonstick sauté pan. Bring the mixture to a simmer over medium heat, stirring often. When the mixture is smooth and bubbling add the pineapple and banana and stir gently to coat. Remove the pan from the heat and cool slightly.

Arrange the pineapple and banana pieces in the base of a 9 inch round cake pan. Pour in the remaining brown sugar mixture over the fruit, making sure that it reaches all edges of the pan. Set the pan aside while you are the cake batter.

Combine the egg yolks and ½ cup sugar in a large bowl and beat with an electric mixer until light, fluffy and doubled in volume. While still mixing, add the vanilla, orange zest and juice.

In a separate bowl, mix together the flour, baking powder and salt. Add to the egg yolk mixture and beat to combine.

Place the reserved egg whites in a bowl and beat with an electric mixer. When very soft peaks begin to form, slowly pour in the remaining ½ cup of sugar while still beating. Continue beating until smooth, glossy peaks form. Fold the egg whites into the cake batter.

Bake 30 minutes or until a toothpick inserted in center comes out clean. Cool on a rack 5 minutes before inverting onto serving plate.

Sweet and Sour Plum Upside Down Cake

Ingredient Inspiration- lime, honey, vinegar, pound cake, cornmeal

Compound- gamma- decalactone

Aroma- apricot, fat, peach, pleasant, sweet

The only thing better than one up-side down cake is... two up-side down cakes. This one brings in some unexpected flavors that tamp down the sweetness and give this a cake a slightly savory edge. Plums and fennel are an incredible pairing, one that we explored in the Flavor Matrix and just couldn't get enough of, so we came back for more. Just like you'll do with this cake.

2 ounces butter

½ cup honey

2 tablespoons red wine vinegar

1 teaspoon fennel seeds (optional)

2 plums, pit removed and cut into 1/8ths

3 eggs, separated

1 cup sugar, divided

1 teaspoon vanilla extract

Zest, finely grated and juice of 1 orange

¾ cup all-purpose flour

¼ cup cornmeal

1 teaspoon baking powder

1/4 teaspoon salt

Preheat oven to 375°.

Combine the butter, honey, vinegar and fennel seeds in a large, nonstick sauté pan. Bring the mixture to a simmer over medium heat, stirring often. When the mixture is smooth and bubbling add the plums and stir gently to coat. Remove the pan from the heat and cool slightly.

Arrange the plums in the base of a 9 inch round cake pan. Pour in the remaining honey syrup over the fruit, making sure that it reaches all edges of the pan. Set the pan aside while you are the cake batter.

Combine the egg yolks and $\frac{1}{2}$ cup sugar in a large bowl and beat with an electric mixer until light, fluffy and doubled in volume. While still mixing, add the vanilla, orange zest and juice.

In a separate bowl, mix together the flour, cornmeal, baking powder and salt. Add to the egg yolk mixture and beat to combine.

Place the reserved egg whites in a bowl and beat with an electric mixer. When very soft peaks begin to form, slowly pour in the remaining $\frac{1}{2}$ cup of sugar while still beating. Continue beating until smooth, glossy peaks form. Fold the egg whites into the cake batter.

Bake 30 minutes or until a toothpick inserted in center comes out clean. Cool on a rack 5 minutes before inverting onto serving plate.

Spicy Chocolate Crinkle Cookie

Inspiration Ingredients- chile, coffee, cocoa

Compound- 2,3- butanediol

Aroma- cream, floral, fruit, herb,

We're both suckers for spicy and sweet combos. But these go beyond that simple pleasure. Fruity aromas are secondary in peppers, cocoa and coffee, but when they come together the flavors explode. Feel free to add as much or as little ancho powder as you like.

- 1 cup all-purpose flour
- 1/2 cup unsweetened cocoa powder
- 1 teaspoon baking powder
- 1 teaspoon ancho chile powder
- 1/4 teaspoon salt
- 1 cup granulated sugar
- 2 fl ounces vegetable oil or coconut oil
- 2 large eggs
- 2 teaspoons vanilla extract
- 1 cup powdered sugar, for coating the cookies
- 1 tablespoon espresso powder

Combine the flour, cocoa powder, baking powder, chile powder powder, and salt in a bowl and whisk to combine.

In a separate bowl, combine the sugar and oil. Beat vigorously with a whisk or electric mixer until light and fluffy. Add the eggs and vanilla and beat again.

Continue mixing gently or on low speed adding 1/3 of the flour mixture at a time, incorporating each addition before adding the next.

With the final addition of flour a firm ball of dough should form. Remove the dough from the bowl, wrap tightly in plastic wrap and press into a disk. Refrigerate for a minimum of 2 hours, ideally overnight.

When ready to bake, combine the powdered sugar and espresso powder in a bowl and whisk to combine. Use a tablespoon measure to scoop the dough making large rounded scoops that are approximately 2 tablespoons in size. Roll each scoop between your palms to make a ball. Roll each ball in the powdered sugar mixture then place on a parchment lined baking sheet. Press each ball of dough to partially flatten.

Bake in a 350°F oven for 10 minutes. Cool the cookies 2-3 minutes on the baking pan, then transfer to a cooling rack to cool completely.

Lime & The Coconut Popcorn Mousse

Ingredient Inspiration- popcorn, coconut, lime

Compound- ethyl acetate

Aroma- brandy, contact glue, grape, sweet

Popcorn milk is an ingredient and a recipe that Jamie has been obsessed with for quite some time. He's made popcorn gelato with peanut butter swirls, popcorn gelato, and even a savory popcorn cream to add an elegant finishing touch to pureed corn soup. But this popcorn mouse is probably the easiest, and most fun to eat of all of them. Coconut, lime and corn are an extraordinary pairing that capitalize on the strong aromas shared by all three.

Mousse

4 tablespoons butter

8 cups popped popcorn,

2 cups milk

1 cup sugar

1 tablespoon vanilla extract

1 teaspoon salt

1 tablespoon powdered gelatin

1 cup heavy whipping cream

1 cup powdered sugar

2 cups popped popcorn

2 tablespoon butter, melted

½ teaspoon fine sea salt

¼ cup Sweetened coconut flakes

Zest of 1 lime

Optional- sprinkle of chile powder or chili seasoning (like Tajin)

For the mousse, mix the gelatin with 2 tablespoons of water, in a small bowl and set aside to hydrate.

Melt the butter in a large sauce pot over medium heat. Add the popcorn and stir well to coat. Continue cook 3-5 minutes until the popcorn is lightly toasted.

Pour in the milk, sugar, vanilla extract and salt. Bring the mixture to a boil, stirring often. Reduce the heat and simmer 2-3 minutes. Cover the pot and whisk in the gelatin, stirring to completely dissolve. Remove the pot from the heat. Set aside to infuse for 15-20 minutes.

In a separate clean bowl, combine the whipping cream and powdered sugar. Whisk or beat on high speed with an electric mixer until stiff peaks form. Refrigerate until ready to use.

Strain the milk into a glass or metal bowl, through a fine mesh strainer. Press the popcorn with the back of a wooden spoon to extract as much liquid as possible. Discard the pressed popcorn bits. Place the milk into the refrigerator to chill or place the bowl over another bowl half filled with ice water and whisk chilled. Let the chilled milk stand 15-30 minutes over the ice to begin to set.

Add 1/3 of the whipped cream to the milk and gently stir with a whisk to combine. Fold another 1/3 into the mixture. When completely incorporated, add the final 1/3 and gently fold to combine. Divide among smaller serving dishes and place in the refrigerator.

Preheat oven to 325°F

For the garnish, combine the popcorn, butter, salt and coconut flakes in a bowl and toss well to combine. Spread the mixture into an even layer on baking sheet and toast in the oven until the coconut is lightly browned and aromatic. Remove from the oven and finely grate lime zest over

the tray. Cool completely on the baking tray. If not serving immediately, store in an air tight container or zip top bag at room temperature.

To serve, garnish the mouse with toasted popcorn and sprinkle with chile seasoning.

Carrot Cake Creme Brulee

Inspiration Ingredients- carrots, walnut

Compound- Maltol

Aroma- caramel, cotton candy, malt, toasted bread, toasted nut

We LOVE carrot cake. When looking at the strength of pairings were between carrot cake ingredients- particularly carrots and walnuts- we decided it was time for a new take on the classic. Both carrots and walnuts also have a string affinity for heavy roasted and caramelized aromas, thus carrot cake crème brulee was born.

2 tablespoon butter

1 cup carrots, peeled and thinly sliced

¼ teaspoon kosher salt

¼ cup brown sugar, packed

2 cups milk

1 tablespoon walnut oil (optional)

2 egg

2 egg yolks

Granulated sugar, as needed

Preheat oven to 315°F

Melt the butter in a large sauce pot over medium heat. Add the carrots and salt. Sauté, stirring often until completely tender. *□The goal here is get the carrot tender and concentrate their flavor. This is done by cooking as much of the water out of the carrot as possible. Try to avoid browning throughout the cooking process, if the bottom of the pot begins to brown a bit, add water, 1 tablespoon at a time, to help the carrots cook without browning. When the carrots are cooked, it should be easy to crush the pieces with the back of a spoon.*

When the carrots are tender, add the sugar and milk. Bring the mixture to a boil and simmer 5 minutes. Transfer the contents to a blender and process until smooth. With the blender running, add the walnut oil, eggs and yolks.

Strain the blended mixture through a fine mesh strainer into a liquid measuring cup or small pitcher.

Place 4 brulee dishes of oven proof ramekins inside a larger baking dish. Pour the custard mixture equally into each baking dish. Fill the larger baking dish with water so that the water comes halfway up the side of the brulee dish.

Cover the large baking dish with plastic wrap. Carefully transfer to the oven and bake for 45 minutes. Remove the pan from the oven, remove the plastic wrap and cool on the counter top for 15-20 minutes. Transfer the dishes to the refrigerator to cool completely. When chilled wrap each brulee individually until ready to serve.

TO serve, cover the top of each brulee with a thin layer of granulated sugar and burn with a torch. Serve immediately.

Rhubarb and Bourbon Zabaglione

Ingredient Inspiration- rhubarb, bourbon

Compound- Mesifurane

Aroma- bread crust, butter, caramel

Think of this dessert as strawberries and cream that spent the summer traveling through Europe and came back with an accent. But not in an annoying way. Mesifurane, plays an unexpectedly important role in the flavor of both strawberries and rhubarb, making both a great match for the roasted and woody aromas of bourbon. Strawberries are a great substitute if you can't find rhubarb at your local market.

Poached rhubarb

2 stalks rhubarb

1 orange

2 cups water

1 cup sugar

2 sprigs fresh mint, plus more for garnish

To prepare the rhubarb, trim 1 inch from the thick, stalk end and cut ¼ inch from the top of the stalk. □ *You are just trimming away the dry, tough ends from the rhubarb. Larger stalks may have very woody stalks that require more trimming. The knife should trim the end fairly easily, if it is tough to cut or very fibrous, cut away more.*

Peel the stalks and cut into pieces that are approximately 4 inches long and 1-2 inches wide.

Finely grate the zest of the orange and reserve a small amount for garnish. After zest, juice the orange.

In a wide bottomed sauce pot, bring the water and sugar to a boil. Reduce the heat to a simmer and add the orange zest, juice, mint, and the cut pieces of rhubarb. Gently simmer until the rhubarb pieces are tender through and easily pierced with the tip of a knife, 3-5 minutes. Remove the pot from heat and let the rhubarb cool in the syrup. Reserve at room temperature if serving within 6 hours. Refrigerate if holding for longer. *□save the rhubarb syrup, its great for cocktails (particularly gin and bourbon), or can be mixed with soda water for a refreshing non alcoholic drink.*

Bourbon Zabaglione

6 egg yolks

2 fl oz bourbon

1 tablespoon vanilla extract

¼ cup + 2 tablespoons sugar

1 cup heavy whipping cream

½ cup powdered sugar

Set a pot of water on the stove and bring to a simmer.

Combine the egg yolks, bourbon, vanilla and sugar in a heat proof bowl. Whisk or beat with an electric mixer to combine.

Set the bowl over the pot of simmer water and continue whisking vigorously or beat with an electric mixer on low speed for 6-8 minutes or until thick ribbons form and the mixture begins to hold soft peaks. *□This is the same process for the first stage of making hollandaise. Because it has a much more liquid mixed with the eggs than hollandaise would it will take longer for them to reach the ribbon stage. However, patience is key here. The eggs yolks must be gently cooked to just the right doneness- if overcooked, they will curdle and if undercook, the zabaglione will be soupy and loose.*

Remove the bowl from the heat and set over a pot of ice water. Continue beating or whisking until chilled.

Place a clean bowl over the ice and add the cream and powdered sugar and beat or whisk until stiff peaks form. Fold the whipped cream into the cooled zabaglione.

Store in covered in the refrigerator until ready to serve. To serve, spoon the zabaglione over the rhubarb and garnish with fresh orange zest and mint.

The King of Brownies

Ingredient Inspiration- peanut butter, banana, chocolate

Compound- 3- hydroxy-2- methyl-4- pyrone

Aroma- caramel, cotton candy, malt, toasted bread, toasted nut

The flavors here are more familiar, it's the technique that makes these brownies amazing.

Bananas and peanut butter work double duty as they not only make the brownies taste great, them make them impossibly gooey and rich tasting without overloading on fat and sugar.

2 ripe bananas

1 egg

1 cup brown sugar, divided

1 teaspoon vanilla extract

½ cup unsalted butter

¾ cup all-purpose flour

¼ cup cocoa powder

½ cup creamy peanut butter

Preheat oven to 350°F

Grease a 9x9 baking pan.

Combine the bananas, egg, ½ cup brown sugar and vanilla in a bowl and beat with an electric mixer until smooth.

Combine the remaining ½ cup brown sugar and butter in a small sauce pot. Cook over medium heat, stirring often until the sugar is melted. Remove from the heat and add the chocolate chips, stir until smooth.

Beat the butter/chocolate mixture into the bananas. Add the flour and cocoa powder and stir to combine.

Transfer the batter to the prepared baking pan. Drop spoonfuls of the peanut butter over the top of the batter and swirl with the back of a spoon.

Bake for 25 minutes, or until the edges are set and a pick inserted into the center comes out slightly wet. Cool completely on a rack before cutting.

Not-That-Kind-of-Mushroom Brownies

Ingredient Inspiration- mushroom, cocoa

Compound- 3-octanol

Aroma- citrus, moss, mushroom, nut, oil

This is a recipe we came up with when promoting The Flavor Matrix. Mushrooms and cocoa are such an interesting combination, and we wanted to simple way to showcase it. In The Flavor Matrix you can find our recipe for a Porcini, Hazelnut and Chocolate Torte, which makes an elegant dessert, but these fudgy brownies are unique and snackable.

4 eggs

1 cup Dutch Process Cocoa Powder

1 tablespoon mushroom powder (porcini is best)

1 teaspoon salt

1 teaspoon baking powder

1 tablespoon vanilla extract

2 sticks unsalted butter

2 cups sugar

¼ cup brown sugar

1½ cups whole wheat flour or all-purpose flour

2 cups semi-sweet chocolate chips

Preheat oven to 350°F

Grease a 9 x 13 making pan.

In a bowl, combine the eggs, cocoa powder, mushroom powder, salt, baking powder, and vanilla. Whisk or beat with an electric mixer until smooth.

Combine the butter and sugars in a small sauce pot and stir over medium heat until the butter is melted and the sugar dissolved. *□melting the butter and sugar together will help produce a shiny crust on top of the baked brownies.*

Remove the pot from the heat and add the flour and chocolate chips. Whisk until smooth.

Pour the chocolate/butter mixture into the eggs and beat again until smooth. Transfer the batter to the greased pan and smooth the top with a spatula.

Bake for 30 minutes, or until a pick inserted into the center of the pan come out clean. Place on a rack and cool completely before cutting.

Brown Butter

The next three recipes incorporate brown butter. Making brown butter is one of the simplest things you can do in your kitchen to amp up roasted, nutty aromas in food. These are particularly desirable in desserts where the aromas pair particularly well with vanilla and generally enhance flavor.

Brown butter is made by toasting the milk solids in whole butter. To do this, you must first melt the butter and boil all of the water out of the butter (a process known as clarifying). Once the butter is clarified, you will see the milk solids collected at the bottom reduce the heat under the pot and watch carefully as the solids begin to toast. Brown butter can be used at any stage, when it has a light amber color, it will have a very delicate flavor that becomes more nutty and robust as the butter darkens. At it's deepest and richest the butter will look nearly black before straining, but be careful. At this stage it quickly turns from brown butter to burnt butter.

Always cool the butter in the pot, remember that the butter will continue to cook after being remove from the it will turn 1-2 shades darker. Or strain immediately into a separate heat proof container. Do not pour it directly into any plastic container, the butter is extremely hot and melt right through the container.

Brown butter can be made in large batches and stored in the refrigerator or freezer indefinitely.

Brown Butter Espresso Carrot Cake

Inspiration Ingredients- walnut, espresso, carrot

Compound- 5-methylfurfural

Aroma- almond, caramel, cooked, roasted garlic, spice

We explored the pairing of toasted nuts, espresso, carrot in The Flavor Matrix coffee and 5 spice roasted carrots with toasted almonds and herbed yogurt. Now we're taking it for a spin on the sweet side. Though not necessary, whole wheat flour adds a little extra nuttiness to the cake.

Brown butter made from 2 sticks of butter

3 eggs

1 cup granulated sugar

½ cup brown sugar

2 fl oz milk

½ teaspoon kosher salt

2 cups whole wheat flour

¼ cup all purpose flour

1 teaspoon baking powder

½ teaspoon baking soda

½ teaspoon ground cinnamon

3 cups grated carrots

1 cup chopped walnuts

Icing

½ stick butter

4 ounces cream cheese

½ cup powdered sugar

1 teaspoon espresso powder or ground coffee

½ teaspoon vanilla extract

Preheat oven to 350°F

Grease a 9 inch loaf pan and set aside.

Brown the butter, then set aside to cool.

In a bowl, combine the eggs, sugar, brown butter, milk and salt. Whisk or beat with an electric mixer to combine.

In a separate bowl, mix together the flour, baking powder, baking soda, and cinnamon. Add to the egg mixture and stir until just combined.

Add the carrots and chopped nuts and stir by hand to combine.

Pour the batter into the greased pan and bake for 50 minutes, or until a pick inserted into the center of the cake comes out clean. Cool in pan.

For the icing, combine the butter, cream cheese, sugar, espresso powder and vanilla in a clean bowl and beat together until smooth.

When the cake is cooled, spread the cream cheese over the loaf in a thick layer.

Brown Butter Ice Cream Sundae

Ingredient Inspiration- brown butter, vanilla

Compound- anisaldehyde

Aroma- almond, anise, caramel, mint, sweet

Vanilla loves roasted, nutty aromas. This recipe brings a triple threat of flavor with nutty quinoa, roasted butter and caramel. Not only that, quinoa in dessert means you can tell yourself that your ice cream indulgences are healthy. Score.

4 oz stick butter

2 cups water

1 cup quinoa, rinsed

½ teaspoon salt

2 cups sugar

2 tablespoons water

¼ cup sesame seeds

2 teaspoons vanilla extract

1 teaspoon large flake sea salt

Cook the butter to medium brown on the stove top and set aside to cool slightly.

Bring the water to a boil. Add the quinoa and salt and stir well. Place a lid on the pot and turn the heat to low. Cook, covered, without stirring until all of the liquid is absorbed, about 15 minutes. Remove the quinoa from the heat and spread into a thin layer on a baking sheet to steam dry.

Line a baking sheet with wax paper and grease the paper with nonstick cooking spray.

In a heavy saucepot, stir together the sugar and water. Place the pot over medium-high heat and cook, swirling the pot occasionally as the sugar melts; do not stir. If you notice crystals forming around the side of the pot, wipe the inside with a brush moistened with water to wash the crystals back into the caramel. *□ Making caramel can be a bit tricky, if any small solid get introduced to the pan, the sugar will begin to crystalize around it and ruin the caramel. This is why you should avoid stirring and only use water to clean down the edges of the pot.*

Continue cooking the caramel at a simmer until it has a deep brown color, 12 to 15 minutes. The temperature should register between 330°F and 350°F on a candy or deep- fry thermometer. As soon as the caramel is ready, reduce the heat and stir in the quinoa, brown butter and vanilla. Mix gently to avoid splattering or spilling the caramel. Cook 3-5 minutes more to expel any remaining moisture from the quinoa. Remove the pot from the heat and stir in the sesame seeds. *□ it is important to make sure you cook out all of the moisture in this step or the brittle will not get crunchy. If you find your caramel a bit soft, spreading it into a thin layer on a silicon baking mat and placing in the oven at 325°F for 10-20 minutes may help eliminate the excess water.*

Nutella and Brown Butter Squash Panini

Ingredient Inspiration- vanilla, squash, Nutella

Compound- 2,3 pentanedione

Aroma- bitter, butter, caramel, fruit, sweet

Veggies for dessert? We say, 'why not!?' The combination is based on squash's string affinity roasted and nutty aromas. You can make these crazy good sandwiches with roasted butternut squash or sweet potatoes, though we prefer the way less sweet butternut squash taste. Either way, these are a completely unexpected sweet treat that everyone is going to love.

2 ounces butter, plus more as needed

2 cups butternut squash peeled and cut into ¼-1/2 inch slices

1 tablespoon vanilla extract

½ cup powdered sugar, plus more as needed

1 cup chocolate hazelnut spread

8 slices white Pullman loaf or cinnamon raisin bread

Brown the butter in a large sauté pan. Add the sliced butternut squash to the hot pan and season with salt. Sauté, stirring often, until just tender.

Add the vanilla extract and powdered sugar to the pan and stir until the sugar is completely dissolved.

Set aside to cool.

Spread 1 to 2 tablespoons of the chocolate hazelnut spread on the one side of each of the slices of bread. Make sandwiches by filling one side of the chocolate covered bread with cooked slices of squash, then closing with a second piece of bread.

Spread the outside of each sandwich with a thin layer of butter. Cook each sandwich in a sandwich press or nonstick saute pan until toasted and golden on the outside.

Cut the finished sandwiches diagonally and arrange on plates. Dust with powdered sugar immediately before serving.

Chocolate and Red Wine Bread Pudding

Ingredient Inspiration- merlot, chocolate, whiskey

Compound- 2-propenal

Aroma- burnt, sweet, pungent

These are a ~~few~~ ALL of Brooke's favorite things. We have always enjoyed nibbling on a little dark chocolate while sipping red wine or a little bourbon at the end of a meal. It's clear why- these three ingredients are a very strong match based on their aroma profiles. They all develop similar toasted aromas in different ways- bourbon and wine draw them from the wooden barrels they are aged in while coca beans develop them in the drying and roasting process. How ever it happens, the result is magical.

1 cup sugar

4 eggs

¼ cup cocoa powder

2 teaspoons vanilla extract

2 cups milk

Pinch nutmeg

⅛ teaspoon cinnamon

1 (12-inch) loaf French bread, crusts trimmed and cut into 1-inch cubes

4 ounces butter, cut into cubes

2 cups bittersweet chocolate

1 cup shelled pistachios (optional)

Merlot Sauce

8 ounces merlot

1 cup sugar

1 cup cream

Grease a 13x9 baking pan. Preheat oven to 350° F.

In a large bowl, combine the sugar, eggs, cocoa powder and vanilla, whisk until smooth. Pour in the milk, nutmeg and cinnamon and continue whisking to combine.

Stir in the bread, mix well and set aside to soften, 30 minutes. *□The bread should soak up nearly all of the custard, occasionally stir the bowl from the bottom to evenly distribute the custard.*

When the bread is completely softened, add the butter, chocolate chips and pistachio. Stir well. Transfer the mixture into the prepared baking dish.

Bake, uncovered for 45 minutes or until the center of the pan is just set. The surface should be browned and firm to the touch with a bit a give. When you lightly press in the center, no liquid should rise to the surface. Remove from the oven and cool on a rack at least 15 minutes before serving.

To prepare the sauce, combine the wine and sugar in a small saucepot and bring to a boil. Simmer until reduced by half, about 8 minutes. Add the cream and cook 2 minutes more.

To cut the bread pudding and place on plates, then drizzle with generous amount of the Merlot sauce. Store any excess bread pudding tightly wrapped in the refrigerator and reheat to serve.

Vernaccia-Poached Pears with Pecorino

Ingredient Inspiration- Vernaccia, pear, Pecorino

Compound- 2-phenylethanol

Aroma- honey, lilac, and rose

When researching THE FLAVOR MATRIX, we learned that there really is something to the adage “what grows together, goes together” because many flavor-creating aromas are imprinted into foods by their environment. With our pears, we bring together the best of Tuscany—Vernaccia di San Gimignano and the Val d’Orcia’s best sheeps-milk cheese, Pecorino. 2-phenylethanol is a major compound found in Vernaccia grapes with its aromas of honey, lilac, and rose. Pear, being a member of the Rosaceae family, carries many of these floral aromas as well. Most impressively, this recipe is all-at-once salty, tart, piquant, and sweet.

Serves 8

1 bottle Vernaccia

1 whole cinnamon stick

2 teaspoons whole black peppercorns

2 teaspoons vanilla extract

2 pieces star anise

1 cup sugar

Zest of 1 lemon

4 Bosc pears, peeled, halved, and cored

¼ lb Pecorino, finely grated

Combine the wine, cinnamon, peppercorns, vanilla, star anise, sugar, and lemon zest in a saucepot. Add the pears and enough water to cover completely.

Bring the mixture to a boil. Reduce the heat to a simmer, cover the pot and cook until the pears are just tender to the tip of a knife. Remove from the heat and cool in the poaching liquid. Serve with whipped cream, garnished with finely grated pecorino cheese.